Statistics

Chapter 14 — Instructor Solutions Manual

October 24, 2026 • Total Exercises: 35

PROBLEM STATEMENT

Chapter 14, Page 517

Solution A

Step 1

Given:

$$\begin{aligned}\hat{y} &= 64,100 + 0.394x_1 + 9.6x_2 - 11,600x_3\\ x_1 &= \text{regional population}\\ x_2 &= \text{per capita income}\\ x_3 &= \text{regional unemployment rate}\\ y &= \text{monthly sales}\end{aligned}$$

(a) The full name of the given equation is the **multiple regression equation**.

Step 2

(b) The number 64,100 is the constant of the given equation, which represents the estimated y-value when $x_1 = x_2 = x_3 = 0$.

The estimate monthly sales for a regional population of 0 with a 0 per capita income and a regional unemployment rate of 0% is $\$64,100$.

Note that this interpretation does not make any sense as a population of 0 would correspond with an estimated monthly sales of $\$0$.

Step 3

(c) Evaluate the multiple regression equation at $x_1 = 796,000$, $x_2 = 6940$, and $x_3 = 6.0$.

$$\begin{aligned}\hat{y} &= 64,100 + 0.394(796,000) + 9.6(6940) - 11,600(6.0)\\ &= 374,748\end{aligned}$$

Thus the estimate monthly sales is $\$374,748$.

Result

(a) Multiple regression equation

(b) The estimate monthly sales for a regional population of 0 with a 0 per capita income and a regional unemployment rate of 0% is $\$64,100$.

(c) $\$374,748$

Solution B

Step 1

Given:

$$\begin{aligned}\hat{y} &= 64,100 + 0.394x_1 + 9.6x_2 - 11,600x_3 \\ x_1 &= \text{regional population} \\ x_2 &= \text{per capita income} \\ x_3 &= \text{regional unemployment rate} \\ y &= \text{monthly sales}\end{aligned}$$

(a) The full name of the given equation is the **multiple regression equation**.

Step 2

(b) The number 64,100 is the constant of the given equation, which represents the estimated y-value when $x_1 = x_2 = x_3 = 0$.

The estimate monthly sales for a regional population of 0 with a 0 per capita income and a regional unemployment rate of 0% is $\$64,100$.

Note that this interpretation does not make any sense as a population of 0 would correspond with an estimated monthly sales of $\$0$.

Step 3

(c) Evaluate the multiple regression equation at $x_1 = 796,000$, $x_2 = 6940$, and $x_3 = 6.0$.

$$\begin{aligned}\hat{y} &= 64,100 + 0.394(796,000) + 9.6(6940) - 11,600(6.0) \\ &= 374,748\end{aligned}$$

Thus the estimate monthly sales is $\$374,748$.

Result

(a) Multiple regression equation

(b) The estimate monthly sales for a regional population of 0 with a 0 per capita income and a regional unemployment rate of 0% is $\$64,100$.

(c) $\$374,748$

PROBLEM STATEMENT

Chapter 14, Page 518

Solution A

Step 1

Given:

$$\begin{aligned}\hat{y} &= 11.6 + 0.4x_1 + 0.286x_2 + 0.112x_3 + 0.002x_4\\ x_1 &= \text{length of time}\\ x_2 &= \text{mechanical aptitude test score}\\ x_3 &= \text{prior on-the-job rating}\\ x_4 &= \text{age}\\ y &= \text{performance}\end{aligned}$$

(a) The full name of the given equation is the **multiple regression equation**.

Step 2

(b) A **response variable** (or dependent variable) is a variable that measures an outcome or result of a study, while an **explanatory variable** (or independent variable) is a variable that we think explains or causes changes in the response variables.

The dependent variable is the y-variable and thus there is 1 dependent variable.

The independent variables are the x-variables. In this case, there are then 4 independent variables: x_1, x_2, x_3, x_4.

Step 3

(c) The number 0.286 is the coefficient of x_2.

The coefficient of x_2 represents the average increase (or decrease when negative) of the y-variable per unit of x_2.

The performance on the new machine increases on average by 0.286 per point on the mechanical aptitude score.

Step 4

(d) The coefficient of x_4 is 0.002.

The coefficient of x_4 represents the average increase (or decrease when negative) of the y-variable per unit of x_4.

This then implies that the performance on the new machine increases on average by 0.002 when the age increases by one year.

Step 5

(e) Evaluate the multiple regression equation at $x_1 = 6$, $x_2 = 280$, $x_3 = 97$ and $x_4 = 35$.

$$\begin{aligned}\hat{y} &= 11.6 + 0.4(6) + 0.286(280) + 0.112(97) + 0.002(35) \\ &= 105.014\end{aligned}$$

Thus the estimate performance is 105.014.

🎯 Result

(a) Multiple regression equation

(b) 1 dependent variable and 4 independent variables.

(c) Coefficient of x_2

(d) 0.002

(e) 105.014

PROBLEM STATEMENT

Chapter 14, Page 518

Solution A

Step 1

$\mathbf{a.})$ The given multiple linear model is

$$\hat{y} = 16.24 + 0.017x_1 + 0.0028x_2 + 42x_3 + 0.0012x_4$$

where:

- $\hat{y}$ is the index of satisfaction of a person,
- x_1 is the age at time of first marriage, x_2 is the annual income,
- x_3 is the number of children,
- x_4 is the value of all asset,
- x_5 is the status of health in the form of index,
- and x_6 is the average number of the of social activities per week.

Step 2

Substitute when x_1 is 18, x_2 is $26,500$, x_3 is 3, x_4 is $156,000$, x_5 is 141 and x_6 is 2.5 into the given multiple linear model to measure the index of satisfaction of a person, thus

$$\begin{aligned}\hat{y} &= 16.24 + 0.017x_1 + 0.0028x_2 + 42x_3 + 0.0012x_4 \\ &\quad +0.19x_5 + 26.8x_6 \\ &= 16.24 + 0.017(18) + 0.0028(26,500) + 42(3) \\ &\quad +0.0012(156,000) + 0.19(141) + 26.8(2.5) \\ &= 497.736\end{aligned}$$

Step 3

In conclusion for (a), the index of satisfaction for the person using the multiple linear regression is

$$\hat{y} = 497.736$$

Step 4

b.) To identify which statements will increase the satisfaction of the person, compare the two results where the annual income of the first result is increased by $10,000$ while the social activities per week of the second result is increased by 2.

Step 5

Substituting x_2 is when $26,500$ is increased by $10,000$ and the given values into the model to find the satisfaction index For additional $\$10,000$ annual income, thus

$$\begin{aligned}\hat{y} &= 16.24 + 0.017x_1 + 0.0028x_2 + 42x_3 + 0.0012x_4 \\ &\quad +0.19x_5 + 26.8x_6 \\ &= 16.24 + 0.017(18) + 0.0028(26,500 + 10,000) + 42(3) \\ &\quad +0.0012(156,000) + 0.19(141) + 26.8(2.5) \\ &= 525.736.\end{aligned}$$

Substituting x_6 is when 2.5 is increased by 2 and the given values into the model to find the satisfaction index for additional two social activities per week, thus

$$\begin{aligned}\hat{y} &= 16.24 + 0.017x_1 + 0.0028x_2 + 42x_3 + 0.0012x_4 \\ &\quad +0.19x_5 + 26.8x_6 \\ &= 16.24 + 0.017(18) + 0.0028(26,500) + 42(3) \\ &\quad +0.0012(156,000) + 0.19(141) + 26.8(2.5 + 2) \\ &= 551.336.\end{aligned}$$

Step 6

By comparing the two results in the original result, an additional annual income $\$10,000$ have an increase of 28 satisfaction while the additional two social activities per week have an increase of 53.6. Therefore, having two more social activities will yield more satisfaction.

Step 7

In conclusion for (b), having two more activities per week will increase the satisfaction by 53.6.

Result

$$a.)\ \hat{y} = 497.736$$
$$b.)\ \text{Two more social activities a week}$$

Solution B

Step 1

a.) The given multiple linear model is

$$\hat{y} = 16.24 + 0.017x_1 + 0.0028x_2 + 42x_3 + 0.0012x_4$$

where:

- $\hat{y}$ is the index of satisfaction of a person,
- x_1 is the age at time of first marriage, x_2 is the annual income,
- x_3 is the number of children,
- x_4 is the value of all asset,
- x_5 is the status of health in the form of index,
- and x_6 is the average number of the of social activities per week.

Step 2

Substitute when x_1 is 18, x_2 is $26,500$, x_3 is 3, x_4 is $156,000$, x_5 is 141 and x_6 is 2.5 into the given multiple linear model to measure the index of satisfaction of a person, thus

$$\begin{aligned}\hat{y} &= 16.24 + 0.017x_1 + 0.0028x_2 + 42x_3 + 0.0012x_4 \\ &\quad +0.19x_5 + 26.8x_6 \\ &= 16.24 + 0.017(18) + 0.0028(26,500) + 42(3) \\ &\quad +0.0012(156,000) + 0.19(141) + 26.8(2.5) \\ &= 497.736\end{aligned}$$

Step 3

In conclusion for (a), the index of satisfaction for the person using the multiple linear regression is

$$\hat{y} = 497.736$$

Step 4

$\mathbf{b.})$ To identify which statements will increase the satisfaction of the person, compare the two results where the annual income of the first result is increased by $10,000$ while the social activities per week of the second result is increased by 2.

Step 5

Substituting x_2 is when $26,500$ is increased by $10,000$ and the given values into the model to find the satisfaction index For additional $\$10,000$ annual income, thus

$$\begin{aligned}\hat{y} &= 16.24 + 0.017x_1 + 0.0028x_2 + 42x_3 + 0.0012x_4 \\ &\quad +0.19x_5 + 26.8x_6 \\ &= 16.24 + 0.017(18) + 0.0028(26,500 + 10,000) + 42(3) \\ &\quad +0.0012(156,000) + 0.19(141) + 26.8(2.5) \\ &= 525.736.\end{aligned}$$

Substituting x_6 is when 2.5 is increased by 2 and the given values into the model to find the satisfaction index for additional two social activities per week, thus

$$\begin{aligned}\hat{y} &= 16.24 + 0.017x_1 + 0.0028x_2 + 42x_3 + 0.0012x_4 \\ &\quad +0.19x_5 + 26.8x_6 \\ &= 16.24 + 0.017(18) + 0.0028(26,500) + 42(3) \\ &\quad +0.0012(156,000) + 0.19(141) + 26.8(2.5 + 2) \\ &= 551.336.\end{aligned}$$

Step 6

By comparing the two results in the original result, an additional annual income $\$10,000$ have an increase of 28 satisfaction while the additional two social activities per week have an increase of 53.6. Therefore, having two more social activities will yield more satisfaction.

Step 7

In conclusion for (b), having two more activities per week will increase the satisfaction by 53.6.

🎯 Result

$$a.)\ \hat{y} = 497.736$$
$$b.)\ \text{Two more social activities a week}$$

PROBLEM STATEMENT

Chapter 14, Page 518

Solution A

Step 1

Given:

$$\begin{aligned}\hat{y} &= 62.65 - 1.86x_1 - 0.52x_2\\ x_1 &= \text{thickness of insulation}\\ x_2 &= \text{outdoor temperature}\\ y &= \text{monthly natural gas consumption}\end{aligned}$$

(a) Evaluate the multiple regression equation at $x_1 = 6$ and $x_2 = 40$.

$$\begin{aligned}\hat{y} &= 62.65 - 1.86(6) - 0.52(40)\\ &= 30.69\end{aligned}$$

Thus the estimates natural gas used is 30.69 cubic feet.

Step 2

(b) The coefficient of the thickness of insulation x_1 is -1.86.

The coefficient of x_1 represents the average increase (or decrease when negative) of the y-variable per unit of x_1.

This then implies that the monthly natural gas consumption is expected to decrease by 1.86 cubic feet when the insulation increases by 1 inch.

Step 3

(c) The coefficient $b_1 = -1.86$ is negative, because the monthly natural gas consumption tends to decrease as the thickness of insulation increases. This makes sense, because more insulation will result in less temperature changes indoor and thus we will require less gas consumption to heat the house.

The coefficient $b_2 = -0.52$ is negative, because the monthly natural gas consumption tends to decrease as the outdoor temperature increases. This makes sense, because a higher temperature will result in a higher indoor temperature and thus we will require less gas to heat the house.

🎯 Result

(a) 30.69 cubic feet

(b) Decrease of 1.86 cubic feet

(c) The monthly natural gas consumption tends to decrease as the thickness of insulation increases. The monthly natural gas consumption tends to decrease as the outdoor temperature increases. Both make sense.

PROBLEM STATEMENT

Chapter 14, Page 523

Solution A

Step 1

(a) The error sum of squares is given in the row "Residual error" and in the column "SS" of the ANOVA table:

$$SSE = 583.693$$

The error degrees of freedom is given in the row "Residual error" and in the column "DF" of the ANOVA table:

$$df_E = 62$$

The standard error of the estimate is the square root of the error sum of squares divided by the error degrees of freedom:

$$s = \sqrt{\frac{SSE}{n-(k+1)}} = \sqrt{\frac{SSE}{df_E}} = \sqrt{\frac{583.693}{62}} \approx 3.0683$$

95% of the residuals are expected to be two standard errors of the estimate from 0 (as the expected value of the residuals is 0).

$$2s = 2(3.0683) = 6.1366$$

Thus we expect 95% of the residuals to be between -6.1366 and 6.1366.

Step 2

(b) The regression sum of squares is given in the row "Regression" and in the column "SS" of the ANOVA table:

$$SSR = 77.907$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table:

$$SS\ total = 661.600$$

The coefficient of multiple determination is the regression sum of squares divided by the total sum of squares:

$$r^2 = \frac{SSR}{SS\ total} = \frac{77.907}{661.600} \approx 0.1178 = 11.78\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

11.78% of the variation in the dependent variable is explained by the variation in the independent variables.

Step 3

(c) The error sum of squares is given in the row "Residual error" and in the column "SS" of the ANOVA table:

$$SSE = 583.693$$

The error degrees of freedom is given in the row "Residual error" and in the column "DF" of the ANOVA table:

$$n - (k+1) = df_E = 62$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table:

$$SS\ total = 661.600$$

The total degrees of freedom is given in the row "Total" and in the column "DF" of the ANOVA table:

$$n - 1 = df_{total} = 64$$

Let us next determine the adjusted coefficient of determination:

$$R^2_{adj} = 1 - \frac{\frac{SSE}{n-(k+1)}}{\frac{SS\ total}{n-1}} = 1 - \frac{\frac{583.693}{62}}{\frac{661.600}{64}} = 0.0893 = 8.93\%$$

🎯 **Result**

(a) $s = 3.0683$, $(-6.1366, 6.1366)$

(b) 11.78% of the variation in the dependent variable is explained by the variation in the independent variables.

(c) 8.93%

PROBLEM STATEMENT

Chapter 14, Page 523

Solution A

Step 1

(a) The error sum of squares is given in the row "Residual error" and in the column "SS" of the ANOVA table:

$$SSE = 2647.38$$

The error degrees of freedom is given in the row "Residual error" and in the column "DF" of the ANOVA table:

$$df_E = 46$$

The standard error of the estimate is the square root of the error sum of squares divided by the error degrees of freedom:

$$s = \sqrt{\frac{SSE}{n-(k+1)}} = \sqrt{\frac{SSE}{df_E}} = \sqrt{\frac{2647.38}{46}} \approx 7.5863$$

95% of the residuals are expected to be two standard errors of the estimate from 0 (as the expected value of the residuals is 0).

$$2s = 2(7.5863) = 15.1726$$

Thus we expect 95% of the residuals to be between -15.1726 and 15.1726.

Step 2

(b) The regression sum of squares is given in the row "Regression" and in the column "SS" of the ANOVA table:

$$SSR = 3710.00$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table:

$$SS\ total = 6357.38$$

The coefficient of multiple determination is the regression sum of squares divided by the total sum of squares:

$$r^2 = \frac{SSR}{SS\ total} = \frac{3710.00}{6357.38} \approx 0.5836 = 58.36\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

58.36% of the variation in the dependent variable is explained by the variation in the independent variables.

Step 3

(c) The error sum of squares is given in the row "Residual error" and in the column "SS" of the ANOVA table:

$$SSE = 2647.38$$

The error degrees of freedom is given in the row "Residual error" and in the column "DF" of the ANOVA table:

$$n - (k+1) = df_E = 46$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table:

$$SS\ total = 6357.38$$

The total degrees of freedom is given in the row "Total" and in the column "DF" of the ANOVA table:

$$n - 1 = df_{total} = 51$$

Let us next determine the adjusted coefficient of determination:

$$R^2_{adj} = 1 - \frac{\frac{SSE}{n-(k+1)}}{\frac{SS\ total}{n-1}} = 1 - \frac{\frac{2647.38}{46}}{\frac{6357.38}{51}} = 0.5383 = 53.83\%$$

🎯 **Result**

(a) $s = 7.5863$, $(-15.1726, 15.1726)$

(b) 58.36% of the variation in the dependent variable is explained by the variation in the independent variables.

(c) 53.83%

PROBLEM STATEMENT

Chapter 14, Page 530

Solution A

Step 1

(a) The estimate of the constant a is given in the row "Constant" and in the column "Coef" of the computer output:

$$a = 84.998$$

The estimate of the coefficient b_1 of the variable x_1 is given in the row "X_1" and in the column "Coef" of the computer output:

$$b_1 = 2.391$$

The estimate of the coefficient b_2 of the variable x_2 is given in the row "X_2" and in the column "Coef" of the computer output:

$$b_2 = -0.4086$$

Combining these estimates, we then obtain the multiple regression equation:

$$\hat{y} = a + b_1 x + b_2 x_2 = 84.998 + 2.391 x_1 - 0.4086 x_2$$

Step 2

(b) Evaluate the multiple regression equation from part (a) at $x_1 = 4$ and $x_2 = 11$:

$$\hat{y} = a + b_1 x + b_2 x_2 = 84.998 + 2.391(4) - 0.4086(11) = 90.0674$$

Thus the value of the dependent variable is estimated to be 90.0674.

Step 3

(c) The total degrees of freedom is given in the row "Total" and in the column "DF" of the ANOVA table:

$$df_{total} = 64$$

The total degrees of freedom is the sample size decreased by 1, thus the sample size is obtained by increasing the total degrees of freedom increased by 1.

$$n = df_{total} + 1 = 64 + 1 = 65$$

The independent variables are the x-variables. Since the multiple regression equation in part (a) contains 2 x-variables, there are 2 independent variables.

Step 4

(d)

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$H_0 : \text{All coefficients are zero}$$
$$H_1 : \text{At least one of the coefficients is nonzero}$$

The P-value of the global test is given in the row "Regression" and in the column "P" of the ANOVA table:

$$P = 0.021$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.021 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

Step 5

(e)

$$\alpha = \text{Significance level} = 0.05$$

$\textbf{Independent variable X_1}$

The P-value for the independent variable x_1 is given in the row "X_1" and in the column "P" of the regression output:

$$P = 0.051$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.051 > 0.05 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the coefficient of X_1 is nonzero.

$\textbf{Independent variable X_2}$

The P-value for the independent variable x_2 is given in the row "X_2" and in the column "P" of the regression output:

$$P = 0.020$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.020 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the coefficient of X_2 is nonzero.

Since the coefficient of X_1 appears to be approximately zero, we would consider eliminating variable X_1.

Step 6

(f) We start by deleting the independent variable with the highest P-value, which is the independent variable X_1.

We determine the regression output for the model with independent variable X_1 deleted. If any of the independent variables still has a P-value greater than 0.05 in the new regression model, then we proceed with deleting this variable from the model as well.

🎯 **Result**

(a) $\hat{y} = 84.998 + 2.391x_1 - 0.4086x_2$

(b) 90.0674

(c) $n = 65$, 2 independent variables

(d) There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

(e) X_1

(f) Delete X_1 from the model

PROBLEM STATEMENT

Chapter 14, Page 531

Solution A

Step 1

(a) The estimate of the constant a is given in the row "Constant" and in the column "Coef" of the computer output:

$$a = 7.987$$

The estimate of the coefficient b_1 of the variable x_1 is given in the row "X_1" and in the column "Coef" of the computer output:

$$b_1 = 0.12242$$

The estimate of the coefficient b_2 of the variable x_2 is given in the row "X_2" and in the column "Coef" of the computer output:

$$b_2 = -0.12166$$

The estimate of the coefficient b_3 of the variable x_3 is given in the row "X_3" and in the column "Coef" of the computer output:

$$b_3 = -0.06281$$

The estimate of the coefficient b_4 of the variable x_4 is given in the row "X_4" and in the column "Coef" of the computer output:

$$b_4 = 0.5235$$

The estimate of the coefficient b_5 of the variable x_5 is given in the row "X_5" and in the column "Coef" of the computer output:

$$b_5 = -0.06472$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned}\hat{y} &= a + b_1 x + b_2 x_2 + b_3 x_3 + b_4 x_4 + b_5 x_5 \\ &= 7.987 + 0.12242 x_1 - 0.12166 x_2 \\ &\quad - 0.06281 x_3 + 0.5235 x_4 - 0.06472 x_5\end{aligned}$$

Step 2

(b) The total degrees of freedom is given in the row "Total" and in the column "DF" of the ANOVA table:

$$df_{total} = 51$$

The total degrees of freedom is the sample size decreased by 1, thus the sample size is obtained by increasing the total degrees of freedom increased by 1.

$$n = df_{total} + 1 = 51 + 1 = 52$$

The independent variables are the x-variables. Since the multiple regression equation in part (a) contains 5 x-variables, there are 5 independent variables.

Step 3

(c)

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$H_0 : \text{All coefficients are zero}$$
$$H_1 : \text{At least one of the coefficients is nonzero}$$

The P-value of the global test is given in the row "Regression" and in the column "p-value" of the ANOVA table:

$$P = 0.000$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.000 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

Step 4

(e)

$$\alpha = \text{Significance level} = 0.05$$

$\textbf{Independent variable } X_1$

The P-value for the independent variable x_1 is given in the row "X_1" and in the column "p-value" of the regression output:

$$P = 0.000$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.000 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the coefficient of X_1 is nonzero.

$\textbf{Independent variable } X_2$

The P-value for the independent variable x_2 is given in the row "X_2" and in the column "p-value" of the regression output:

$$P = 0.028$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.028 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the coefficient of X_2 is nonzero.

Step 5

$\textbf{Independent variable X_3}$

The P-value for the independent variable x_3 is given in the row "X_3" and in the column "p-value" of the regression output:

$$P = 0.114$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.114 > 0.05 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the coefficient of X_3 is nonzero.

$\textbf{Independent variable X_2}$

The P-value for the independent variable x_4 is given in the row "X_4" and in the column "p-value" of the regression output:

$$P = 0.001$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.001 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the coefficient of X_4 is nonzero.

Step 6

$\textbf{Independent variable X_5}$

The P-value for the independent variable x_5 is given in the row "X_5" and in the column "p-value" of the regression output:

$$P = 0.112$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.112 > 0.05 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the coefficient of X_5 is nonzero.

Since the coefficients of X_3 and X_5 appear to be approximately zero, we would consider eliminating variable X_3 and variable X_5. However, X_3 has the highest P-value and thus we would consider eliminating variable X_3 first.

Step 7

(f) We start by deleting the independent variable with the highest P-value, which is the independent variable X_3.

We determine the regression output for the model with independent variable X_3 deleted. If any of the independent variables still has a P-value greater than 0.05 in the new regression model, then we proceed with deleting this variable from the model as well.

🎯 Result

(a) $\hat{y} = 7.987 + 0.12242x_1 - 0.12166x_2 - 0.06281x_3 + 0.5235x_4 - 0.06472x_5$

(b) $n = 52$, 5 independent variables

(c) There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

(e) X_3

(f) Delete X_3 from the model

PROBLEM STATEMENT

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Solution A

Step 1

(a) I will use Excel to determine the multiple regression equation. The regression output is given in the image below.

The estimate of the constant a is given in the row "Intercept" and in the column "Coefficients" of the computer output:

$$a \approx 29.2762$$

The estimate of the coefficient b_1 of the variable aptitude test score x_1 is given in the row "X Variable 1" and in the column "Coefficients" of the computer output:

$$b_1 \approx 5.2225$$

The estimate of the coefficient b_2 of the variable union membership x_2 is given in the row "X Variable 2" and in the column "Coefficients" of the computer output:

$$b_2 \approx 22.1351$$

Combining these estimates, we then obtain the multiple regression equation:

$$\hat{y} = a + b_1 x + b_2 x_2 = 29.2762 + 5.2225 x_1 + 22.1351 x_2$$

SUMMARY OUTPUT

Regression Statistics

Multiple R	0.7300651691
R Square	0.5329951511
Adjusted R Squa	0.4551610096
Standard Error	16.91656565
Observations	15

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	3919.291011	1959.645506	6.847832338	0.01037357232
Residual	12	3434.042322	286.1701935		
Total	14	7353.333333			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95%*	*Upper 95%*
Intercept	29.27618014	12.76659949	2.293185446	0.04069635615	1.460149758	57.09221052	1.460149758	57.09221052
X Variable 1	5.222463375	1.702483522	3.067555902	0.009761027486	1.513070485	8.931856265	1.513070485	8.931856265
X Variable 2	22.13510581	8.852212713	2.50051671	0.02788894996	2.847791432	41.42242018	2.847791432	41.42242018

Step 2

(b) The coefficient of determination is given to the right of "R Square" in the regression output:

$$R^2 \approx 0.5330 = 53.30\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

53.30% of the variation in the job performance score is explained by the variation in the aptitude test score and union membership.

We note that the coefficient of determination is not very small (not very close to 0%) and thus the variables are effective in predicting performance.

Step 3

(c)

$$\alpha = \text{Significance level} = 0.05 \text{ (assumption)}$$

$\textbf{Independent variable X_2}$

The P-value for the independent variable union membership x_2 is given in the row "X_2" and in the column "P" of the regression output:

$$P \approx 0.0279$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.0279 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the coefficient of X_2 is nonzero and thus union membership should be included as an independent variable.

Step 4

(d) I will use Excel to determine the multiple regression equation. I will now include a variable x_3, which is the product of the variables x_1 and x_2. The regression output is given in the image below.

The estimate of the constant a is given in the row "Intercept" and in the column "Coefficients" of the computer output:

$$a \approx 38.6889$$

The estimate of the coefficient b_1 of the variable aptitude test score x_1 is given in the row "X Variable 1" and in the column "Coefficients" of the computer output:

$$b_1 \approx 3.8017$$

The estimate of the coefficient b_2 of the variable union membership x_2 is given in the row "X Variable 2" and in the column "Coefficients" of the computer output:

$$b_2 \approx -0.1007$$

The estimate of the coefficient b_2 of the interaction term $x_3 = x_1 x_2$ is given in the row "X Variable 3" and in the column "Coefficients" of the computer output:

$$b_3 \approx 3.6101$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned} \hat{y} &= a + b_1 x + b_2 x_2 + b_3 x_1 x_2 \\ &= 38.6889 + 3.8017 x_1 - 0.1007 x_2 + 3.6101 x_1 x_2 \end{aligned}$$

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.7581292648					
R Square	0.5747599822					
Adjusted R Square	0.4587854319					
Standard Error	16.86020489					
Observations	15					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	3	4226.401736	1408.800579	4.95591473	0.02045550247	
Residual	11	3126.931598	284.2665089			
Total	14	7353.333333				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	38.68893528	15.61766138	2.477255355	0.03072310438	4.314694629	73.06317593
X Variable 1	3.801670146	2.178913977	1.744754582	0.1088634994	-0.9940871443	8.597427437
X Variable 2	-0.1006999877	23.14073331	-0.00435163339	0.9966058272	-51.03311019	50.83171021
X Variable 3	3.61009456	3.473233124	1.039404621	0.3209243714	-4.034439943	11.25462906

Step 5

The coefficient of determination is given to the right of "R Square" in the regression output:

$$R^2 \approx 0.5748 = 57.48\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

58.48% of the variation in the job performance score is explained by the variation in the aptitude test score and union membership and their interaction.

We note that the coefficient of determination is not very small (not very close to 0%) and thus the variables are effective in predicting performance.

Step 6

$$\alpha = \text{Significance level} = 0.05 \text{ (assumption)}$$

$\textbf{Independent variable } X_2\}$

The P-value for the independent variable union membership x_2 is given in the row "X_2" and in the column "P" of the regression output:

$$P \approx 0.9966$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.9966 > 0.05 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the coefficient of X_2 is nonzero and thus union membership should not be included as an independent variable.

Result

(a) $\hat{y} = 29.2762 + 5.2225x_1 + 22.1351x_2$

(b) Yes

(c) Union membership should be included as an independent vairable.

(d) $\hat{y} = 38.6889 + 3.8017x_1 - 0.1007x_2 + 3.6101x_1x_2$

Exercise 10

PROBLEM STATEMENT

Chapter 14, Page 545

Solution A

Step 1

I will use Excel to determine the multiple regression equation. I will now include a variable x_3, which is the product of the variables x_1 and x_2. The regression output is given in the image below.

The estimate of the constant a is given in the row "Intercept" and in the column "Coefficients" of the computer output:

$$a \approx -314.6362$$

The estimate of the coefficient b_1 of x_1 is given in the row "X Variable 1" and in the column "Coefficients" of the computer output:

$$b_1 \approx 2.1503$$

The estimate of the coefficient b_2 of x_2 is given in the row "X Variable 2" and in the column "Coefficients" of the computer output:

$$b_2 \approx 0.0941$$

The estimate of the coefficient b_2 of the interaction term $x_3 = x_1 x_2$ is given in the row "X Variable 3" and in the column "Coefficients" of the computer output:

$$b_3 \approx -0.0006$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned}\hat{y} &= a + b_1 x + b_2 x_2 + b_3 x_1 x_2 \\ &= -314.6362 + 2.1503 x_1 + 0.0941 x_2 - 0.0006 x_1 x_2\end{aligned}$$

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.9208097504					
R Square	0.8478905965					
Adjusted R Square	0.8261606817					
Standard Error	4.234700062					
Observations	25					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	3	2099.173623	699.724541	39.01950858	0.000000009083	
Residual	21	376.5863769	17.93268461			
Total	24	2475.76				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	-314.6362433	162.6498568	-1.934439104	0.06664624329	-652.8851361	23.61264947
X Variable 1	2.150299742	1.158717162	1.855758948	0.07757785987	-0.2593845016	4.559983986
X Variable 2	0.09410407069	0.05944247753	1.583111516	0.1283409122	-0.02951332809	0.2177214695
X Variable 3	-0.00055643870	0.000421524018	-1.320064063	0.2010227453	-0.00143304588	0.000320168473

Step 2

$$\alpha = \text{Significance level} = 0.05 \text{ (assumption)}$$

The P-value for the interaction $x_3 = x_1 x_2$ is given in the row "X Variable 3" and in the column "P-value" of the regression output:

$$P \approx 0.2010$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.2010 > 0.05 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the coefficient of $x_1 x_2$ is nonzero and thus there is no significant interaction between the number of sales calls and the miles driven.

🎯 Result

$$\hat{y} = -314.6362 + 2.1503 x_1 + 0.0941 x_2 - 0.0006 x_1 x_2$$

No

Exercise 11

PROBLEM STATEMENT

Chapter 14, Page 545

Solution A

Step 1

(a) I will use Excel to determine the multiple regression equation. The regression output is given in the image below.

The estimate of the constant a is given in the row "Intercept" and in the column "Coefficients" of the computer output:

$$a \approx 3080.0530$$

The estimate of the coefficient b_1 of Bidders x_1 is given in the row "X Variable 1" and in the column "Coefficients" of the computer output:

$$b_1 \approx -54.1885$$

The estimate of the coefficient b_2 of Age x_2 is given in the row "X Variable 2" and in the column "Coefficients" of the computer output:

$$b_2 \approx 16.2891$$

Combining these estimates, we then obtain the multiple regression equation:

$$\hat{y} = a + b_1 x + b_2 x_2 = 3080.0530 - 54.1885 x_1 + 16.2891 x_2$$

It is surprising that there is an inverse relationship between the number of bidders and the price of a painting, because we expect the price of a painting to increase as the number of bidders increase.

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.804861447							
R Square	0.6478019489							
Adjusted R Square	0.6157839443							
Standard Error	222.0678517							
Observations	25							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	2	1995489.123	997744.5617	20.23242723	0.000010343235			
Residual	22	1084910.877	49314.13075					
Total	24	3080400						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95%*	*Upper 95%*
Intercept	3080.053036	343.884898	8.956639427	0.000000008615	2366.879411	3793.226661	2366.879411	3793.226661
X Variable 1	-54.18854049	12.28149085	-4.412211933	0.000220413369	-79.65879348	-28.71828749	-79.65879348	-28.71828749
X Variable 2	16.2891028	3.783917929	4.304824552	0.000286551051	8.441737354	24.13646825	8.441737354	24.13646825

Step 2

(b) I will use Excel to determine the multiple regression equation. I will now include a variable x_3, which is the product of the variables x_1 and x_2. The regression output is given in the image below.

The estimate of the constant a is given in the row "Intercept" and in the column "Coefficients" of the computer output:

$$a \approx 3971.6782$$

The estimate of the coefficient b_1 of Bidders x_1 is given in the row "X Variable 1" and in the column "Coefficients" of the computer output:

$$b_1 \approx -184.9945$$

The estimate of the coefficient b_2 of Age x_2 is given in the row "X Variable 2" and in the column "Coefficients" of the computer output:

$$b_2 \approx 6.3530$$

The estimate of the coefficient b_2 of the interaction term $x_3 = x_1 x_2$ is given in the row "X Variable 3" and in the column "Coefficients" of the computer output:

$$b_3 \approx 1.4621$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned}\hat{y} &= a + b_1 x + b_2 x_2 + b_3 x_1 x_2 \\ &= 3971.6782 - 184.9945 x_1 + 6.3530 x_2 + 1.4621 x_1 x_2\end{aligned}$$

The interaction variable is not significant, because the corresponding P-value of about 0.2650 is greater than 0.05.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.8176209402
R Square	0.6685040018
Adjusted R Square	0.6211474306
Standard Error	220.5124198
Observations	25

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	3	2059259.727	686419.909	14.11639367	0.000029157491
Residual	21	1021140.273	48625.72728		
Total	24	3080400			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	3971.678169	850.1752699	4.671599269	0.000130601852	2203.641914	5739.714425
X Variable 1	-184.9944896	114.8713471	-1.610449379	0.1222296897	-423.8825325	53.89355325
X Variable 2	6.352997654	9.455044781	0.6719161888	0.5089623928	-13.3098443	26.01583961
X Variable 3	1.462103382	1.276735319	1.145189109	0.2650149661	-1.193013053	4.117219816

Step 3

(c) In part (b), we determine that the interaction variable was not significant and thus we remove the interaction variable from the model.

We then obtain the model determine in part (a). All P-values in the column "P-value" of this model are small (less than 0.05), which implies that all coefficients are significantly different from 0 and thus we select this model. This then implies that we selected the variables bidders and age.

Result

(a) $\hat{y} = 3080.0530 - 54.1885x_1 + 16.2891x_2$, Yes

(b) $\hat{y} = 3971.6782 - 184.9945x_1 + 6.3530x_2 + 1.4621x_1x_2$, Not significant

(c) Bidders and age

PROBLEM STATEMENT

Chapter 14, Page 546

Solution A

Step 1

I will use Excel to determine the multiple regression equation.Initially we use all variables. The regression output is given in the image below.

X Variable 1=Income

X Variable 2=Family Size

X Variable 3= Senior Parent

X Variable 4=Education

The highest P-value in the column "P-value" is approximately 0.7186, which is given in the row "X Variable 3." Since this P-value is large (greater than 0.05), we delete the variable "Senior Parent" from the model.

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.9557337249							
R Square	0.9134269528							
Adjusted R Square	0.8441685151							
Standard Error	298.412012							
Observations	10							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	4	4697791.356	1174447.839	13.18867394	0.007241069113			
Residual	5	445248.6445	89049.7289					
Total	9	5143040						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95%*	*Upper 95%*
Intercept	879.3937802	439.1036569	2.002702019	0.1015883612	-249.3581037	2008.145664	-249.3581037	2008.145664
X Variable 1	16.32862748	7.397759946	2.207239435	0.07837629178	-2.687919854	35.34517481	-2.687919854	35.34517481
X Variable 2	274.6823214	153.651833	1.787693098	0.133862072	-120.2922894	669.6569322	-120.2922894	669.6569322
X Variable 3	-83.04632698	217.7625551	-0.3813618321	0.7186026889	-642.8227954	476.7301415	-642.8227954	476.7301415
X Variable 4	-38.60445866	42.87597613	-0.9003750387	0.4091897545	-148.820664	71.61174671	-148.820664	71.61174671

Step 2

We use Excel to recalculate the new multiple regression equation. The regression output is given in the image below.

X Variable 1=Income

X Variable 2=Family Size

X Variable 3=Education

The highest P-value in the column "P-value" is approximately 0.3913, which is given in the row "X Variable 3." Since this P-value is large (greater than 0.05), we delete the variable "Education" from the model.

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.9544154078							
R Square	0.9109087706							
Adjusted R Square	0.8663631559							
Standard Error	276.3451213							
Observations	10							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	3	4684840.244	1561613.415	20.44889889	0.001494230447			
Residual	6	458199.7563	76366.62605					
Total	9	5143040						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95%*	*Upper 95%*
Intercept	829.8497157	388.4284211	2.136428929	0.07652682472	-120.6003894	1780.299821	-120.6003894	1780.299821
X Variable 1	15.81726453	6.737237144	2.347737535	0.05723517821	-0.6681608553	32.30268991	-0.6681608553	32.30268991
X Variable 2	292.1974005	135.7844345	2.15192118	0.0749073738	-40.05514089	624.4499419	-40.05514089	624.4499419
X Variable 3	-36.30787977	39.31181088	-0.9235870585	0.3913307892	-132.5004155	59.88465597	-132.5004155	59.88465597

Step 3

We use Excel to recalculate the new multiple regression equation. The regression output is given in the image below.

X Variable 1=Income

X Variable 2=Family Size

The highest P-value in the column "P-value" is approximately 0.0586, which is given in the row "X Variable 1." Since this P-value is large (greater than 0.05), we delete the variable "Income" from the model.

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.947756706							
R Square	0.8982427737							
Adjusted R Square	0.8691692804							
Standard Error	273.4283623							
Observations	10							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	2	4619698.515	2309849.257	30.89559161	0.000336107661			
Residual	7	523341.4853	74763.06932					
Total	9	5143040						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95%*	*Upper 95%*
Intercept	712.7474139	363.2769993	1.961994333	0.09055321135	-146.266183	1571.761011	-146.266183	1571.761011
X Variable 1	12.14703038	5.383032745	2.256540311	0.05862978241	-0.5818193081	24.87588007	-0.5818193081	24.87588007
X Variable 2	371.9715918	103.6648332	3.588213865	0.008878270962	126.8432148	617.0999688	126.8432148	617.0999688

Step 4

We use Excel to recalculate the new multiple regression equation. The regression output is given in the image below.

X Variable 1=Family Size

We note that all P-values are small (less than 0.05) and thus we have obtained an appropriate regression model.

The coefficient of determination is given to the right of "R Square" in the regression output, which is thus about 0.8242. Since this value is large (close to 1), the model is useful as well.

The estimate of the constant a is given in the row "Intercept" and in the column "Coefficients" of the computer output:

$$a \approx 1271.5862$$

The estimate of the coefficient b_1 of the variable family size x_2 is given in the row "X Variable 1" and in the column "Coefficients" of the computer output:

$$b_1 \approx 540.6897$$

Combining these estimates, we then obtain the multiple regression equation:

$$\hat{y} = a + b_1 x_2 = 1271.5862 + 540.6897 x_2$$

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.9078667498							
R Square	0.8242220353							
Adjusted R Square	0.8022497897							
Standard Error	336.160881							
Observations	10							
ANOVA								
	df	*SS*	*MS*	*F*	*Significance F*			
Regression	1	4239006.897	4239006.897	37.51196172	0.000281708946			
Residual	8	904033.1034	113004.1379					
Total	9	5143040						
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95%*	*Upper 95%*
Intercept	1271.586207	326.7559133	3.891547651	0.004599169524	518.0857202	2025.086694	518.0857202	2025.086694
X Variable 1	540.6896552	88.28017231	6.12470095	0.000281708946	337.1152129	744.2640974	337.1152129	744.2640974

Result

$\hat{y} = 1271.5862 + 540.6897 x_2$ with x_2 representing the family size

PROBLEM STATEMENT

Chapter 14, Page 553

Solution A

Step 1

(a) The total degrees of freedom is the sum of the regression degrees of freedom and the error degrees of freedom, which are the two values in the column "DF" of the partial ANOVA table:

$$df_{total} = 4 + 35 = 39$$

The total degrees of freedom is the sample size decreased by 1, thus the sample size is obtained by increasing the total degrees of freedom increased by 1.

$$n = df_{total} + 1 = 39 + 1 = 40$$

Step 2

(b) The regression degrees of freedom is given in the row "Regression" and in the column "DF" of the ANOVA table:

$$df_R = 4$$

The regression degrees of freedom is equal to the number of independent variables and thus there are 4 independent variables.

Step 3

(c) The regression sum of squares is given in the row "Regression" and in the column "Sum of squares" of the ANOVA table:

$$SSR = 750$$

The total sum of squares is sum of the regression sum of squares and the error sum of squares, which are the two values in the column "Sum of squares" of the partial ANOVA table:

$$SS\ total = SSR + SSE = 750 + 500 = 1250$$

The coefficient of multiple determination is the regression sum of squares divided by the total sum of squares:

$$r^2 = \frac{SSR}{SS\ total} = \frac{750}{1250} \approx 0.6 = 60\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

60% of the variation in the dependent variable is explained by the variation in the independent variables.

Step 4

(d) The error sum of squares is given in the row "Error" and in the column "Sum of squares" of the ANOVA table:

$$SSE = 500$$

The error degrees of freedom is given in the row "Error" and in the column "df" of the ANOVA table:

$$df_E = 35$$

The standard error of the estimate is the square root of the error sum of squares divided by the error degrees of freedom:

$$s = \sqrt{\frac{SSE}{n-(k+1)}} = \sqrt{\frac{SSE}{df_E}} = \sqrt{\frac{500}{35}} \approx 3.7996$$

Step 5

(e) The MSR is the SSR divided by df_R:

$$MSR = \frac{SSR}{df_R} = \frac{750}{4} = 187.5$$

The MSE is the SSE divided by df_E:

$$MSE = \frac{SSE}{df_E} = \frac{500}{35} \approx 14.2857$$

The value of the F test statistic is the ratio of the MSR and the MSE.

$$F = \frac{MSR}{MSE} = \frac{187.5}{14.2857} \approx 13.1250$$

The F- critical value is given in the row with $dfd = df_E = 35$ and in the column with $dfn = df_T = 4$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 2.64$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 2.64.

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$14.2857 > 2.64 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to reject the claim that all regression coefficients are equal to zero.

Result

(a) $n = 40$

(b) 4 independent variables

(c) 60%

(d) 3.7996

(e) There is sufficient evidence to reject the claim that all regression coefficients are equal to zero.

PROBLEM STATEMENT

Chapter 14, Page 554

Solution A

Step 1

Given:

$$b_1 = 2.676$$
$$s_{b_1} = 0.56$$
$$b_2 = -0.880$$
$$s_{b_2} = 0.71$$
$$n = \text{Sample size} = 25$$
$$\alpha = \text{Significance level} = 0.05$$

Hypothesis test first independent variable

Given claim: Slope is zero

The null hypothesis or the alternative hypothesis states the given claim The null hypothesis includes that the slope is zero. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \beta_1 = 0$$

$$H_1 : \beta_1 \neq 0$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Two-tailed

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value t_0 that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 25 - 2 = 23$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.069$$

The rejection region then contains all values smaller than -2.069 and all values greater than 2.069, thus we reject H_0 when $t < -2.069$ or $t > 2.069$.

Compute the value of the test statistic:

$$t = \frac{b_1 - 0}{s_{b_1}} = \frac{2.676 - 0}{0.56} \approx 4.7786$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = 4.7786 > 2.069 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the coefficient b_1 is nonzero.

Step 3

Hypothesis test second independent variable

Given claim: Slope is zero

The null hypothesis or the alternative hypothesis states the given claim The null hypothesis includes that the slope is zero. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \beta_1 = 0$$

$$H_1 : \beta_1 \neq 0$$

Step 4

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Two-tailed

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value t_0 that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 2 = 25 - 2 = 23$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.069$$

The rejection region then contains all values smaller than -2.069 and all values greater than 2.069, thus we reject H_0 when $t < -2.069$ or $t > 2.069$.

Compute the value of the test statistic:

$$t = \frac{b_2 - 0}{s_{b_2}} = \frac{-0.880 - 0}{0.71} \approx -1.2394$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$-2.069 < -1.2394 < 2.069 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the coefficient b_2 is nonzero.

This then implies that we would consider deleting the second independent variable (corresponding to b_2) from the regression equation.

Result

Yes

PROBLEM STATEMENT

Chapter 14, Page 554

Solution A

Step 1

(a) The total degrees of freedom is given in the row "Total" and in the column "DF" of the ANOVA table:

$$df_{total} = 25$$

The total degrees of freedom is the sample size decreased by 1, thus the sample size is obtained by increasing the total degrees of freedom increased by 1.

$$n = df_{total} + 1 = 25 + 1 = 26$$

Step 2

(b) The regression sum of squares is given in the row "Regression" and in the column "SS" of the ANOVA table:

$$SSR = 100$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table:

$$SS\ total = 140$$

The coefficient of multiple determination is the regression sum of squares divided by the total sum of squares:

$$R^2 = \frac{SSR}{SS\ total} = \frac{100}{140} \approx 0.7143 = 71.43\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

71.43% of the variation in the dependent variable is explained by the variation in the independent variables.

Step 3

(c) The error sum of squares is given in the row "Error" and in the column "SS" of the ANOVA table:

$$SSE = 40$$

The error degrees of freedom is given in the row "Eerror" and in the column "DF" of the ANOVA table:

$$df_E = 20$$

The standard error of the estimate is the square root of the error sum of squares divided by the error degrees of freedom:

$$s = \sqrt{\frac{SSE}{n-(k+1)}} = \sqrt{\frac{SSE}{df_E}} = \sqrt{\frac{40}{20}} \approx 1.4142$$

Step 4

(d)

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$H_0 : \text{All coefficients are zero}$$
$$H_1 : \text{At least one of the coefficients is nonzero}$$

The value of the F-test statistic is the ratio of the two values in the column "MS":

$$F = \frac{MSR}{MSE} = \frac{20}{2} = 10$$

The F- critical value is given in the row with $dfd = df_E = 20$ and in the column with $dfn = df_R = 5$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 2.71$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 2.71.

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$10 > 2.71 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that at least one of the regression coefficients is significant.

Step 5

(e) If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Two-tailed

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value t_0 that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = df_E = 20$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.086$$

The rejection region then contains all values smaller than -2.086 and all values greater than 2.086, thus we reject H_0 when $t < -2.086$ or $t > 2.086$.

When the value in the column "t-ratio" is less than -2.086 or greater than 2.086, then the coefficient is significantly different from 0. We note that the t-ratios of X_1 and X_5 are then not significantly different from 0 (as their t-ratio is between -2.086 and 2.086), which implies that we would consider omitting varaibles x_1 and x_5.

Result

(a) $n = 26$

(b) 71.43% of the variation in the dependent variable is explained by the variation in the independent variables.

(c) 1.4142

(d) There is sufficient evidence to support the claim that at least one of the regression coefficients is significant.

(e) Yes, x_1 and x_5

PROBLEM STATEMENT

Chapter 14, Page 554

Solution A

Step 1

Given:

$$\begin{aligned} k &= \text{Number of independent variables} = 5 \\ n &= \text{Sample size} = 20 \\ MSE &= 5.10 \\ SS_{total} &= 519.68 \\ \alpha &= \text{Significance level} = 0.05 \end{aligned}$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$\begin{aligned} H_0 &: \text{All coefficients are zero} \\ H_1 &: \text{At least one of the coefficients is nonzero} \end{aligned}$$

Step 2

df_R is the number of independent variables

$$df_R = k = 5$$

df_E is the total sample size decreased by the number of groups k and 1.

$$df_E = n - k - 1 = 20 - 5 - 1 = 14$$

The F- critical value is given in the row with $dfd = df_E = 14$ and in the column with $dfn = df_R = 5$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 2.96$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 2.96.

Step 3

MSE is SSE divided by df_E, which implies that SSE is the product of MSE and df_E:

$$SSE = MSE \times df_E = 5.10 \times 14 = 71.4$$

Since the sum $SS_{total} = SSR + SSE$ holds:

$$SSR = SS_{total} - SSE = 519.68 - 71.4 = 448.28$$

MSR is SSR divided by df_R:

$$MSR = \frac{SSR}{d.f._R} = \frac{448.28}{5} = 89.656$$

The value of the test statistic F is then MSR divided by MSE:

$$F = \frac{MSR}{MSE} = \frac{89.656}{5.10} \approx 17.58$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$17.58 > 2.96 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that at least one of the regression coefficients is nonzero.

🎯 Result

Yes

💡 Solution B

Step 1

Given:

$$\begin{aligned} k &= \text{Number of independent variables} = 5 \\ n &= \text{Sample size} = 20 \\ MSE &= 5.10 \\ SS_{total} &= 519.68 \\ \alpha &= \text{Significance level} = 0.05 \end{aligned}$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$\begin{aligned} H_0 &: \text{All coefficients are zero} \\ H_1 &: \text{At least one of the coefficients is nonzero} \end{aligned}$$

Step 2

df_R is the number of independent variables

$$df_R = k = 5$$

df_E is the total sample size decreased by the number of groups k and 1.

$$df_E = n - k - 1 = 20 - 5 - 1 = 14$$

The F- critical value is given in the row with $dfd = df_E = 14$ and in the column with $dfn = df_R = 5$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 2.96$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 2.96.

Step 3

MSE is SSE divided by df_E, which implies that SSE is the product of MSE and df_E:

$$SSE = MSE \times df_E = 5.10 \times 14 = 71.4$$

Since the sum $SS_{total} = SSR + SSE$ holds:

$$SSR = SS_{total} - SSE = 519.68 - 71.4 = 448.28$$

MSR is SSR divided by df_R:

$$MSR = \frac{SSR}{d.f._R} = \frac{448.28}{5} = 89.656$$

The value of the test statistic F is then MSR divided by MSE:

$$F = \frac{MSR}{MSE} = \frac{89.656}{5.10} \approx 17.58$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$17.58 > 2.96 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that at least one of the regression coefficients is nonzero.

🎯 Result

Yes

PROBLEM STATEMENT

Chapter 14, Page 554

Solution A

Step 1

(a) The estimate of the constant a is given in the row "Constant" and in the column "Coef" of the computer output:

$$a = 14.00$$

The estimate of the coefficient b_1 of the variable x_1 is given in the row "X_1" and in the column "Coef" of the computer output:

$$b_1 = -1.00$$

The estimate of the coefficient b_2 of the variable x_2 is given in the row "X_2" and in the column "Coef" of the computer output:

$$b_2 = 30.00$$

The estimate of the coefficient b_3 of the variable x_3 is given in the row "X_3" and in the column "Coef" of the computer output:

$$b_3 = 0.20$$

Combining these estimates, we then obtain the multiple regression equation:

$$\hat{y} = a + b_1 x + b_2 x_2 + b_3 x_3 = 14.00 - 1.00x_1 + 30.00x_2 - 0.20x_3$$

Evaluate the regression equation at $x_1 = 4$, $x_2 = 0.4$, and $x_3 = 30$:

$$\hat{y} = 14.00 - 1.00(4) + 30.00(0.4) - 0.20(30) = 16$$

Thus the estimated sales are 16 thousand dollars or $\$16,000$.

Step 2

(b) The regression sum of squares is given in the row "Regression" and in the column "SS" of the ANOVA table:

$$SSR = 3050.00$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table:

$$SS\ total = 5250.00$$

The coefficient of multiple determination is the regression sum of squares divided by the total sum of squares:

$$r^2 = \frac{SSR}{SS\ total} = \frac{3050.00}{5250.00} \approx 0.5810 = 58.10\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

58.10% of the variation in the dependent variable is explained by the variation in the independent variables.

Step 3

(c) The error sum of squares is given in the row "Error" and in the column "SS" of the ANOVA table:

$$SSE = 2200.00$$

The error degrees of freedom is given in the row "Error" and in the column "DF" of the ANOVA table:

$$df_E = 26$$

The standard error of the estimate is the square root of the error sum of squares divided by the error degrees of freedom:

$$s = \sqrt{\frac{SSE}{n-(k+1)}} = \sqrt{\frac{SSE}{df_E}} = \sqrt{\frac{2200.00}{26}} \approx 9.1987$$

Step 4

(d)

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$H_0 : \text{All coefficients are zero}$$
$$H_1 : \text{At least one of the coefficients is nonzero}$$

The F- critical value is given in the row with $dfd = d.f._E = 26$ and in the column with $dfn = d.f._R = 3$ (number of independent variables) of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 2.98$$

We reject the null hypothesis H_0 when the value of the test statistic exceeds 2.98.

The value of the F test statistic is the ratio of the regression mean square and the mean square error, which are the two values in the column "MS" of the output:

$$F = \frac{MSR}{MSE} = \frac{1016.67}{84.62} \approx 12.0145$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$F = 12.0145 > 2.98 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that at least one of the regression coefficients is nonzero.

Step 5

(e)

$$\alpha = \text{Significance level} = 0.05$$

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Two-tailed

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value t_0 that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = df_E = 26$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.056$$

The rejection region then contains all values smaller than -2.056 and all values greater than 2.056, thus we reject H_0 when $t < -2.056$ or $t > 2.056$.

The coefficient is then significantly different from zero when its t-ratio is less than -2.056 or larger than 2.056. We note that X_1 is the only independent variable which has a t-ratio between -2.056 and 2.056, which implies that we would consider eliminating X_1 from the model..

Result

(a) 16 thousand dollars

(b) 58.10%

(c) 9.1987

(d) There is sufficient evidence to support the claim that at least one of the regression coefficients is nonzero.

(e) Yes, X_1

PROBLEM STATEMENT

Chapter 14, Page 555

Solution A

Step 1

(a) We note that the highest correlation (in absolute value) given in the correlation matrix is 0.964, which is given in the row "income" and in the column "sales."

This then implies that the single variable "income" has the strongest correlation with the dependent variable.

A strong correlation between two independent variables could pose a problem, which is called **multicollinearity**.

Step 2

(b) The regression sum of squares is given in the row "Regression" and in the column "SS" of the ANOVA table:

$$SSR = 1593.81$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table:

$$SS\ total = 1602.89$$

The coefficient of multiple determination is the regression sum of squares divided by the total sum of squares:

$$r^2 = \frac{SSR}{SS\ total} = \frac{1593.81}{1602.89} \approx 0.9943 = 99.43\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

99.43% of the variation in the dependent variable is explained by the variation in the independent variables.

Step 3

(c)

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$H_0 : \text{All coefficients are zero}$$
$$H_1 : \text{At least one of the coefficients is nonzero}$$

The P-value of the global test is given in the row "Regression" and in the column "P" of the ANOVA table:

$$P = 0.000$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.000 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

Step 4

(d) If the P-value is less than the significance level, then we reject the null hypothesis H_0 of a zero coefficient:

$$\begin{aligned}
\text{outlets: } P = 0.823 > 0.05 &\Rightarrow \text{ Fail to reject } H_0 \\
\text{cars: } P = 0.035 < 0.05 &\Rightarrow \text{ Reject } H_0 \\
\text{income: } P = 0.001 < 0.05 &\Rightarrow \text{ Reject } H_0 \\
\text{age: } P = 0.081 > 0.05 &\Rightarrow \text{ Fail to reject } H_0 \\
\text{bosses: } P = 0.864 > 0.05 &\Rightarrow \text{ Fail to reject } H_0
\end{aligned}$$

Thus we conclude that the coefficients of outlets, age, and bosses are approximately zero, while the coefficients of cars and income are nonzero.

We would then consider eliminating "outlets" and "bosses" as they both have coefficients that are approximately 0.

Step 5

(e) The regression sum of squares is given in the row "Regression" and in the column "SS" of the ANOVA table:

$$SSR = 1593.66$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table:

$$SS\ total = 1602.89$$

The coefficient of multiple determination is the regression sum of squares divided by the total sum of squares:

$$r^2 = \frac{SSR}{SS\ total} = \frac{1593.66}{1602.89} \approx 0.9942 = 99.42\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

99.42% of the variation in the dependent variable is explained by the variation in the independent variables.

We note that the coefficient of determination decreased by 0.01% compared to part (b).

Step 6

(f) We note that the histogram and the stem-and-leaf plot are both approximately normal, because most of the data values lie in the middle of the graphs.

This then implies that the normality assumption appears reasonable.

Step 7

(g) The residuals appear to be randomly scattered in the residual plot, while there is no strong pattern present in the scatterplot.

This then implies that there are no violations of the assumption.

🎯 Result

(a) Income, Yes, multicolinnearity

(b) 99.43%

(c) There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

(d) Yes

(e) 0.01%

(f) Yes

(g) No

PROBLEM STATEMENT

Chapter 14, Page 557

Solution A

Step 1

(a) We note that the highest correlation (in absolute value) given in the correlation matrix is 0.911, which is given in the row "gpa" and in the column "legal."

This then implies that the single variable "high school GPA" has the strongest correlation with the dependent variable Paralegal GPA.

A strong correlation between two independent variables could pose a problem, which is called $\textbf{multicollinearity}$. We consider the rule of thumb that a correlation between two independent variables poses a problem when the correlation is closer to -0.7 or 0.7. Since all correlations between two independent variables are quite a bit lower than 0.7, none of the correlations appear to pose a problem in this case.

Step 2

(b) The regression sum of squares is given in the row "Regression" and in the column "SS" of the ANOVA table:

$$SSR = 4.3595$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table:

$$SS\ total = 5.0631$$

The coefficient of multiple determination is the regression sum of squares divided by the total sum of squares:

$$r^2 = \frac{SSR}{SS\ total} = \frac{4.3595}{5.0631} \approx 0.8610 = 86.10\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

86.10% of the variation in the dependent variable is explained by the variation in the independent variables.

Step 3

(c)

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$H_0 : \text{All coefficients are zero}$$
$$H_1 : \text{At least one of the coefficients is nonzero}$$

The P-value of the global test is given in the row "Regression" and in the column "P" of the ANOVA table:

$$P = 0.014$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.014 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

Step 4

(d) If the P-value is less than the significance level, then we reject the null hypothesis H_0 of a zero coefficient:

$$\text{GPA: } P = 0.010 < 0.05 \Rightarrow \text{ Reject } H_0$$
$$\text{Verbal: } P = 0.482 > 0.05 \Rightarrow \text{ Fail to reject } H_0$$
$$\text{Math: } P = 0.393 > 0.05 \Rightarrow \text{ Fail to reject } H_0$$

Thus we conclude that the coefficients of verbal and math are approximately zero, while the coefficient high school GPA is nonzero.

We would then consider eliminating "verbal" and "math" as they both have coefficients that are approximately 0.

Step 5

(e) The regression sum of squares is given in the row "Regression" and in the column "SS" of the ANOVA table:

$$SSR = 4.2061$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table:

$$SS\ total = 5.0631$$

The coefficient of multiple determination is the regression sum of squares divided by the total sum of squares:

$$r^2 = \frac{SSR}{SS\ total} = \frac{4.2061}{5.0631} \approx 0.8307 = 83.07\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

83.07% of the variation in the dependent variable is explained by the variation in the independent variables.

We note that the coefficient of determination decreased by $86.10\% - 83.07\% = 3.03\%$ compared to part (b).

Step 6

(f) We note that the histogram and the stem-and-leaf plot are slightly positively skewed, because most of the data values lie near the top of the graphs.

However, the skewness does not appear to be strong and thus the normality assumption appears reasonable.

Step 7

(g) The residuals appear to be randomly scattered in the residual plot, while there is no strong pattern present in the scatterplot.

This then implies that there are no violations of the assumption.

🎯 Result

(a) High school GPA, No

(b) 86.10%

(c) There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

(d) Yes

(e) 3.03%

(f) Yes

(g) No

PROBLEM STATEMENT

Chapter 14, Page 559

Solution A

Step 1

(a) I will use Excel to determine the correlation matrix of the given data, which results in the following table:

	Y	X_1	X_2	X_3
Y	1			
X_1	0.8676	1		
X_2	0.5472	0.1868	1	
X_3	0.3106	0.2075	0.4578	1

We note that the largest correlation of 0.8676 belongs to the variable X_1 (years of experience).

Moreover, we note that none of the correlation between two independent variables (X_i) are large (less than -0.7 or larger than 0.7) and thus it is safe to assume that there are no problems with multicollinearity.

Step 2

(b) I will use Excel to determine the regression equation. The regression output is given in the image below.

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.9529597183					
R Square	0.9081322247					
Adjusted R Square	0.8909070168					
Standard Error	2.389667843					
Observations	20					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	3	903.1938016	301.0646005	52.72111845	0.000000016231	
Residual	16	91.36819837	5.710512398			
Total	19	994.562				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	19.91519148	1.916260316	10.39273804	0.000000016029	15.85290111	23.97748185
X Variable 1	0.8993793668	0.08767876384	10.25766477	0.000000019257	0.7135086919	1.085250042
X Variable 2	0.1539160021	0.03144064388	4.895446882	0.000161688594	0.08726481493	0.2205671892
X Variable 3	-0.66730799	1.213932825	-0.5497075097	0.590111373	-3.240730601	1.906114621

The estimate of the constant a is given in the row "Constant" and in the column "Coef" of the computer output:

$$a = 19.9152$$

The estimate of the coefficient b_1 of x_1 is given in the row "X Variable 1" and in the column "Coef" of the computer output:

$$b_1 = 0.8994$$

The estimate of the coefficient b_2 of x_2 is given in the row "X Variable 2" and in the column "Coef" of the computer output:

$$b_2 = 0.1539$$

The estimate of the coefficient b_3 of x_3 is given in the row "X Variable 3" and in the column "Coef" of the computer output:

$$b_3 = -0.6673$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned} \hat{y} &= a + b_1 x_1 + b_2 x_2 + b_3 x_3 \\ &= 19.9152 + 0.8994 x_1 + 0.1539 x_2 - 0.6673 x_3 \end{aligned}$$

Evaluate the multiple regression equation at $x_1 = 5$, $x_2 = 60$, and $x_3 = 0$:

$$\hat{y} = 19.9152 + 0.8994(5) + 0.1539(60) - 0.6673(0) = 33.6462$$

Thus the estimate salary is 33.6462 thousand dollars or $\$33,646.20$.

Step 3

(c)

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$H_0 : \text{All coefficients are zero}$$
$$H_1 : \text{At least one of the coefficients is nonzero}$$

The P-value of the global test is given in the row "Regression" and in the column "Significance F" of the ANOVA table:

$$P \approx 0.0000$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.0000 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

Step 4

(d)

$$\alpha = \text{Significance level} = 0.05$$

The P-values of the coefficients of the variables are given in the column "P-value" of the output.

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$x_1 \; P = 0.0000 < 0.05 \Rightarrow \text{ Reject } H_0$$
$$x_2 \; P = 0.0002 < 0.05 \Rightarrow \text{ Reject } H_0$$
$$x_3 \; P = 0.5901 > 0.05 \Rightarrow \text{ Fail to reject } H_0$$

There is sufficient evidence to support the claim that the coefficient of x_1 and x_2 are nonzero, while there is not sufficient to support the claim that the coefficient of x_3 is nonzero.

As it is likely that the coefficient of x_3 is zero, we consider deleting variable X_3 from the model.

Step 5

(e) I will use Excel to determine the regression equation. We delete variable x_3 from the model. The regression output is given in the image below.

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.9520489464					
R Square	0.9063971964					
Adjusted R Square	0.8953851019					
Standard Error	2.340107986					
Observations	20					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	2	901.4682084	450.7341042	82.3092458	0.000000001802	
Residual	17	93.09379156	5.476105386			
Total	19	994.562				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	20.11570487	1.842208157	10.91934415	0.000000004202	16.22898544	24.0024243
X Variable 1	0.8926481736	0.08501892485	10.49940558	0.000000007543	0.7132739232	1.072022424
X Variable 2	0.1463804083	0.0277079702	5.282971191	0.000060883512	0.08792170171	0.204839115

The estimate of the constant a is given in the row "Constant" and in the column "Coef" of the computer output:

$$a = 20.1157$$

The estimate of the coefficient b_1 of x_1 is given in the row "X Variable 1" and in the column "Coef" of the computer output:

$$b_1 = 0.8926$$

The estimate of the coefficient b_2 of x_2 is given in the row "X Variable 2" and in the column "Coef" of the computer output:

$$b_2 = 0.1464$$

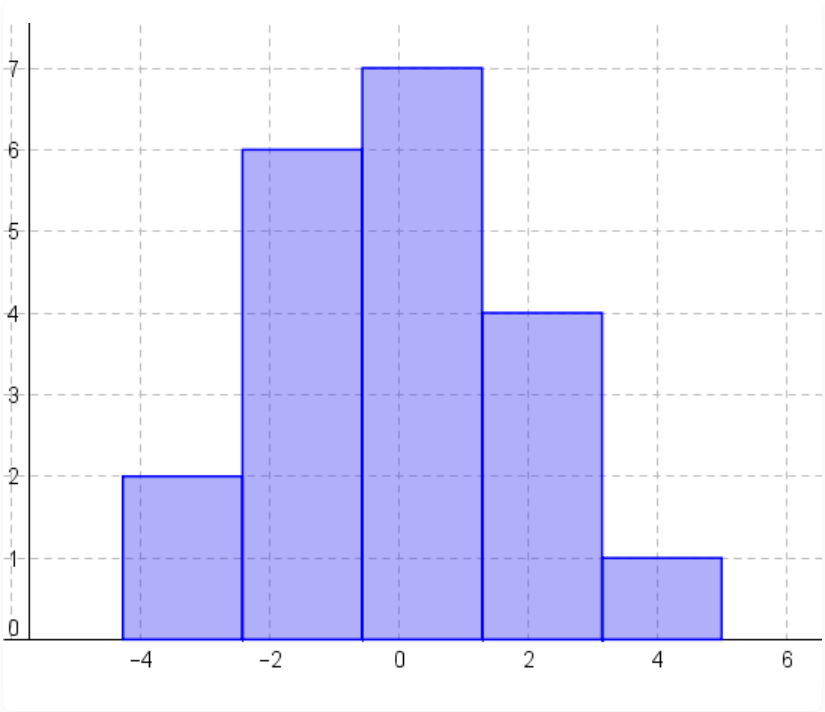
Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned}\hat{y} &= a + b_1 x_1 + b_2 x_2 \\ &= 20.1157 + 0.8926 x_1 + 0.1464 x_2\end{aligned}$$

We note that all P-values are now small and thus all coefficients are now significantly different from zero. This then implies that we found an appropriate model.

Step 6

(f)

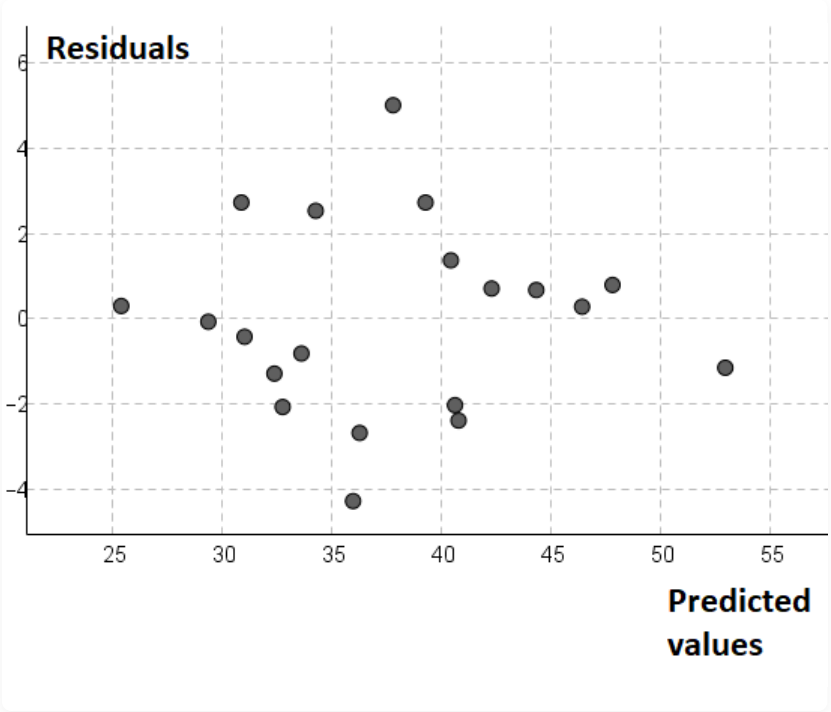


The histogram of the residuals is approximately normal, because the highest bars are roughly in the middle of the histogram.

This then implies that it is appropriate to assume that the residuals follow a normal distribution.

Step 7

(g)



The points appears to be randomly scatter in the residual plot with no obvious pattern.

However, we note that the vertical spread in the points in the middle in the residual plot is greater than the vertical spread in the points on either side of the residual plot, which indicates that the equal variance assumption for the residuals does not appear to be satisfied.

Result

(a) Years of experience X_1, No

(b) 33.6462 thousand dollars

(c) There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

(d) Yes, x_3

(e) $\hat{y} = 20.1157 + 0.8926x_1 + 0.1464x_2$

(f) It is appropriate to assume that the residuals follow a normal distribution.

(g) Yes, no equal variances

PROBLEM STATEMENT

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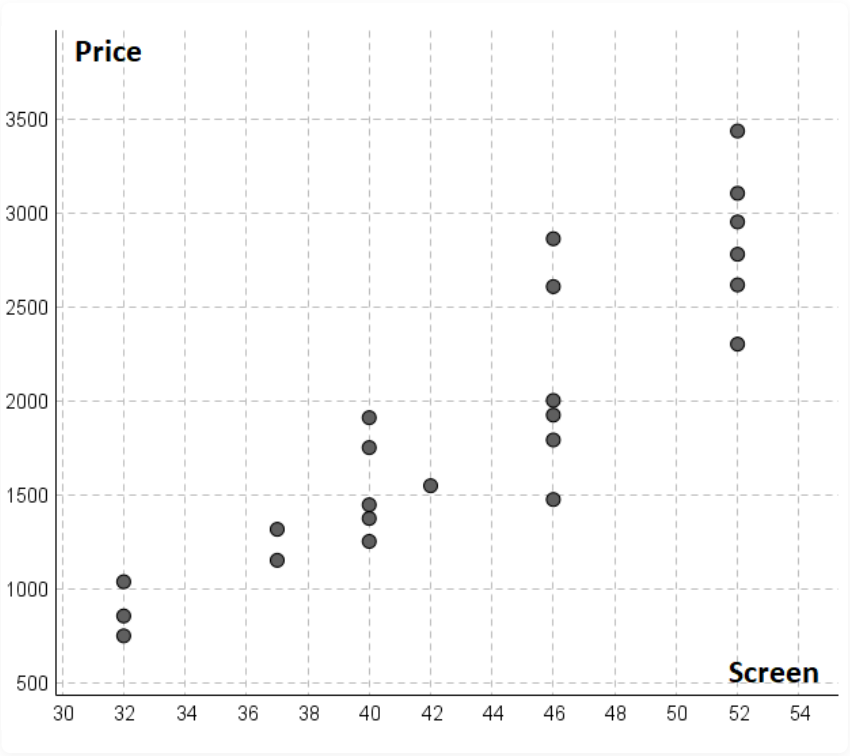
Solution A

Step 1

(a)

Scatter diagram

Screen is on the horizontal axis and price is on the vertical axis.



We note that there is no strong curvature present in the scatter diagram, thus there appears to be a linear relationship between the screen size and the price.

Step 2

(b) A **response variable** (or dependent variable) is a variable that measures an outcome or result of a study, while an **explanatory variable** (or independent variable) is a variable that we think explains or causes changes in the response variables.

We expect the price to increase as the screen size increases, which implies that the independent variable is the screen size and the dependent variable is the price.

Step 3

(c) I will use Excel to determine the regression equation. The regression output is given in the image below.

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.8925310211					
R Square	0.7966116236					
Adjusted R Square	0.7869264628					
Standard Error	356.133106					
Observations	23					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	1	10431926.04	10431926.04	82.25073816	0.000000010417	
Residual	21	2663446.574	126830.7892			
Total	22	13095372.61				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	-2484.321462	491.4825047	-5.054750552	0.000052674474	-3506.41528	-1462.227645
X Variable 1	101.3423936	11.1743239	9.06921927	0.000000010417	78.10411503	124.5806722

The estimate of the constant a is given in the row "Constant" and in the column "Coef" of the computer output:

$$a = 101.3424$$

The estimate of the coefficient b_1 of the variable screen size x_1 is given in the row "X_1" and in the column "Coef" of the computer output:

$$b_1 = 11.1743$$

Combining these estimates, we then obtain the multiple regression equation:

$$\hat{y} = a + b_1 x = 101.3424 + 11.1743 x_1$$

Step 4

(d) I will use Excel to determine the regression equation. The regression output is given in the image below.

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.9209700524					
R Square	0.8481858374					
Adjusted R Square	0.8242151801					
Standard Error	323.4731769					
Observations	23					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	3	11107309.58	3702436.527	35.38433795	0.000000056008	
Residual	19	1988063.027	104634.8962			
Total	22	13095372.61				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	-1926.72926	511.1692295	-3.769259081	0.001297689365	-2996.618737	-856.8397837
X Variable 1	94.11911435	10.83068305	8.690044196	0.000000048010	71.45023456	116.7879941
X Variable 2	-366.4073711	172.3576538	-2.125854948	0.0468470259	-727.1560808	-5.658661372
X Variable 3	-381.4403427	168.8088242	-2.259599548	0.03578748944	-734.7612667	-28.11941873

The estimate of the constant a is given in the row "Constant" and in the column "Coef" of the computer output:

$$a = -1926.7293$$

The estimate of the coefficient b_1 of the variable screen size x_1 is given in the row "X_1" and in the column "Coef" of the computer output:

$$b_1 = 94.1191$$

The estimate of the coefficient b_2 of the variable manufacturer x_2 (1=Sharp, 0=not sharp) is given in the row "X_2" and in the column "Coef" of the computer output:

$$b_2 = -366.4074$$

The estimate of the coefficient b_3 of the variable manufacturer x_3 (1=Samsung, 0=not Samsung) is given in the row "X_2" and in the column "Coef" of the computer output:

$$b_3 = -381.4403$$

Note that $x_2 = x_3 = 0$ implies that the manufacturer is Sony.

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned}\hat{y} &= a + b_1 x + b_2 x_2 + b_3 x_3 \\ &= -1926.7293 + 94.1191 x_1 - 366.4074 x_2 - 381.4403 x_3\end{aligned}$$

Some manufacturers can command a premium price, because the coefficients of x_2 and x_3 are negative and thus the remaining manufacturer Sony will be able to command a premium price.

Step 5

(e)

$$\alpha = \text{Significance level} = 0.05$$

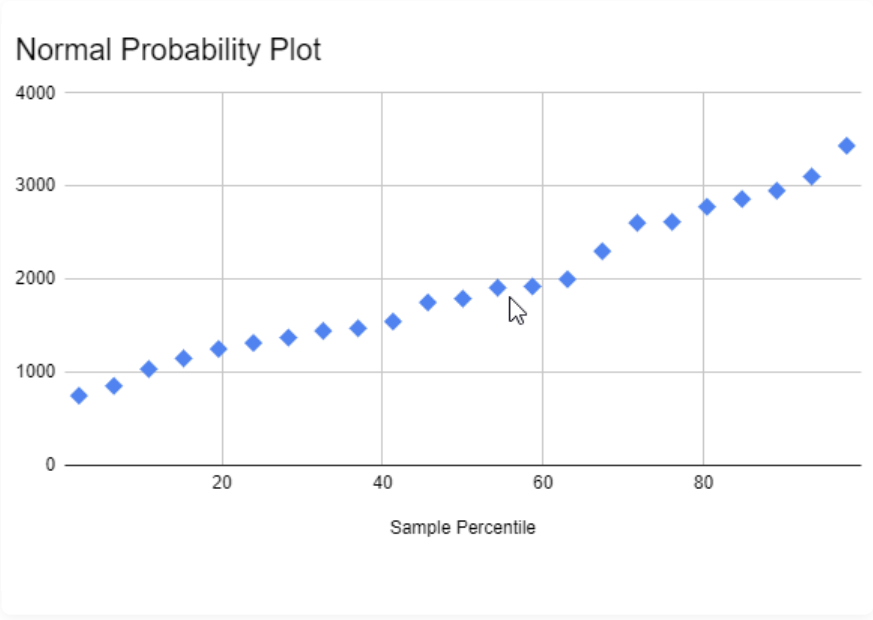
If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$x_1\ P = 0.0000 < 0.05 \Rightarrow \text{ Reject } H_0$$
$$x_2\ P = 0.0468 < 0.05 \Rightarrow \text{ Reject } H_0$$
$$x_3\ P = 0.0358 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that each coefficient is nonzero.

Step 6

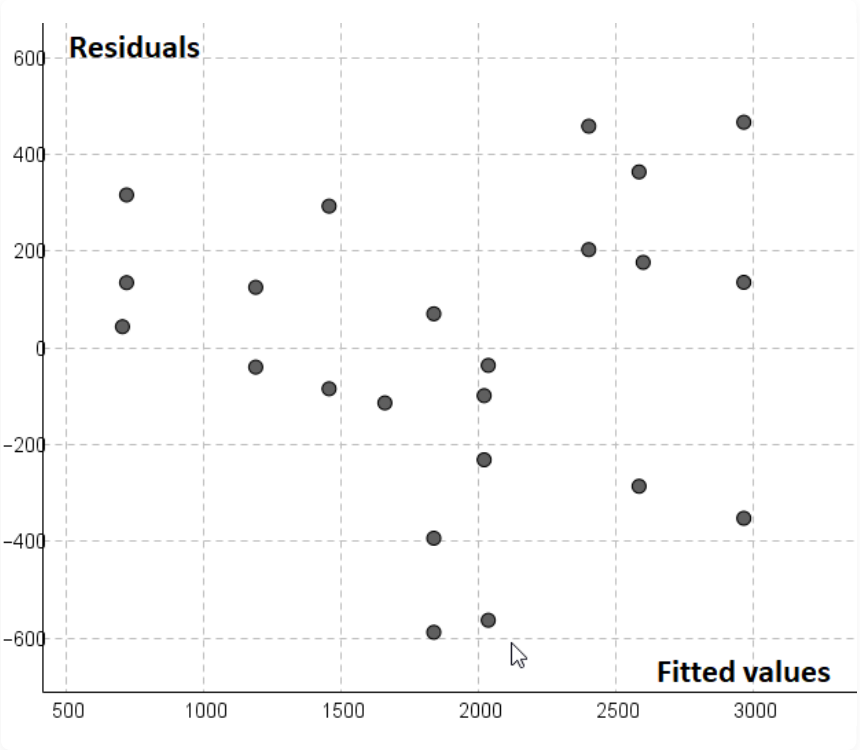
(f)



The pattern in the normal probability plot of the residuals does not show strong curvature, which implies that it is appropriate to assume that the residuals follow a normal distribution.

Step 7

(g)



We note that the vertical spread in the points to the left in the residual plot is less than the vertical spread in the points to the right in the residual plot.

This then implies that the data values do not appear to have the same amount of variation everywhere in the residual plot.

Result

(a) Yes

(b) Price

(c) $\hat{y} = 101.3424 + 11.1743x_1$

(d) Yes

(e) All are significant

(f) It is appropriate to assume that the residuals follow a normal distribution.

(g) No

PROBLEM STATEMENT

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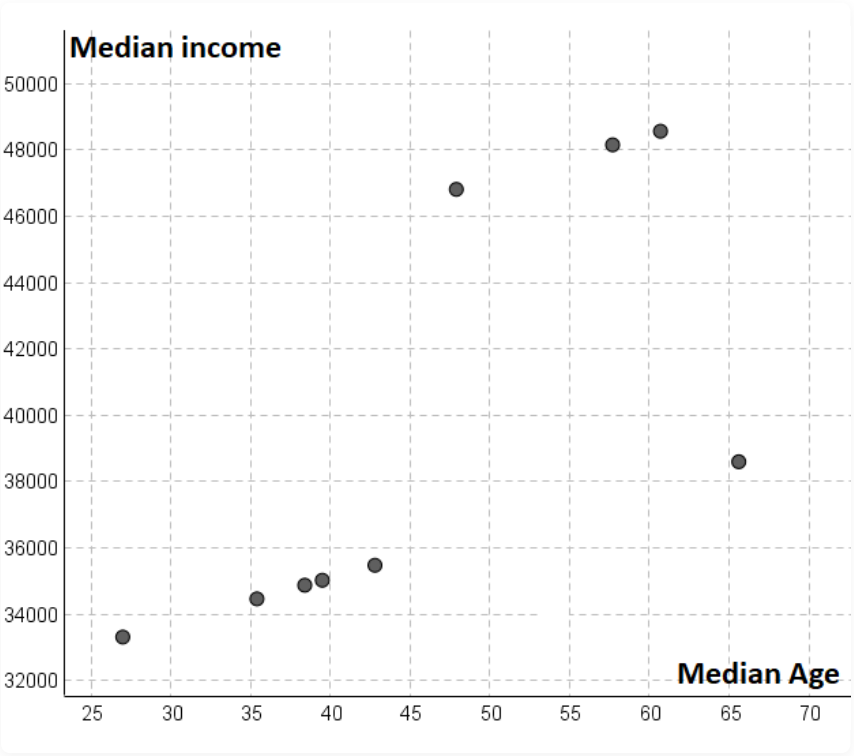
Solution A

Step 1

(a)

Scatter diagram

Screen is on the horizontal axis and price is on the vertical axis.



We note that there is no strong curvature present in the scatter diagram, thus there appears to be a linear relationship between the median income and the median age.

Step 2

(b) A **response variable** (or dependent variable) is a variable that measures an outcome or result of a study, while an **explanatory variable** (or independent variable) is a variable that we think explains or causes changes in the response variables.

We expect the median income to increase as the median age increases, which implies that the independent variable is the median age and the dependent variable is the median income.

Step 3

(c) I will use Excel to determine the regression equation. The regression output is given in the image below.

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.7214069411					
R Square	0.5204279747					
Adjusted R Square	0.4519176853					
Standard Error	4774.122002					
Observations	9					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	1	173137785.3	173137785.3	7.59634764	0.02825244065	
Residual	7	159545686.2	22792240.89			
Total	8	332683471.6				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	22804.66424	6255.241555	3.645688825	0.008224800106	8013.368454	37595.96002
X Variable 1	361.5855949	131.1924083	2.756147246	0.02825244072	51.36484656	671.8063432

The estimate of the constant a is given in the row "Constant" and in the column "Coef" of the computer output:

$$a = 22804.6642$$

The estimate of the coefficient b_1 of the variable median age x_1 is given in the row "X_1" and in the column "Coef" of the computer output:

$$b_1 = 361.5856$$

Combining these estimates, we then obtain the multiple regression equation:

$$\hat{y} = a + b_1 x = 22804.6642 + 361.5856 x_1$$

Step 4

(d) I will use Excel to determine the regression equation. The regression output is given in the image below.

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.9999999995					
R Square	0.999999999					
Adjusted R Square	0.9999999987					
Standard Error	0.23156469					
Observations	9					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	2	332683471.2	166341735.6	3102105434	0	
Residual	6	0.321733234	0.05362220567			
Total	8	332683471.6				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	29619.20002	0.3281190729	90269.66875	0	29618.39714	29620.0029
X Variable 1	136.887012	0.007580349645	18058.13958	0	136.8684635	136.9055604
X Variable 2	10639.69662	0.1950558809	54546.9153	0	10639.21933	10640.1739

The estimate of the constant a is given in the row "Constant" and in the column "Coef" of the computer output:

$$a = 29619.2000$$

The estimate of the coefficient b_1 of the variable screen size x_1 is given in the row "X_1" and in the column "Coef" of the computer output:

$$b_1 = 136.8870$$

The estimate of the coefficient b_2 of the variable Coastal x_2 is given in the row "X_2" and in the column "Coef" of the computer output:

$$b_2 = 10639.6966$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned}\hat{y} &= a + b_1 x + b_2 x_2 \\ &= 29619.2000 + 136.8870 x_1 + 10639.6966 x_2\end{aligned}$$

We note that all P-values in the regression output are 0, which indicate that all coefficients are nonzero and thus the dummy variable has a significant influence on the incomes.

Step 5

(e)

$$\alpha = \text{Significance level} = 0.05$$

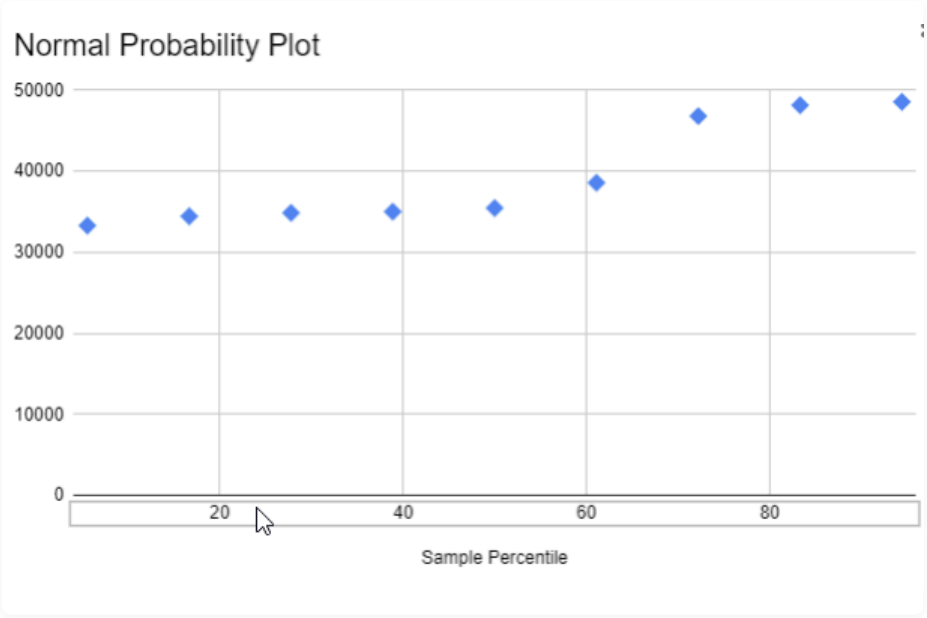
If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$x_1\ P = 0 < 0.05 \Rightarrow \text{ Reject } H_0$$
$$x_2\ P = 0 < 0.05 \Rightarrow \text{ Reject } H_0$$
$$x_3\ P = 0 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that each coefficient is nonzero.

Step 6

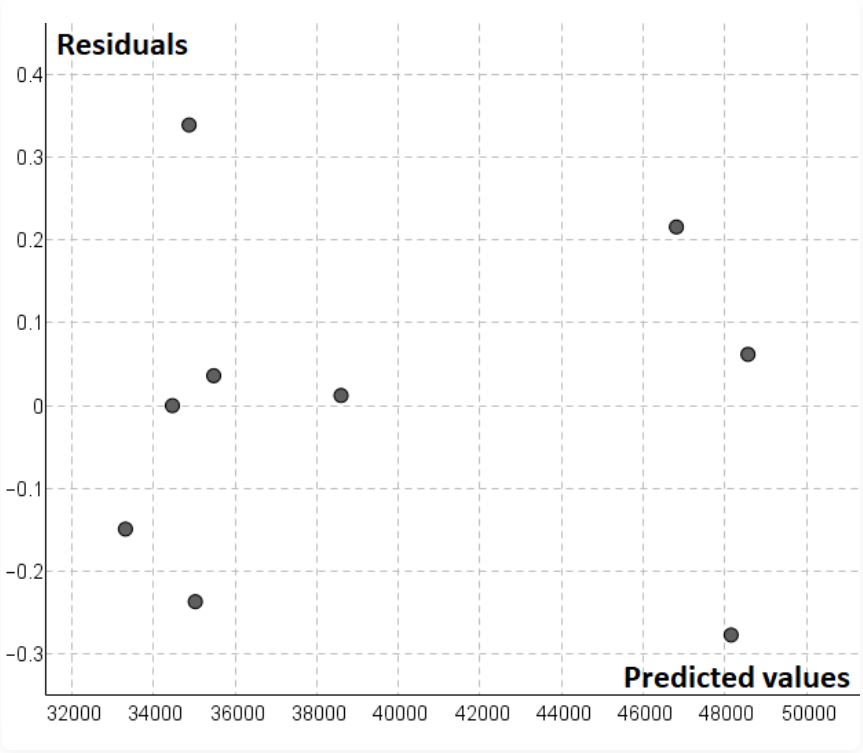
(f)



The pattern in the normal probability plot of the residuals shows strong curvature (between 50 and 80 on the horizontal axis), which implies that it is not appropriate to assume that the residuals follow a normal distribution.

Step 7

(g)



We note that the vertical spread in the points to the left in the residual plot is about the same as the vertical spread in the points to the right in the residual plot.

This then implies that the data values appear to have the same amount of variation everywhere in the residual plot.

Result

(a) Yes

(b) Median income

(c) $\hat{y} = 22804.6642 + 361.5856x_1$

(d) Yes

(e) All are significant

(f) It is not appropriate to assume that the residuals follow a normal distribution.

(g) Yes

PROBLEM STATEMENT

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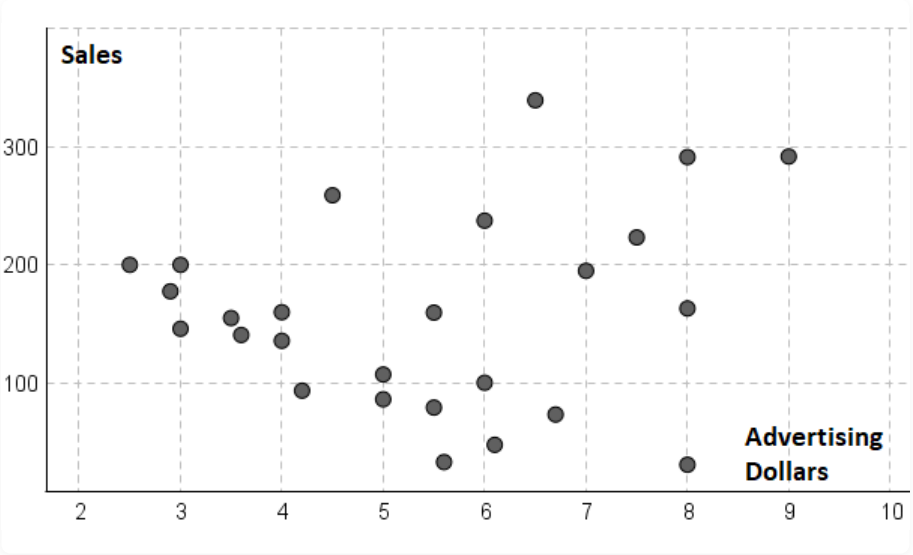
Solution A

Step 1

(a)

Scatter diagram

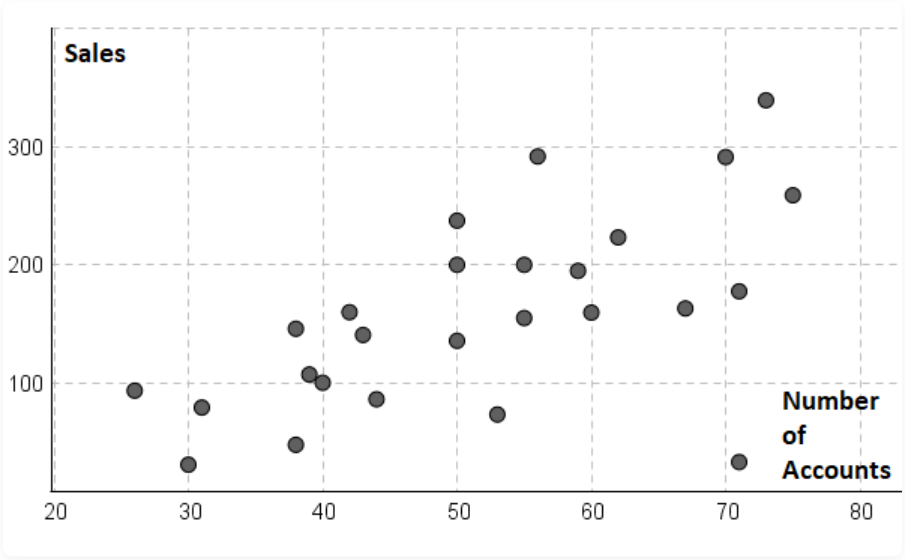
Advertising dollars is on the horizontal axis and Sales is on the vertical axis.



We note that there is no strong curvature present in the scatter diagram between advertising dollars and sales, thus there appears to be a linear relationship between advertising dollars and sales.

Step 2

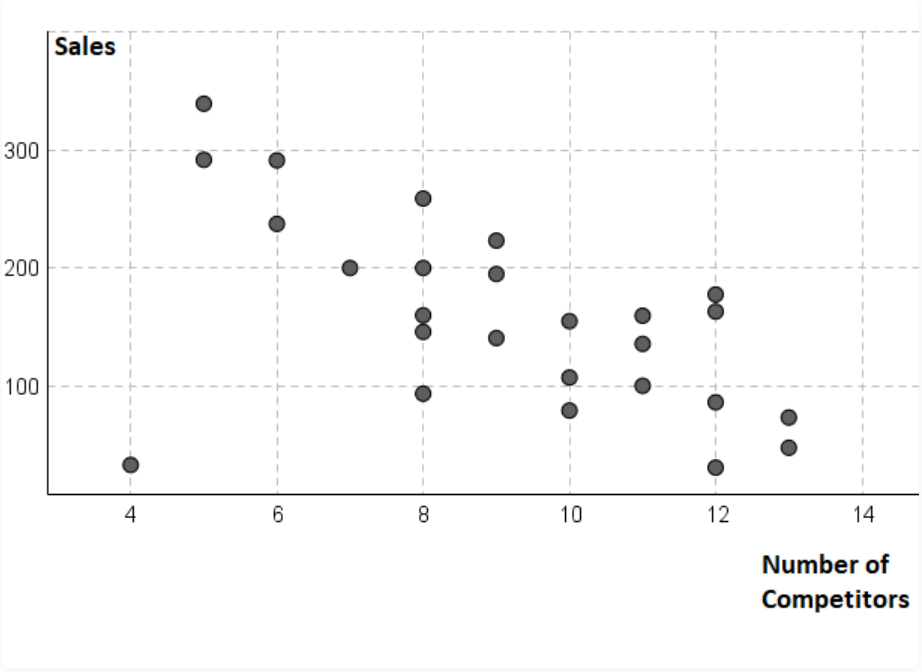
Number of accounts is on the horizontal axis and Sales is on the vertical axis.



We note that there is no strong curvature present in the scatter diagram between number of accounts and sales, thus there appears to be a linear relationship between number of accounts and sales.

Step 3

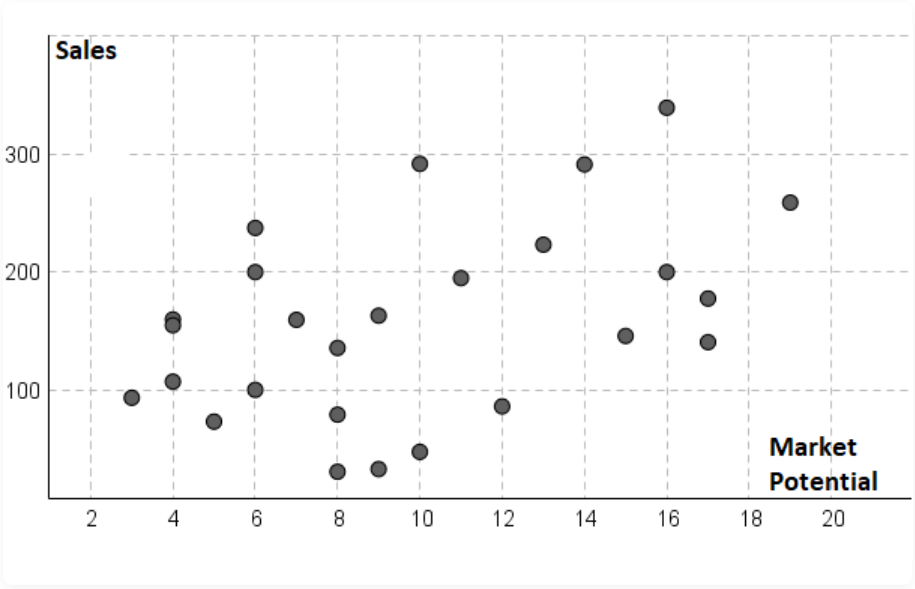
Number of Competitors is on the horizontal axis and Sales is on the vertical axis.



We note that there is no strong curvature present in the scatter diagram between number of competitors and sales, thus there appears to be a linear relationship between number of competitors and sales.

Step 4

Market potential is on the horizontal axis and Sales is on the vertical axis.



We note that there is no strong curvature present in the scatter diagram between market potential and sales, thus there appears to be a linear relationship between market potential and sales.

We note that advertising potential appears to contribute the least to the sales, because the other scatterplots slope upwards or downwards much stronger.

Step 5

(b) We need to determine the correlation between each pair of variables and combine the results in a matrix:

	Sales	Advertising	Accounts	Competitors	Potential
Sales	1				
Advertising	0.1485	1			
Accounts	0.6111	0.1726	1		
Competitors	-0.5706	-0.0383	-0.3243	1	
Potential	0.4470	-0.0706	0.4682	-0.2021	1

We note that the largest correlation 0.6111 belongs to the variables Number of accounts and Sales.

Moreover, we note that none of the correlations between two independent variables (variables excluding Sales) are large (less than -0.7 or larger than 0.7) and thus it is safe to assume that there are no problems with multicollinearity.

The variable Advertising dollars appears to be redundant, because all correlations including "Advertising" are small.

Step 6

(c) I will use Excel to determine the regression equation. The regression output is given in the image below.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.7483733348
R Square	0.5600626482
Adjusted R Squa	0.4762650573
Standard Error	59.3564952
Observations	26

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	4	94189.29948	23547.32487	6.683517302	0.001239463496
Residual	21	73987.06398	3523.193523		
Total	25	168176.3635			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	111.3722521	80.09648073	1.390476225	0.178946086	-55.1974975	277.9420017
X Variable 1	3.687776713	6.680965529	0.5519825984	0.5867872014	-10.20605165	17.58160507
X Variable 2	2.153498627	1.006692518	2.139182113	0.04432594709	0.05996693744	4.247030317
X Variable 3	-12.92528255	4.66956676	-2.654298249	0.014833801729	-22.65210094	-2.798464111
X Variable 4	3.39881653	2.890845887	1.175716957	0.2528656811	-2.613026577	9.410659637

Step 7

(d)

$\alpha = \text{Significance level} = 0.05$

If the P-value is less than the significance level, then we reject the null hypothesis H_0.

$x_1\ P = 0.5868 > 0.05 \Rightarrow$ Fail to reject H_0

$x_2\ P = 0.0443 < 0.05 \Rightarrow$ Reject H_0

$x_3\ P = 0.0148 < 0.05 \Rightarrow$ Reject H_0

$x_3\ P = 0.2529 > 0.05 \Rightarrow$ Fail to reject H_0

There is sufficient evidence to support the claim that the coefficient of the number of accounts and the coefficient of the number of competitors are both nonzero, while there is not sufficient evidence to support the claim that the coefficient of Advertising dollars and market potential are both nonzero.

This then implies that we should drop Advertising dollars and market potential from the model.

The estimate of the constant a is given in the row "Intercept" and in the column "Coef" of the computer output:

$$a = 111.3723$$

The estimate of the coefficient b_1 of the variable Advertising dollars x_1 is given in the row "X Variable 1" and in the column "Coef" of the computer output:

$$b_1 = 3.6878$$

The estimate of the coefficient b_2 of the variable Number of accounts x_2 is given in the row "X Variable 2" and in the column "Coef" of the computer output:

$$b_2 = 2.1535$$

The estimate of the coefficient b_3 of the variable Number of competitors x_3 is given in the row "X Variable 3" and in the column "Coef" of the computer output:

$$b_3 = -12.9253$$

The estimate of the coefficient b_4 of the variable manufacturer x_4 is given in the row "X Variable 4" and in the column "Coef" of the computer output:Market potential

$$b_4 = 3.3988$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned}\hat{y} &= a + b_1 x + b_2 x_2 + b_3 x_3 + b_4 x_4 \\ &= 111.3723 + 3.6878 x_1 + 2.1535 x_2 - 12.9253 x_3 + 3.3988 x_4\end{aligned}$$

The P-value of the global test is given in the row "Regression" and in the column "Significant F" of the output. If the P-value is less than the significant level, then we reject the null hypothesis H_0. We assume that the

significance level is 0.05.

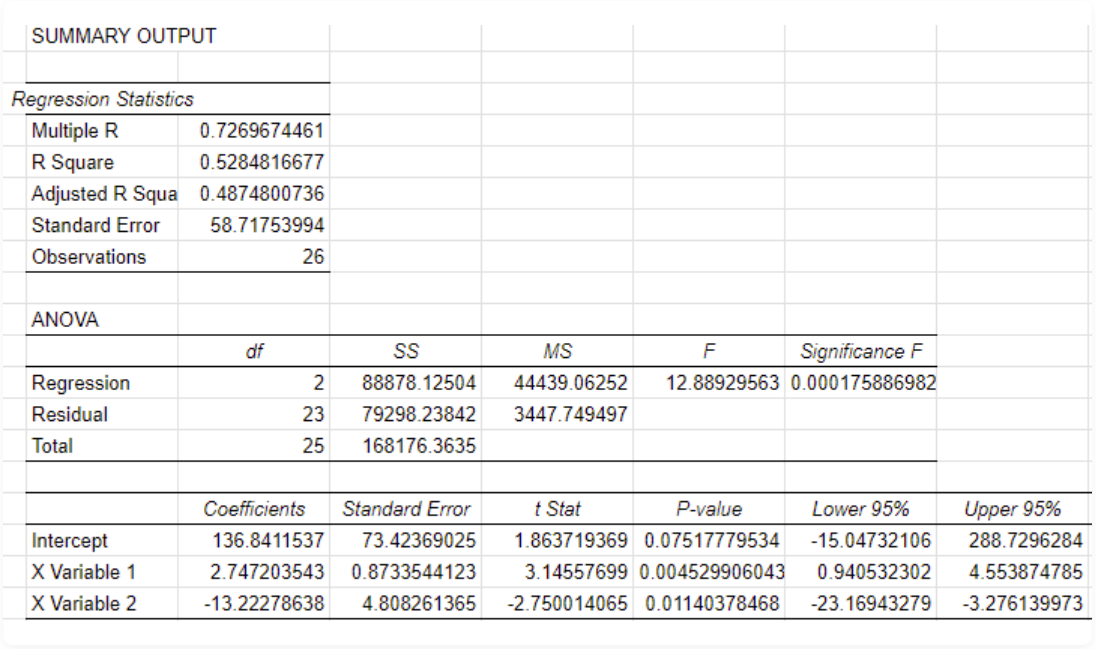
$$P = 0.001239 < 0.05 \Rightarrow \text{Reject } H_0$$

Thus we can conclude that some of the independent variables are useful in explaining the variation in the dependent variable.

Step 8

(e) As stated in part (d), we will drop Advertising dollars and market potential from the model.

I will use Excel to determine the regression equation. The regression output is given in the image below.



SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.7269674461
R Square	0.5284816677
Adjusted R Squa	0.4874800736
Standard Error	58.71753994
Observations	26

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	88878.12504	44439.06252	12.88929563	0.000175886982
Residual	23	79298.23842	3447.749497		
Total	25	168176.3635			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	136.8411537	73.42369025	1.863719369	0.07517779534	-15.04732106	288.7296284
X Variable 1	2.747203543	0.8733544123	3.14557699	0.004529906043	0.940532302	4.553874785
X Variable 2	-13.22278638	4.808261365	-2.750014065	0.01140378468	-23.16943279	-3.276139973

The estimate of the constant a is given in the row "Intercept" and in the column "Coef" of the computer output:

$$a = 136.8411$$

The estimate of the coefficient b_1 of the variable Number of accounts x_2 is given in the row "X Variable 1" and in the column "Coef" of the computer output:

$$b_1 = 2.7472$$

The estimate of the coefficient b_2 of the variable Number of competitors x_3 is given in the row "X Variable 2" and in the column "Coef" of the computer output:

$$b_2 = -13.2228$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned}\hat{y} &= a + b_1 x_2 + b_2 x_3 \\ &= 136.8411 + 2.7472 x_2 - 13.2228 x_3\end{aligned}$$

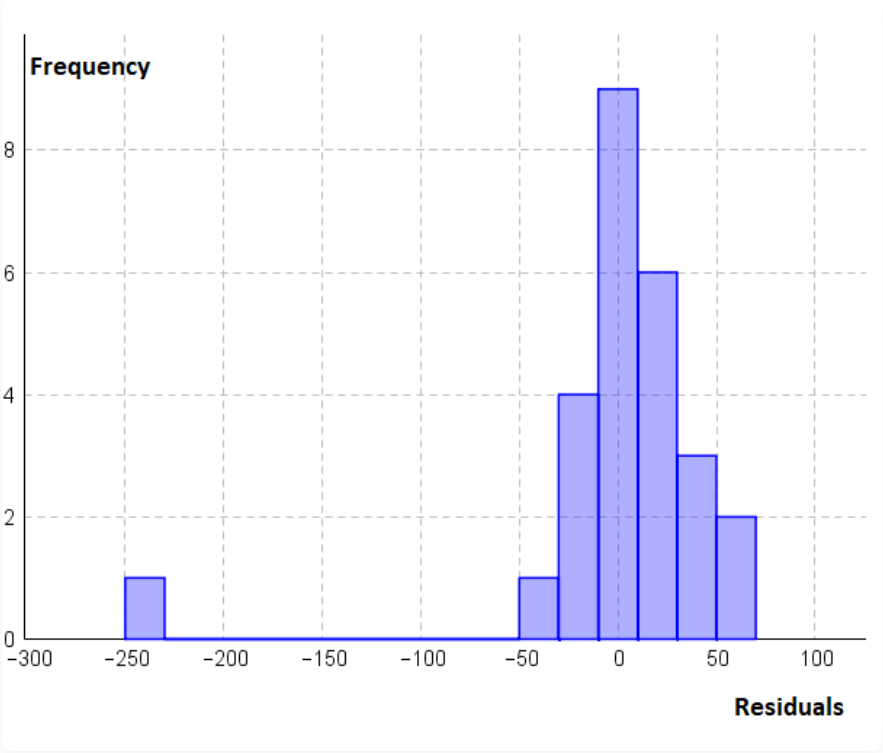
Since all P-values (except the intercept) are now less than 0.05, all remaining variables are significant.

Step 9

(f)

Histogram of the residuals

The residuals are on the horizontal axis, while the bars of the histogram need to have the same width and the height needs to be equal to the frequency.



We note that there is a strong outlier present the histogram (due to one bar in the histogram being far removed from the other bars in the histogram).

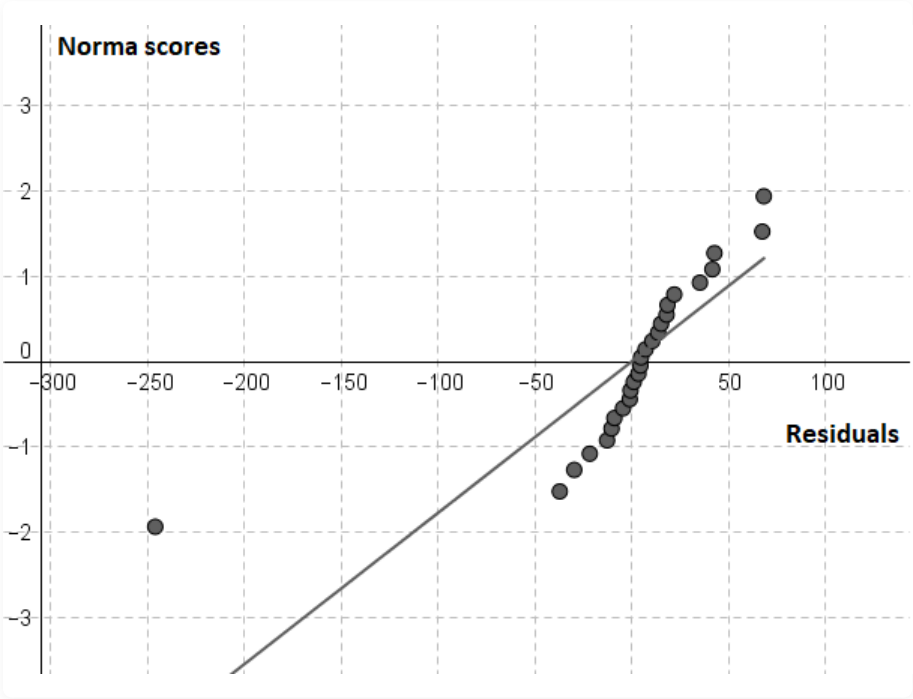
Step 10

Normal probability plot

The data values are on the horizontal axis and the standardized normal scores are on the vertical axis.

If the data contains n data values, then the standardized normal scores are the z-scores in the normal probability table of the appendix corresponding to an area of $\frac{j-0.5}{n}$ (or the closest area) with $j \in \{1, 2, 3, ..., n\}$.

The smallest standardized score corresponds with the smallest data value, the second smallest standardized score corresponds with the second-smallest data value, and so on.



Step 11

We note that there is a strong outlier present the histogram (due to one bar in the histogram being far removed from the other bars in the histogram).

Moreover, the pattern in the normal probability plot of the residuals shows strong curvature due to this outlier, which implies that it is not appropriate to assume that the residuals follow a normal distribution.

Step 12

(g) We need to determine the variance inflation factor $VIF = \frac{1}{1-R_j^2}$ for each of the independent variables, where R_j^2 represents the coefficient of determination when the independent variable is used as dependent variable and the remaining independent variables are used as independent variables.

$$\text{Advertising}VIF = \frac{1}{1-R_j^2} = \frac{1}{1-0.0592} \approx 1.0629$$
$$\text{Accounts}VIF = \frac{1}{1-R_j^2} = \frac{1}{1-0.3118} \approx 1.4531$$
$$\text{Competitors}VIF = \frac{1}{1-R_j^2} = \frac{1}{1-0.1085} \approx 1.1217$$
$$\text{Potential}VIF = \frac{1}{1-R_j^2} = \frac{1}{1-0.2454} \approx 1.3252$$

Since none of the VIF-values are greater than 10, there are no problems in this case.

Result

(a) Advertising dollars appears to contribute the least to the sales

(b) No, Yes

(c) $\hat{y} = 111.3723 + 3.6878x_1 + 2.1535x_2 - 12.9253x_3 + 3.3988x_4$, Yes

(d) Yes

(e) $\hat{y} = 136.8411 + 2.7472x_2 - 13.2228x_3$

(f) Yes, one outlier among the residuals

(g) No

PROBLEM STATEMENT

Chapter 14, Page 561

Solution A

Step 1

(a) I will use Excel to determine the regression equation. The regression output is given in the image below.

The estimate of the constant a is given in the row "Constant" and in the column "Coef" of the computer output:

$$a = -6.7410$$

The estimate of the coefficient b_1 of the variable Popul x_1 is given in the row "X Variable 1" and in the column "Coef" of the computer output:

$$b_1 = 0.0076$$

The estimate of the coefficient b_2 of the variable Adv x_2 is given in the row "X Variable 2" and in the column "Coef" of the computer output:

$$b_2 = 0.0531$$

The estimate of the coefficient b_3 of the variable Income x_3 is given in the row "X Variable 3" and in the column "Coef" of the computer output:

$$b_3 = 1.1242$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned}\hat{y} &= a + b_1x_1 + b_2x_2 + b_3x_3 \\ &= -6.7410 + 0.0076x_1 + 0.0531x_2 + 1.1242x_3\end{aligned}$$

Regression Statistics						
Multiple R	0.9124484585					
R Square	0.8325621895					
Adjusted R Squa	0.8061246404					
Standard Error	0.3385800217					
Observations	23					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	3	10.83027303	3.610091009	31.49165562	0.000000140864	
Residual	19	2.178092191	0.1146364311			
Total	22	13.00836522				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	-6.741026173	8.939133198	-0.7541028893	0.4600320204	-25.45084669	11.96879434
X Variable 1	0.007554484007	0.001877843469	4.022957254	0.000727049050	0.003624112517	0.0114848555
X Variable 2	0.05308397956	0.01524928793	3.481079235	0.002500406075	0.02116685361	0.08500110551
X Variable 3	1.124209302	0.2597405859	4.328200376	0.000362340104	0.5805660161	1.667852588

Step 2

(b)

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$H_0 : \text{All coefficients are zero}$$
$$H_1 : \text{At least one of the coefficients is nonzero}$$

The P-value of the global test is given in the row "Regression" and in the column "Significance F" of the ANOVA table:

$$P \approx 0.0000$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.0000 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

Step 3

(c) The P-values of the coefficients of the variables are given in the column "P-value".

If the P-value is less than the significance level, then we reject the null hypothesis H_0 of a zero coefficient:

$$\begin{aligned} \text{Popul: } P = 0.0007 < 0.05 &\Rightarrow \text{ Reject } H_0 \\ \text{Adv: } P = 0.0025 < 0.05 &\Rightarrow \text{ Reject } H_0 \\ \text{Income: } P = 0.0004 < 0.05 &\Rightarrow \text{ Reject } H_0 \end{aligned}$$

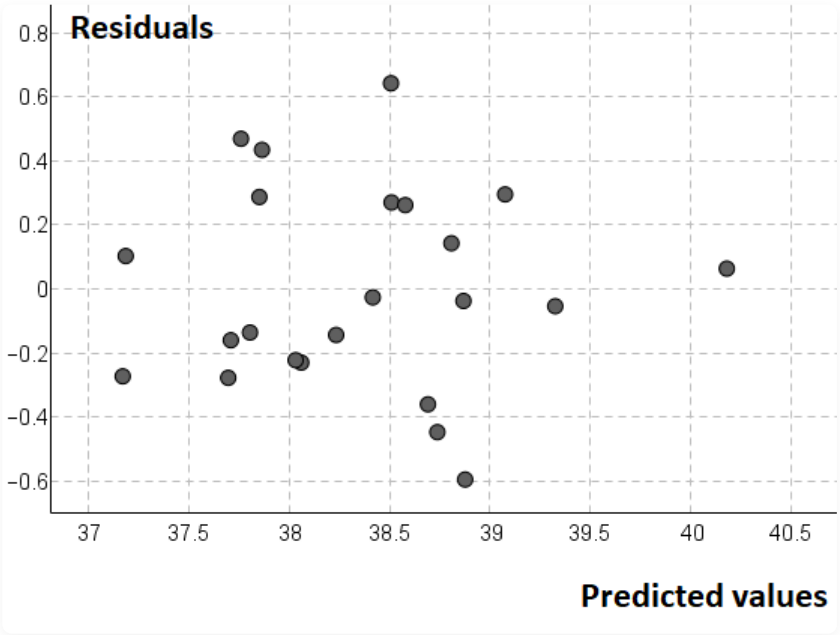
Thus we conclude that all coefficients are nonzero and thus we would not consider deleting any coefficients.

Step 4

(d) The residuals appear to be randomly scattered in the residual plot, while there is no strong pattern present in the scatterplot.

Moreover, the vertical spread of the points is approximately the same everywhere in the graph.

This then implies that there are no violations of the assumption.



Step 5

(e) We note that most of the data values in the stem-and-leaf plot are centered in the middle of the stem-and-leaf plot, which indicates that the residuals have approximately a normal distribution.

Thus we can conclude that there are no problems with the normality assumption.

-6	0			
-5				
-4	5			
-3	6			
-2	8	7	3	2
-1	6	4	4	
-0	5	4	3	
0	6			
1	0	4		
2	6	7	9	
3	0			
4	3	7		
5				
6	4			

Key: $3|1$ means 0.31

Result

(a) $\hat{y} = -6.7410 + 0.0076x_1 + 0.0531x_2 + 1.1242x_3$

(b) There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

(c) No

(d) No

(e) No

Exercise 25

PROBLEM STATEMENT

Chapter 14, Page 562

Solution A

Step 1

(a) I will use Excel to determine the regression equation. The regression output is given in the image below.

The estimate of the constant a is given in the row "Constant" and in the column "Coef" of the computer output:

$$a = 5367.7830$$

The estimate of the coefficient b_1 of the variable Length of Service x_1 is given in the row "X Variable 1" and in the column "Coef" of the computer output:

$$b_1 = -54.8342$$

The estimate of the coefficient b_2 of the variable Age x_2 is given in the row "X Variable 2" and in the column "Coef" of the computer output:

$$b_2 = 58.6894$$

The estimate of the coefficient b_3 of the variable Gender x_3 is given in the row "X Variable 3" and in the column "Coef" of the computer output:

$$b_3 = 1416.3358$$

The estimate of the coefficient b_3 of the variable Job x_4 is given in the row "X Variable 4" and in the column "Coef" of the computer output:

$$b_4 = -1159.3822$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned}\hat{y} &= a + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4 \\ &= 5367.7830 - 54.8342x_1 + 58.6894x_2 + 1416.3358x_3 - 1159.3822x_4\end{aligned}$$

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.3449168798
R Square	0.118967654
Adjusted R Squa	-0.02199752141
Standard Error	2940.086684
Observations	30

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	4	29180808.67	7295202.168	0.8439506683	0.5106115774
Residual	25	216102742.8	8644109.712		
Total	29	245283551.5			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	5367.782984	4292.142905	1.250606772	0.2226568019	-3472.050729	14207.6167
X Variable 1	-54.83417928	63.7077657	-0.8607142108	0.3975752949	-186.0427778	76.37441921
X Variable 2	58.68944783	78.92413632	0.7436184996	0.464039226	-103.8578523	221.236748
X Variable 3	1416.335813	1122.013615	1.262316067	0.2184841405	-894.4944648	3727.166092
X Variable 4	-1159.382204	1113.085666	-1.041592969	0.3075691064	-3451.825028	1133.060619

Step 2

(b) The value of R^2 is given to the right of "R Square" in the regression output:

$$R^2 \approx 0.1190 = 11.90\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

11.90% of the variation in the monthly salary is explained by the variation in the independent variables.

Step 3

(c)

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$H_0 : \text{All coefficients are zero}$$
$$H_1 : \text{At least one of the coefficients is nonzero}$$

The P-value of the global test is given in the row "Regression" and in the column "Significance F" of the ANOVA table:

$$P \approx 0.5106$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.5106 > 0.05 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

Step 4

(d) The P-values of the coefficients of the variables are given in the column "P-value".

If the P-value is less than the significance level, then we reject the null hypothesis H_0 of a zero coefficient:

$$\begin{aligned}\text{Length of Service: } P = 0.3976 > 0.05 &\Rightarrow \text{ Fail to reject } H_0\\ \text{Age: } P = 0.4640 > 0.05 &\Rightarrow \text{ Fail to reject } H_0\\ \text{Gender: } P = 0.2185 > 0.05 &\Rightarrow \text{ Fail to reject } H_0\\ \text{Job: } P = 0.3076 > 0.05 &\Rightarrow \text{ Fail to reject } H_0\end{aligned}$$

Thus we conclude that all coefficients are not significantly different from zero and thus we would consider deleting all variables from the model.

Step 5

(e) We cannot rerun the regression equation using only the significant independent variables as none of the independent variables are significant.

🎯 Result

(a) $\hat{y} = 5367.7830 - 54.8342x_1 + 58.6894x_2 + 1416.3358x_3 - 1159.3822x_4$

(b) 11.90% of the variation in the monthly salary is explained by the variation in the independent variables.

(c) There is not sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

(d) All independent variables should be dropped

(e) Cannot be executed

PROBLEM STATEMENT

Chapter 14, Page 562

Solution A

Step 1

Let

Y=Food

X_1=Income

X_2=Size

(a) I will use Excel to determine the correlation matrix of the given data, which results in the following table:

	Y	X_1	X_2
Y	1		
X_1	0.1559	1	
X_2	0.8758	-0.0962	1

We note that the largest correlation of 0.8758 belongs to the variable X_2 (Size).

Moreover, we note that none of the correlation between two independent variables (X_i) are large (less than -0.7 or larger than 0.7) and thus it is safe to assume that there are no problems with multicollinearity.

Step 2

(b) I will use Excel to determine the regression equation. The regression output is given in the image below.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.90841946
R Square	0.8252259152
Adjusted R Squa	0.8093373621
Standard Error	0.4208393538
Observations	25

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	18.39717724	9.198588621	51.93839282	0.000000004647
Residual	22	3.896326758	0.1771057617		
Total	24	22.293504			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	2.841213755	0.2635356767	10.78113518	0.000000000301	2.294674215	3.387753294
X Variable 1	0.006137947395	0.002267877534	2.706472154	0.01288994813	0.001434657277	0.01084123751
X Variable 2	0.4245282859	0.04228003433	10.040869	0.000000001120	0.3368448618	0.51221171

The estimate of the constant a is given in the row "Constant" and in the column "Coef" of the computer output:

$$a = 2.8412$$

The estimate of the coefficient b_1 of the variable x_1 is given in the row "X Variable 1" and in the column "Coef" of the computer output:

$$b_1 = 0.0061$$

The estimate of the coefficient b_2 of the variable x_2 is given in the row "X Variable 2" and in the column "Coef" of the computer output:

$$b_2 = 0.4245$$

Combining these estimates, we then obtain the multiple regression equation:

$$\hat{y} = a + b_1 x_1 + b_2 x_2 = 2.4812 + 0.0061 x_1 + 0.4245 x_2$$

The coefficient of the variable Size (x_2) is 0.4245, which implies that the amount spent on food increases by $\$0.4245$ per additional family member.

Step 3

(c) The value of R^2 is given to the right of "R Square" in the regression output:

$$R^2 \approx 0.8252 = 82.52\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

82.52% of the variation in the dependent variable is explained by the variation in the independent variables.

We can conclude that this value is greater than 0, because it is very high and the P-value 0.000000004647 of the global ANOVA test is extremely small (given underneath "Significance F").

Step 4

(d)

$$\alpha = \text{Significance level} = 0.05$$

The P-values of the coefficients of the variables are given in the column "P-value".

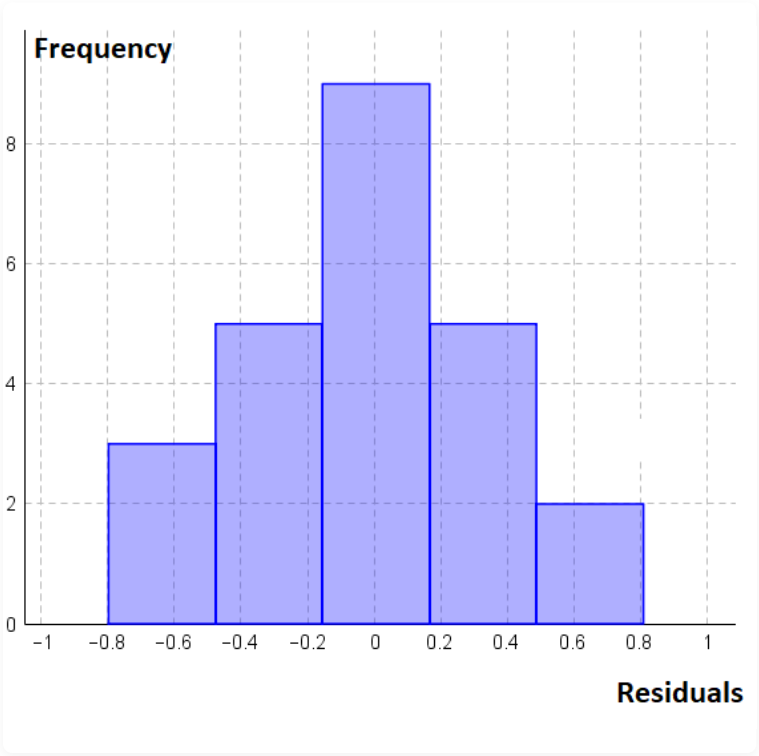
If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$x_1\ P = 0.0129 < 0.05 \Rightarrow \text{ Reject } H_0$$
$$x_2\ P = 0.0000 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that each coefficient is nonzero and thus we would not consider deleting either of the independent variables.

Step 5

(e)

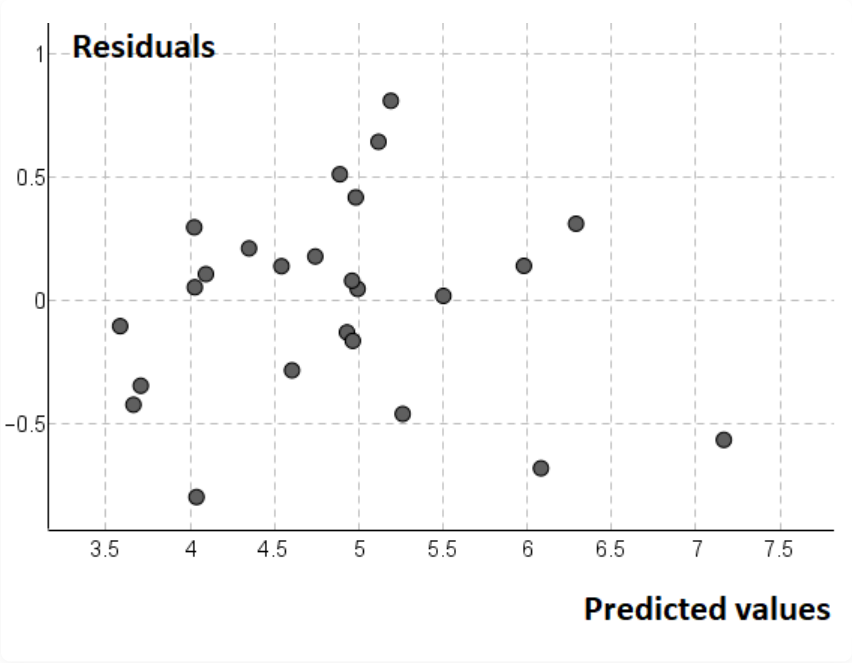


We note that the highest bar in the histogram is in the middle of the histogram, which implies that the histogram is approximately normal.

We can then conclude that there are no problems with the normality assumption in this case.

Step 6

(f)



We note that the points appear to be randomly scatter in the residual plot and there is no obvious pattern in the residual plot.

Moreover, the vertical spread in the points is approximately the same everywhere in the residual plot.

Thus we can then conclude that there are do not appear to be any problems with homoscedasticity.

Result

(a) No

(b) $\hat{y} = 2.4812 + 0.0061x_1 + 0.4245x_2$, $0.4245

(c) 82.52%, Yes

(d) No

(e) No

(f) No

Exercise 27

PROBLEM STATEMENT

Chapter 14, Page 563

Solution A

Step 1

Let

Y=P/E

X_1=EPS

X_2=Yield

(a) I will use Excel to determine the regression equation. The regression output is given in the image below.

The estimate of the constant a is given in the row "Constant" and in the column "Coef" of the computer output:

$$a = 26.6696$$

The estimate of the coefficient b_1 of the variable x_1 is given in the row "X Variable 1" and in the column "Coef" of the computer output:

$$b_1 = -5.2949$$

The estimate of the coefficient b_2 of the variable x_2 is given in the row "X Variable 2" and in the column "Coef" of the computer output:

$$b_2 = 1.3366$$

Combining these estimates, we then obtain the multiple regression equation:

$$\hat{y} = a + b_1 x_1 + b_2 x_2 = 26.6696 - 5.2949 x_1 + 1.3366 x_2$$

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.6195441903
R Square	0.3838350037
Adjusted R Squa	0.3113450042
Standard Error	9.532668267
Observations	20

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	962.3331272	481.1665636	5.295006291	0.01630890091
Residual	17	1544.819993	90.87176428		
Total	19	2507.15312			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	29.66963535	5.758932929	5.151932783	0.000079837365	17.51934905	41.81992164
X Variable 1	-5.294917669	1.631054593	-3.246315415	0.004749427954	-8.736142027	-1.853693311
X Variable 2	1.33660765	1.794056127	0.7450199745	0.4664355917	-2.44851988	5.121735179

Step 2

(b)

$$\alpha = \text{Significance level} = 0.05$$

The P-values of the variables are given in the column "P-value".

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$x_1 \; P = 0.0047 < 0.05 \Rightarrow \text{ Reject } H_0$$
$$x_2 \; P = 0.4664 > 0.05 \Rightarrow \text{ Fail to reject } H_0$$

There is sufficient evidence to support the claim that the coefficient of EPS is nonzero, while there is not sufficient evidence to support the claim that the coefficient of yield is nonzero.

This then implies that EPS is an effective predictor, while yield is not.

Step 3

(c) The constant represents the estimated value when all independent variables are zero, while the coefficients represent the average increase of the dependent variable per unit of the corresponding independent variable.

The estimated P/E ratio is 26.6696 when the earnings per share is $\$0$ and the dividend percentage is 0%.

The P/E ratio decreases on average by 5.2949 per $\$1$ in earnings per share.

The P/E ratio increases on average by 1.3366 per 1% of dividend percentage.

Step 4

(d) A stock is undervalued when the actual P/E ratio is much lower than the predicted P/E ratio.

The two smallest residuals of -17.8095 and -16.2673 corresponding with stock 2 and stock 11, which are thus the two stocks that look particularly undervalued.

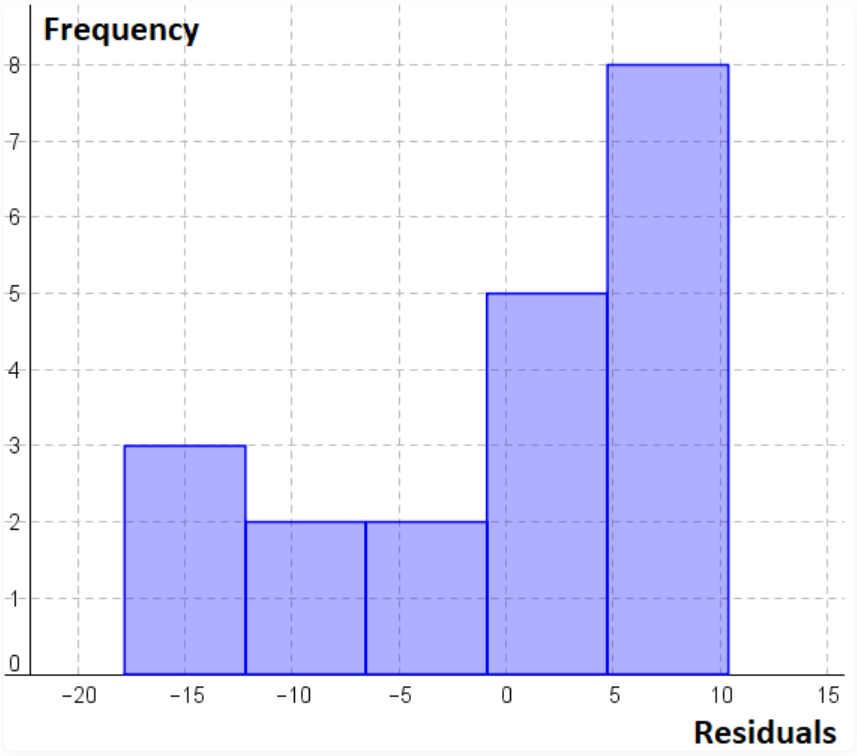
RESIDUAL OUTPUT

Observation	*Predicted Y*	*Residuals*
1	*18.54212075*	*2.247879253*
2	20.8395678	-17.8095678
3	36.71250012	7.747499881
4	33.74984001	7.970159985
5	25.72886701	-6.768867012
6	22.54473189	-4.124731886
7	31.47568292	3.34431708
8	20.55676511	9.873234893
9	24.83751549	5.132484508
10	23.60878486	-12.74878486
11	17.61734039	-16.26734039
12	20.09922773	5.330772269
13	22.77862458	-0.6386245817
14	36.14952495	-11.93952495
15	20.52490895	10.38509105
16	25.47697208	10.31302792
17	16.06072125	2.929278747
18	24.71871162	-4.50871162
19	30.77031707	2.109682929
20	7.767275427	7.422724573

Step 5

(e) The histogram of the residuals shows strong skewness, because the highest bar is far to the right in the histogram.

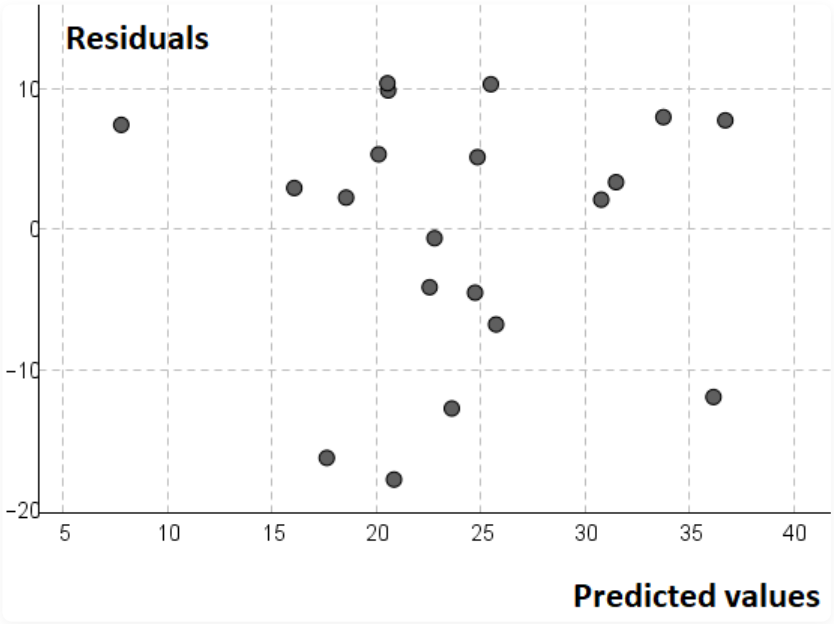
This then implies that the residuals are not approximately normally distributed and thus the normality assumption does not appear to be satisfied.



Step 6

(f) We note that the vertical spread in the points is roughly the same everywhere in the residual plot.

This then implies that the data values appear to have the same amount of variation everywhere in the residual plot and thus there does not appear to be a problem with homoscedasticity.



Step 7

(g) I will use Excel to determine the correlation matrix of the given data, which results in the following table:

	Y	X_1	X_2
Y	1		
X_1	-0.6031	1	
X_2	0.0432	0.1605	

We note that the correlation between the independent variables is 0.1605.

Since this correlation is small (between -0.7 and 0.7), none of the correlations indicate multicollinarity.

🎯 Result

(a) $\hat{y} = 26.6696 - 5.2949x_1 + 1.3366x_2$

(b) Yes

(c) The estimated P/E ratio is 26.6696 when the earnings per share is $\$0$ and the dividend percentage is 0%. The P/E ratio decreases on average by 5.2949 per $\$1$ in earnings per share. The P/E ratio increases on average by 1.3366 per 1% of dividend percentage.

(d) Yes, stock 2 and stock 11

(e) The normality assumption does not appear to be satisfied.

(f) No

(g) No

Exercise 28

PROBLEM STATEMENT

Chapter 14, Page 563

Solution A

Step 1

Let

Y =Amount of Tip

X_1 = Amount of Bill

X_2 = Number of Diners

(a) I will use Excel to determine the regression equation. The regression output is given in the image below.

The estimate of the constant a is given in the row "Constant" and in the column "Coef" of the computer output:

$$a = -0.5856$$

The estimate of the coefficient b_1 of the variable x_1 is given in the row "X Variable 1" and in the column "Coef" of the computer output:

$$b_1 = 0.0985$$

The estimate of the coefficient b_2 of the variable x_2 is given in the row "X Variable 2" and in the column "Coef" of the computer output:

$$b_2 = 0.5979$$

Combining these estimates, we then obtain the multiple regression equation:

$$\hat{y} = a + b_1 x_1 + b_2 x_2 = -0.5856 + 0.0985 x_1 + 0.5979 x_2$$

Since the coefficient of x_2 is 0.5979, an additional dinner adds $\$0.5979$ to the amount of the tips.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.9324883227
R Square	0.8695344719
Adjusted R Squa	0.8598703587
Standard Error	0.791025111
Observations	30

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	112.5992071	56.29960353	89.97560919	0
Residual	27	16.89445961	0.6257207263		
Total	29	129.4936667			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	-0.5855657728	0.3951816656	-1.481763512	0.1499809759	-1.396411565	0.225280019
X Variable 1	0.09850570708	0.02082080337	4.731119416	0.000062739408	0.05578494784	0.1412264663
X Variable 2	0.5979100404	0.2321437755	2.575602292	0.0158009184	0.121590363	1.074229718

Step 2

(b)

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$H_0 : \text{All coefficients are zero}$$
$$H_1 : \text{At least one of the coefficients is nonzero}$$

The P-value of the global test is given in the row "Regression" and in the column "Significance F" of the ANOVA table:

$$P = 0$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

Step 3

(c)

$$\alpha = \text{Significance level} = 0.05$$

The P-values of the coefficients of the variables are given in the column "P-value" of the regression output.

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$x_1 \; P = 0.0001 < 0.05 \Rightarrow \text{ Reject } H_0$$
$$x_2 \; P = 0.0158 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that all coefficients are nonzero and thus none of the variables should be deleted.

Step 4

(d) The value of R^2 is given to the right of "R Square" in the regression output:

$$R^2 \approx 0.8695 = 86.95\%$$

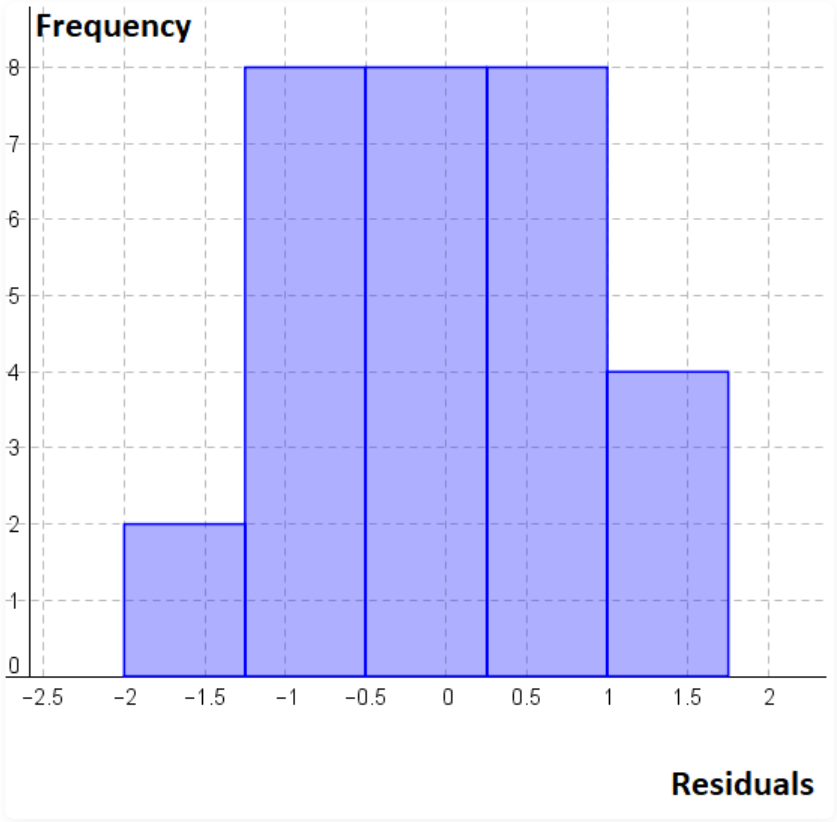
The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

86.95% of the variation in the amount of tip is explained by the variation in the independent variables.

Step 5

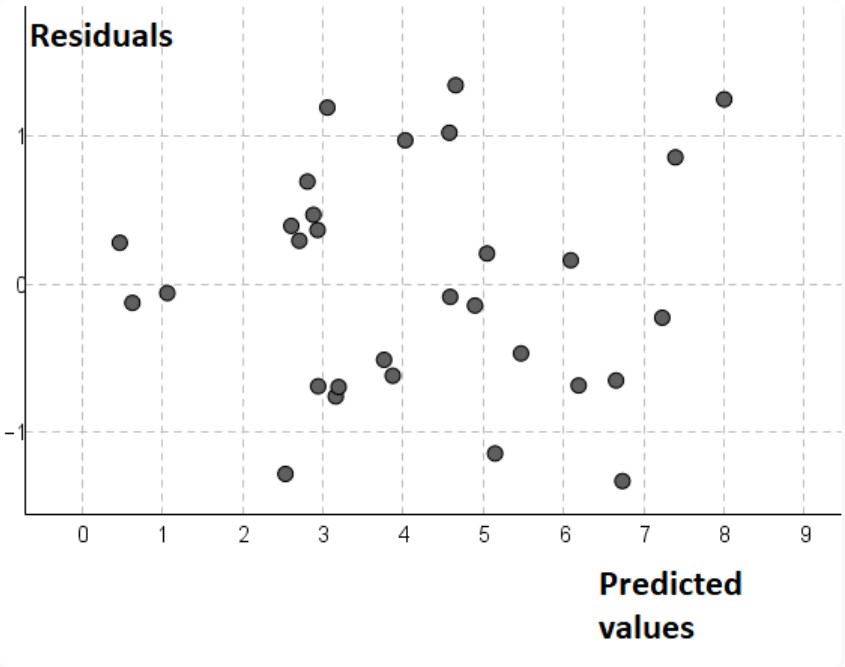
(e) The highest bars are roughly in the middle of the histogram and thus the histogram is approximately normally distributed.

It is then reasonable to assume that the residuals follow a normal distribution.



Step 6

(f) The points in the residual plot appear to be randomly scattered and there is no obvious pattern in the residual plot, thus it is reasonable to conclude that the residuals are random.



Result

(a) $\hat{y} = -0.5856 + 0.0985x_1 + 0.5979x_2, \$0.5979$

(b) There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

(c) No

(d) 86.95% of the variation in the amount of tip is explained by the variation in the independent variables.

(e) Yes

(f) Yes

PROBLEM STATEMENT

Chapter 14, Page 564

Solution A

Step 1

Let

Y=Sales

X=Informercials

(a) The estimate of the constant a is given in the row "Constant" and in the column "Coef" of the computer output:

$$a = 1.0188$$

The estimate of the coefficient b of the variable x is given in the row "X Variable 1" and in the column "Coef" of the computer output:

$$b = 0.0829$$

Combining these estimates, we then obtain the multiple regression equation:

$$\hat{y} = a + bx = 1.0188 + 0.0829x$$

All P-values in the column "P-value" are nonzero and thus none of the coefficients are approximately zero. This then implies that the sales are predictable fro the number of commercials.

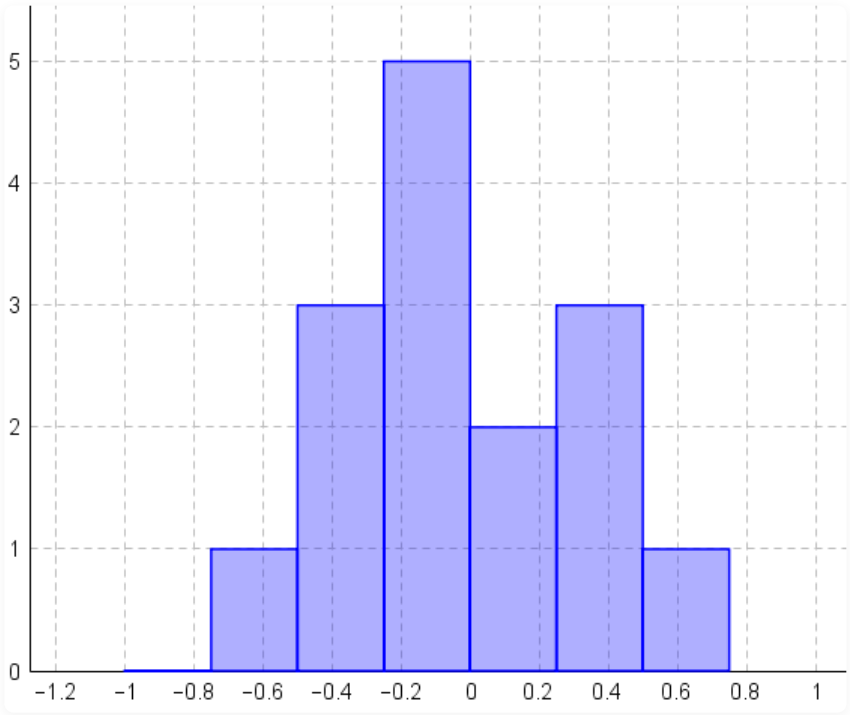
Moreover, $R = 0.6520 = 65.20\%$ is not very small and thus the predictions will be reasonably accurate.

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.8075065786					
R Square	0.6520668746					
Adjusted R Squa	0.625302788					
Standard Error	0.308674826					
Observations	15					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	1	2.321358073	2.321358073	24.36350192	0.000272142709	
Residual	13	1.238641927	0.0952801482			
Total	14	3.56				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	1.018752467	0.3104973659	3.281034171	0.005961341455	0.3479636915	1.689541243
X Variable 1	0.08290564548	0.0167963242	4.935939821	0.000272142710	0.0466193932	0.1191918978

Step 2

(b) We note that the highest bars are approximate in the middle of the histogram, which indicates that the histogram is approximately normally distribution.

This then implies that the normality assumption seems reasonable.



Result

(a) $\hat{y} = 1.0188 + 0.0829x$, Yes

(b) Yes

Exercise 30

PROBLEM STATEMENT

Chapter 14, Page 564

Solution A

Step 1

Let

$X=$4th of July

Y =Fall festival

I will use Excel to determine the regression equation. The regression output is given in the image below.

The estimate of the constant a is given in the row "Constant" and in the column "Coef" of the computer output:

$$a = 1.7208$$

The estimate of the coefficient b of the variable x is given in the row "X Variable 1" and in the column "Coefficients" of the computer output:

$$b = 0.7618$$

Combining these estimates, we then obtain the multiple regression equation:

$$\hat{y} = a + bx = 1.7208 + 0.7618x$$

The P-value of the coefficient of x is given in the row "X Varaible 1" and in the column "P-value", which is thus approximately 0.00002.

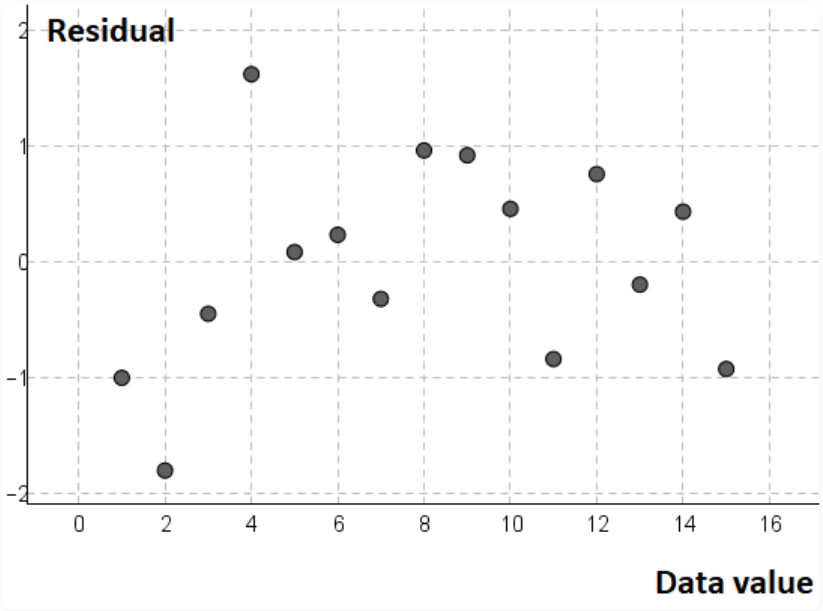
Since the P-value is small, there is sufficient evidence to support the claim that the coefficient of x is nonzero and thus the amount spent on fireworks is related to attendance at the fall festival.

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.8719904567					
R Square	0.7603673566					
Adjusted R Squa	0.7419340764					
Standard Error	0.9429462904					
Observations	15					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	1	36.67707982	36.67707982	41.24970412	0.000022616174	
Residual	13	11.55892018	0.8891477065			
Total	14	48.236				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	1.720779796	1.122725393	1.532680927	0.1493230254	-0.7047209475	4.14628054
X Variable 1	0.7618203684	0.1186156959	6.422593255	0.000022616174	0.5055667373	1.018073999

Step 2

Residual plot

The number of the data value (1st data value is numbered 1, 2nd data value is numbered 2, etc.) is on the horizontal axis and the residuals are on the vertical axis.



Step 3

Since the pattern in the residual plot slopes upwards, there is a pattern over time in the residuals and thus there is a problem with autocorrelation.

🎯 Result

$$\hat{y} = 1.7208 + 0.7618x$$

Yes

There is a problem with autocorrelation.

Exercise 31

PROBLEM STATEMENT

Chapter 14, Page 564

Solution A

Step 1

Let

Y=Auction price

X_1=Loan

X_2=Monthly payments

Step 2

X_3=Payments made

(a) I will use Excel to determine the multiple regression equation. The regression output is given in the image below.

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

The estimate of the constant a is given in the row "Intercept" and in the column "Coefficients" of the computer output:

$$H_0 : \text{All coefficients are zero}$$
$$a \approx -19064.129$$
$$H_1 : \text{At least one of the coefficients is nonzero}$$

The estimate of the coefficient b_1 of the variable x_1 is given in the row "X Variable 1" and in the column "Coefficients" of the computer output:

The P-value of the global test is given in the row "Regression" and in the column "Significance F" of the ANOVA table:

$$b_1 \approx -0.9751$$
$$P = 0.1074$$

The estimate of the coefficient b_2 of the variable x_2 is given in the row "X Variable 2" and in the column "Coefficients" of the computer output:

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.1074 > 0.05 \Rightarrow \text{Fail to reject } H_0$$
$$b_2 \approx 163.0672$$

There is not sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

The estimate of the coefficient b_3 of the variable x_3 is given in the row "X Variable 3" and in the column "Coefficients" of the computer output:

$$b_3 \approx 22.1351$$

Step 3

Combining these estimates, we then obtain the multiple regression equation:

(b) The P-values of the coefficients b_1, b_2, b_3 are given in hte column "P-value" of the regression output.

$$\hat{y} = a + b_1 x_1 + b_2 x_2 + b_3 x_3$$
$$= -19064.129 - 0.9751x_1 + 163.0672x_2 + 1179.7207x_3$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$X_1 \quad P = 0.3416 > 0.05 \Rightarrow \text{Fail to reject } H_0$$
$$X_2 \quad P = 0.0647 > 0.05 \Rightarrow \text{Fail to reject } H_0$$
$$X_3 \quad P = 0.1314 > 0.05 \Rightarrow \text{Fail to reject } H_0$$

There is not sufficient evidence to support the claim that each coefficient is different from 0 and thus we would remove some of the variables (we will start with the variable with the largest P-value, which is x_1).

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.5558706365
R Square	0.3089921645
Adjusted R Squa	0.1794281954
Standard Error	38861.68659
Observations	20

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	3	10805074543	3601691514	2.38486183	0.1074001515
Residual	16	24163690957	1510230685		
Total	19	34968765500			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	-19064.129	108525.0088	-0.1756657679	0.8627605149	-249126.8687	210998.6107
X Variable 1	-0.9751078298	0.9948527142	-0.9801529572	0.3415997932	-3.084101357	1.133885697

X Variable 2	163.0672215	82.20875887	1.983574788	0.06473533238	-11.20756094	337.3420039
X Variable 3	1179.720742	741.9591235	1.590007731	0.1313952785	-393.162325	2752.60381

Step 4

(c) I will use Excel to determine the multiple regression equation. We remove the variable "Loan" from the model. The regression output is given in the image below.

The estimate of the constant a is given in the row "Intercept" and in the column "Coefficients" of the computer output:

$$a \approx -97389.0388$$

The estimate of the coefficient b_1 of the variable x_2 is given in the row "X Variable 1" and in the column "Coefficients" of the computer output:

$$b_1 \approx 149.3238$$

The estimate of the coefficient b_2 of the variable x_3 is given in the row "X Variable 2" and in the column "Coefficients" of the computer output:

$$b_2 \approx 821.3160$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned}\hat{y} &= a + b_1 x_2 + b_2 x_3 \\ &= -97389.0388 + 149.3238 x_2 + 821.3160 x_3\end{aligned}$$

We note that the P-values of the variable are still high (greater than 0.05) and thus we will again remove the variable with the highest P-value, which is now variable x_3 (which has a P-value of about 0.2199).

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.5172054433
R Square	0.2675014706
Adjusted R Squa	0.181325173
Standard Error	38816.74086
Observations	20

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	9354196196	4677098098	3.10411886	0.07093448753
Residual	17	25614569304	1506739371		
Total	19	34968765500			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	-97389.03876	73344.80142	-1.327824698	0.2017905022	-252133.0419	57354.96441
X Variable 1	149.3237801	80.91045654	1.84554366	0.08244913272	-21.38235999	320.0299201
X Variable 2	821.3159625	644.8496265	1.273655018	0.219925142	-539.1978124	2181.829737

Step 5

I will use Excel to determine the multiple regression equation. We remove the variable "Loan" and "Payments made" from the model. The regression output is given in the image below.

The estimate of the constant a is given in the row "Intercept" and in the column "Coefficients" of the computer output:

$$a \approx -96370.0830$$

The estimate of the coefficient b_1 of the variable x_2 is given in the row "X Variable 1" and in the column "Coefficients" of the computer output:

$$b_1 \approx 169.8113$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned} \hat{y} &= a + b_1 x_2 \\ &= -96370.0830 + 169.8113 x_2 \end{aligned}$$

We note that the P-value of the variable is now less than 0.05 (0.0496) and thus we have found an appropriate regression model.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.4445267151
R Square	0.1976040004
Adjusted R Squa	0.1530264449
Standard Error	39481.921
Observations	20

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	6909967952	6909967952	4.432813734	0.04956062743
Residual	18	28058797548	1558822086		
Total	19	34968765500			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	-96370.08302	74597.23093	-1.291872122	0.2127430782	-253093.0477	60352.88168
X Variable 1	169.8113495	80.65419707	2.105424835	0.04956062986	0.3631713117	339.2595277

🎯 **Result**

(a) There is not sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

(b) Yes

(c) $\hat{y} = -96370.0830 + 169.8113x_2$

Exercise 32

PROBLEM STATEMENT

Chapter 14, Page 565

Solution A

Step 1

Let

Y=Auction price

X_1=Loan

X_2=Monthly payments

X_3=Payments made

X_4=X_1X_3

$$\alpha = \text{Significance level} = 0.05$$

I will use Excel to determine the multiple regression equation. The regression output is given in the image below.

Step 2

The estimate of the constant a is given in the row "Intercept" and in the column "Coefficients" of the computer output:

$$a \approx -339580.5233$$

The estimate of the coefficient b_1 of the variable x_1 is given in the row "X Variable 1" and in the column "Coefficients" of the computer output:

$$b_1 \approx 1.7715$$

The estimate of the coefficient b_2 of the variable x_2 is given in the row "X Variable 2" and in the column "Coefficients" of the computer output:

$$b_2 \approx 213.4172$$

The estimate of the coefficient b_3 of the variable x_3 is given in the row "X Variable 3" and in the column "Coefficients" of the computer output:

$$b_3 \approx 13234.9706$$

The estimate of the coefficient b_4 of the variable x_4 is given in the row "X Variable 4" and in the column "Coefficients" of the computer output:

$$b_4 \approx -0.1171$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned}\hat{y} &= a + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_1x_3 \\ &= -339580.5233 + 1.7715x_1 + 213.4172x_2 + 13234.9706x_3 - 0.1171x_4\end{aligned}$$

The P-value corresponding to the interaction term is given in the row "X Variable 3" and in the column "P-value" of the regression output:

$$P \approx 0.0225$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.0225 < 0.05 \Rightarrow \text{Reject } H_0$$

There is sufficient evidence to support the claim that the interaction is significant.

Coefficients

There is sufficient evidence to support the claim that the interaction is significant.

SUMMARY OUTPUT	
Regression Statistics	
Multiple R	0.7191374287
R Square	0.5171586413
Adjusted R Squa	0.3884009457
Standard Error	33550.32662
Observations	20

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	4	18084399254	4521099814	4.01652607	0.02074133382
Residual	15	16884366246	1125624416		
Total	19	34968765500			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	-339580.5233	157047.3269	-2.16228146	0.04716316612	-674318.9754	-4842.07122
X Variable 1	1.771513853	1.379935101	1.283766063	0.2187009985	-1.169748178	4.712775884
X Variable 2	213.4172236	73.68297907	2.896425013	0.0110746706	56.36567209	370.4687751
X Variable 3	13234.97061	4783.61652	2.766729012	0.01439333352	3038.933397	23431.00782
X Variable 4	-0.1170758713	0.04603822859	-2.54301425	0.02250743945	-0.2152040323	-0.01894771038

Exercise 33

PROBLEM STATEMENT

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Solution A

Step 1

Let

Y=Selling price

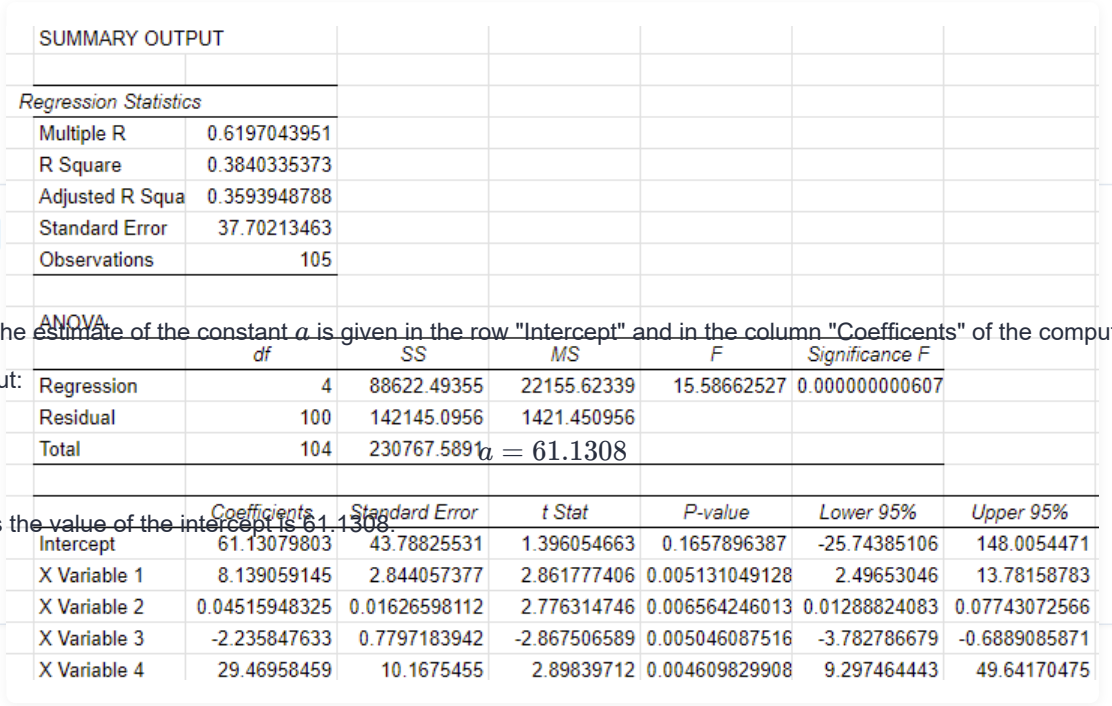
X_1=Number of bedrooms

X_2=Size of house

X_3=Center of the city

X_4=Number of bathrooms

(a) I will use Excel to determine the regression output, which is given in the image below.



SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.6197043951
R Square	0.3840335373
Adjusted R Squa	0.3593948788
Standard Error	37.70213463
Observations	105

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	4	88622.49355	22155.62339	15.58662527	0.000000000607
Residual	100	142145.0956	1421.450956		
Total	104	230767.5891			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	61.13079803	43.78825531	1.396054663	0.1657896387	-25.74385106	148.0054471
X Variable 1	8.139059145	2.844057377	2.861777406	0.005131049128	2.49653046	13.78158783
X Variable 2	0.04515948325	0.01626598112	2.776314746	0.006564246013	0.01288824083	0.07743072566
X Variable 3	-2.235847633	0.7797183942	-2.867506589	0.005046087516	-3.782786679	-0.6889085871
X Variable 4	29.46958459	10.1675455	2.89839712	0.004609829908	9.297464443	49.64170475

Step 2

(b) The estimate of the constant a is given in the row "Intercept" and in the column "Coefficents" of the computer output:

$$a = 61.1308$$

Thus the value of the intercept is 61.1308.

The estimate of the constant a is given in the row "Intercept" and in the column "Coefficents" of the computer output:

$$a = 61.1308$$

The estimate of the coefficient b_1 of the variable x_1 is given in the row "X Variable 1" and in the column "Coefficents" of the computer output:

$$b_1 = 8.1391$$

The estimate of the coefficient b_2 of the variable x_2 is given in the row "X variable 2" and in the column "Coefficents" of the computer output:

$$b_2 = 0.0452$$

The estimate of the coefficient b_3 of the variable x_3 is given in the row "X variable 3" and in the column "Coefficents" of the computer output:

$$b_3 = -2.2358$$

The estimate of the coefficient b_4 of the variable x_4 is given in the row "X variable 4" and in the column "Coefficents" of the computer output:

$$b_4 = 29.4696$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned}\hat{y} &= a + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4 \\ &= 61.1308 + 8.1391x_1 - 0.0452x_2 - 2.2358x_3 + 29.4696x_4\end{aligned}$$

We are not surprised that the coefficient for distance from the center of the city is negative, because the further a house is from the center of the city, the less reachable the house is and thus the cheaper we expect the house to be.

Step 3

(c) I will use Excel to determine the correlation matrix of the given data, which results in the following table:

	Y	X_1	X_2	X_3	X_4
Y	1				
X_1	0.4674	1			
X_2	0.3710	0.3835	1		
X_3	-0.3470	-0.1534	-0.1172	1	
X_4	0.3822	0.3289	0.0244	-0.1950	1

We note that the largest correlation of 0.4674 between the dependent variable and an independent variable belongs to the variable X_1 (number of bedrooms). However, all correlations with the dependent variable are weak as they are all less than 0.5.

Moreover, we note that none of the correlation between two independent variables (X_i) are large (less than -0.7 or larger than 0.7) and thus it is safe to assume that there are no problems with multicollinearity.

Step 4

(d)

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$H_0 : \text{All coefficients are zero}$$
$$H_1 : \text{At least one of the coefficients is nonzero}$$

The P-value of the global test is given in the row "Regression" and in the column "Significance F" of the ANOVA table:

$$P \approx 0.0000$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.0000 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

Step 5

(e)

$$\alpha = \text{Significance level} = 0.05$$

The P-values of the coefficients of the variables are given in the column "P-value" of the regression output.

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$\begin{array}{ll} x_1 & P = 0.0051 < 0.05 \Rightarrow \text{ Reject } H_0 \\ x_2 & P = 0.0066 < 0.05 \Rightarrow \text{ Reject } H_0 \\ x_3 & P = 0.0050 < 0.05 \Rightarrow \text{ Reject } H_0 \\ x_4 & P = 0.0046 < 0.05 \Rightarrow \text{ Reject } H_0 \end{array}$$

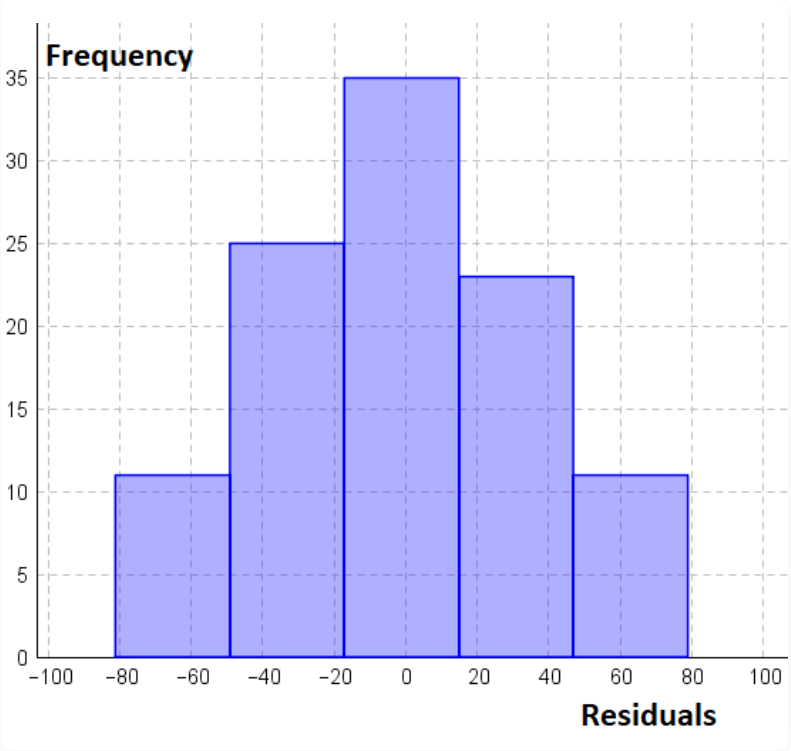
There is sufficient evidence to support the claim that each coefficient is nonzero and thus we do not consider deleting any of the variables.

Step 6

(f) We concluded that we did not want to delete any of the variables and thus we do not need to rerun the analysis.

Step 7

(g)

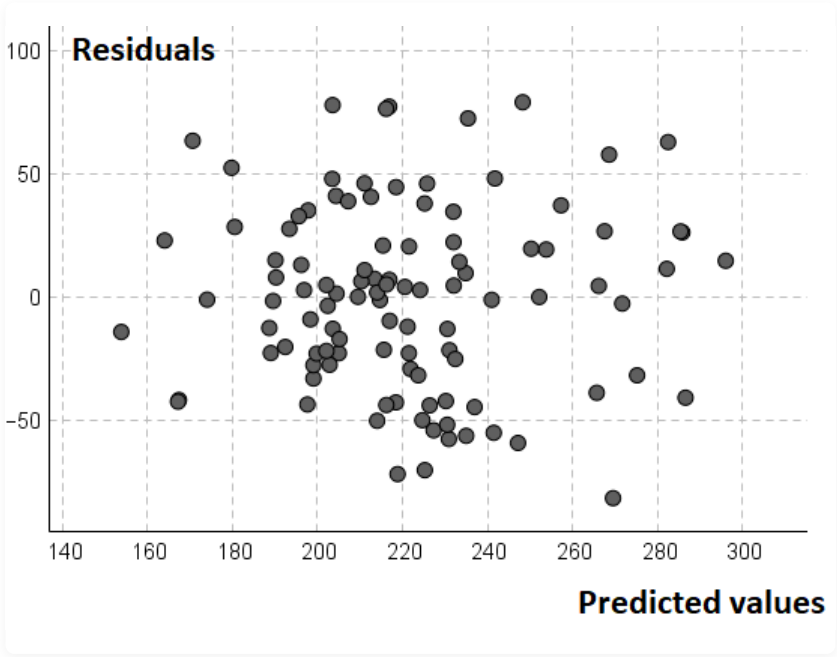


The highest bars in the histogram are roughly in the middle of the histogram, which implies that the histogram is approximately normally distributed.

This then implies that is it reasonable to conclude that the normality assumption has been met.

Step 8

(h)



The residuals appear to be randomly scattered in the residual plot and there is no noticeable pattern in the points.

Moreover, the vertical spread appears to be approximately the same everywhere in the residual plot.

This then indicates that none of the regression assumptions appear to be violated.

Result

(a) $\hat{y} = 8.1391 + 8.1391x_1 - 0.0452x_2 - 2.2358x_3 + 29.4696x_4$

(b) 61.1308

(c) All are weak, No

(d) There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

(e) No

(f) Does not need to be executed.

(g) Yes

(h) None of the regression assumptions appear to be violated.

Exercise 34

PROBLEM STATEMENT

Chapter 14, Page 565

Solution A

Step 1

Let

Y=Number of games won

X_1=Team batting average

X_2=Number of stolen bases

X_3=Number of errors committed

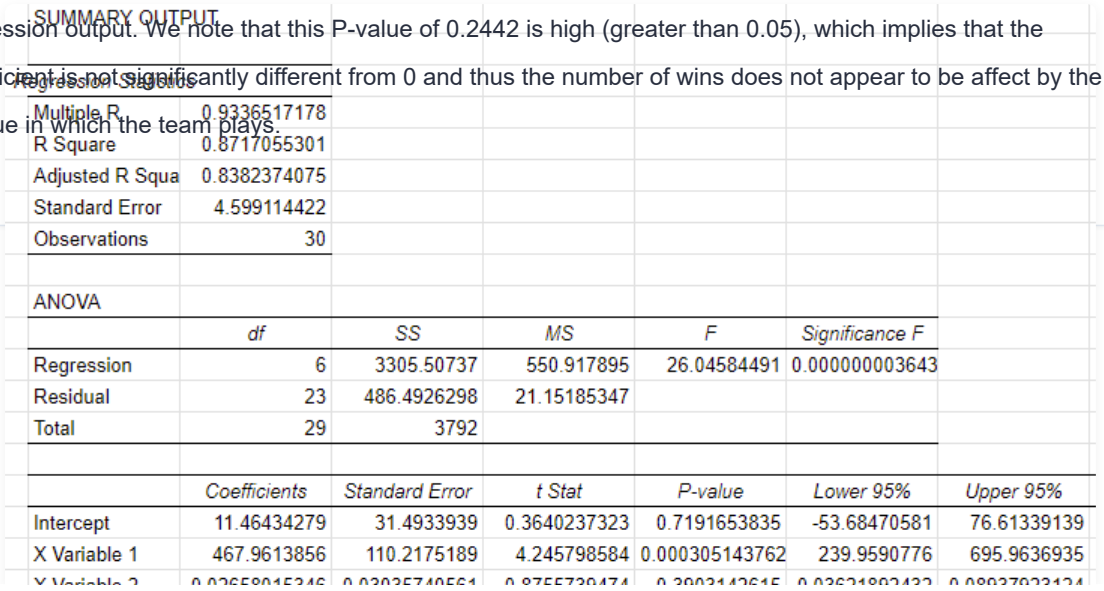
Step 2

X_4=Team ERA

X_5=Number of home runs

X_6=League

(a) I will use Excel to determine the regression output, which is given in the image below.



SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.9336517178					
R Square	0.8717055301					
Adjusted R Squa	0.8382374075					
Standard Error	4.599114422					
Observations	30					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	6	3305.50737	550.917895	26.04584491	0.000000003643	
Residual	23	486.4926298	21.15185347			
Total	29	3792				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	11.46434279	31.4933939	0.3640237323	0.7191653835	-53.68470581	76.61339139
X Variable 1	467.9613856	110.2175189	4.245798584	0.000305143762	239.9590776	695.9636935
X Variable 2	0.026580015346	0.03035740561	0.8755739474	0.3902142615	-0.03621892432	0.08937923124

The estimate of the coefficient b_4 of the variable x_4 is given in the row "X variable 4" and in the column "Coefficients" of the computer output:

$$b_4 = -16.7134$$

The estimate of the coefficient b_5 of the variable x_5 is given in the row "X variable 5" and in the column "Coefficents" of the computer output:

The estimate of the constant a is given in the row "Intercept" and in the column "Coefficents" of the computer output:

$$b_5 = 0.1583$$
$$a = 11.4643$$

The estimate of the coefficient b_6 of the variable x_6 is given in the row "X variable 6" and in the column "Coefficients" of the computer output:

The estimate of the coefficient b_1 of the variable x_1 is given in the row "X Variable 1" and in the column "Coefficients" of the computer output:

$$b_6 = -2.5693$$
$$b_1 = 467.9614$$

Combining these estimates, we then obtain the multiple regression equation:

The estimate of the coefficient b_2 of the variable x_2 is given in the row "X variable 2" and in the column "Coefficients" of the computer output:

$$\begin{aligned} \hat{y} &= a + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4 + b_5x_5 + b_6x_6 \\ &= 11.4643 + 467.9614x_1 + 0.0266x_2 - 0.0963x_3 \\ &\quad - 16.7134x_4 + 0.1583x_5 - 2.5693x_6 \end{aligned}$$

$$b_2 = 0.0266$$

The estimate of the coefficient b_3 of the variable x_3 is given in the row "X variable 3" and in the column "Coefficients" of the computer output:

We are not surprised that the coefficient for ERA is negative, because the lower the team earned run average, the better the team performed and thus the more wins we would expect.

$$b_3 = -0.0963$$

The P-value corresponding to the league (x_6) is given in the row "X Variable 6" and in the column "P-value" of the regression output. We note that this P-value of 0.2442 is high (greater than 0.05), which implies that the coefficient is not significantly different from 0 and thus the number of wins does not appear to be affect by the league in which the team plays.

	X Variable 2	0.026586015346	0.030357405361	0.875573947	0.3903142615	-0.0362692432	0.0693792312
	X Variable 3	-0.09626811934	0.05829659499	-1.651350638	0.1122566193	-0.2168638136	0.02432757487
	X Variable 4	-16.71342704	2.135526654	-7.826372479	0.000000062368	-21.13110048	-12.29575359
	X Variable 5	0.1583171041	0.02924549146	5.413384976	0.000016828231	0.09781819593	0.2188160122
	X Variable 6	-2.56934335	2.149846069	-1.195128985	0.2442148927	-7.016638756	1.877952056

Step 3

(b) The value of R^2 is given to the right of "R Square" in the regression output:

$$R^2 \approx 0.8717 = 87.17\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

87.17% of the variation in the number of wins is explained by the variation in the independent variables.

Step 4

(c) I will use Excel to determine the correlation matrix of the given data, which results in the following table:

	Y	X_1	X_2	X_3	X_4	X_5	X_6
Y	1						
X_1	0.4672	1					
X_2	0.2669	0.2667	1				
X_3	-0.4080	-0.2654	-0.0925	1			
X_4	-0.6348	0.0634	0.0097	0.1808	1		
X_5	0.5135	0.2903	0.2839	-0.1374	0.0658	1	
X_6	0.0773	0.4315	0.3431	-0.0427	0.3063	0.4255	1

We note that the largest correlation in absolute value of -0.6348 between the dependent variable and an independent variable belongs to the variable X_4 (team ERA). This correlation as well as the correlation with the number of homeruns appears stront ($r > 0.5$), while all other correlations with the dependent variable are weak.

Moreover, we note that none of the correlation between two independent variables (X_i) are large (less than -0.7 or larger than 0.7) and thus it is safe to assume that there are no problems with multicollinearity.

Step 5

(d)

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$H_0 : \text{All coefficients are zero}$$
$$H_1 : \text{At least one of the coefficients is nonzero}$$

The P-value of the global test is given in the row "Regression" and in the column "Significance F" of the ANOVA table:

$$P \approx 0.0000$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.0000 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

Step 6

(e)

$$\alpha = \text{Significance level} = 0.05$$

The P-values of the coefficients of the variables are given in the column "P-value" of the regression output.

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$\begin{aligned}
x_1 \quad & P = 0.0003 < 0.05 \Rightarrow \text{ Reject } H_0 \\
x_2 \quad & P = 0.3903 > 0.05 \Rightarrow \text{ Fail to reject } H_0 \\
x_3 \quad & P = 0.1123 > 0.05 \Rightarrow \text{ Fail to reject } H_0 \\
x_4 \quad & P = 0.0000 < 0.05 \Rightarrow \text{ Reject } H_0 \\
x_5 \quad & P = 0.0000 < 0.05 \Rightarrow \text{ Reject } H_0 \\
x_6 \quad & P = 0.2442 > 0.05 \Rightarrow \text{ Fail to reject } H_0
\end{aligned}$$

We note that the coefficients of x_2, x_3, and x_6 are not significantly different from zero and thus we would consider deleting these variables from the model.

Step 7

(f) I will use Excel to determine the regression output, which is given in the image below. We start by deleting the variable x_2 from the model, because this variable had the largest P-value.

The estimate of the constant a is given in the row "Intercept" and in the column "Coefficients" of the computer output:

$$a = 11.3361$$

The estimate of the coefficient b_1 of the variable x_1 is given in the row "X Variable 1" and in the column "Coefficients" of the computer output:

$$b_1 = 478.3458$$

The estimate of the coefficient b_3 of the variable x_3 is given in the row "X variable 2" and in the column "Coefficients" of the computer output:

$$b_3 = -0.0971$$

The estimate of the coefficient b_4 of the variable x_4 is given in the row "X variable 3" and in the column "Coefficients" of the computer output:

$$b_4 = -16.8702$$

The estimate of the coefficient b_5 of the variable x_5 is given in the row "X variable 4" and in the column "Coefficients" of the computer output:

$$b_5 = 0.1619$$

The estimate of the coefficient b_6 of the variable x_6 is given in the row "X variable 5" and in the column "Coefficients" of the computer output:

$$b_6 = -2.1565$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned}\hat{y} &= a + b_1x_1 + b_3x_3 + b_4x_4 + b_5x_5 + b_6x_6 \\ &= 11.3361 + 478.3458x_1 - 0.0971x_3 \\ &- 16.8702x_4 + 0.1619x_5 - 2.1565x_6\end{aligned}$$

We note that the P-values of x_3 (0.1071) and x_6 (0.3118) are still large and thus we next delete variable x_6 from the model.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.9313588211
R Square	0.8674292536
Adjusted R Squa	0.8398103481
Standard Error	4.576699458
Observations	30

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	5	3289.29173	657.858346	31.40708287	0.000000000878

Residual	24	502.7082702	20.94617793			
Total	29	3792				

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	11.33608655	31.33956376	0.3617180709	0.7207279572	-53.34559357	76.01776666
X Variable 1	478.3457886	109.0434949	4.386743006	0.000197560412	253.2910779	703.4004994
X Variable 2	-0.09711396477	0.0580045058	-1.674248637	0.107066227	-0.21682938	0.02260145049
X Variable 3	-16.87017862	2.117637847	-7.966507889	0.000000033990	-21.24076829	-12.49958894
X Variable 4	0.1618849073	0.02881907996	5.617282285	0.000008793493	0.1024052501	0.2213645646
X Variable 5	-2.156546829	2.087291302	-1.033179618	0.3118193229	-6.464504316	2.151410657

Step 8

I will use Excel to determine the regression output, which is given in the image below. We start by deleting the variables x_2 and x_6 from the model, because this variable had the largest P-value.

The estimate of the constant a is given in the row "Intercept" and in the column "Coefficients" of the computer output:

$$a = 25.8274$$

The estimate of the coefficient b_1 of the variable x_1 is given in the row "X Variable 1" and in the column "Coefficients" of the computer output:

$$b_1 = 437.8005$$

The estimate of the coefficient b_3 of the variable x_3 is given in the row "X variable 2" and in the column "Coefficients" of the computer output:

$$b_3 = -0.1002$$

The estimate of the coefficient b_4 of the variable x_4 is given in the row "X variable 3" and in the column "Coefficients" of the computer output:

$$b_4 = -17.5207$$

The estimate of the coefficient b_5 of the variable x_5 is given in the row "X variable 4" and in the column "Coefficients" of the computer output:

$$b_5 = 0.1514$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned}\hat{y} &= a + b_1x_1 + b_3x_3 + b_4x_4 + b_5x_5 \\ &= 25.8274 + 437.8005x_1 - 0.1002x_3 - 17.5207x_4 + 0.1514x_5\end{aligned}$$

We note that the P-value of x_3 (0.0966) is still large and thus we next delete variable x_3 from the model as well.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.9281879322
R Square	0.8615328375
Adjusted R Squa	0.8393780915
Standard Error	4.582870193
Observations	30

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	4	3266.93252	816.73313	38.88705552	0.000000000217
Residual	25	525.06748	21.0026992		
Total	29	3792			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	25.82735734	28.06353354	0.9203173684	0.3662026988	-31.97057144	83.62528612
X Variable 1	437.8005492	101.874371	4.29745524	0.000230106917	227.9863565	647.614742
X Variable 2	-0.1001568696	0.05800779267	-1.726610598	0.09657662476	-0.2196261539	0.01931241481
X Variable 3	-17.52073527	2.024584745	-8.653989571	0.000000005446	-21.69044557	-13.35102497
X Variable 4	0.1513824941	0.02700317838	5.606099104	0.000007838385	0.09576840764	0.2069965806

Step 9

I will use Excel to determine the regression output, which is given in the image below. We start by deleting the variables x_2, x_3 and x_6 from the model, because this variable had the largest P-value.

The estimate of the constant a is given in the row "Intercept" and in the column "Coefficients" of the computer output:

$$a = 7.2235$$

The estimate of the coefficient b_1 of the variable x_1 is given in the row "X Variable 1" and in the column "Coefficients" of the computer output:

$$b_1 = 482.0952$$

The estimate of the coefficient b_4 of the variable x_4 is given in the row "X variable 2" and in the column "Coefficients" of the computer output:

$$b_4 = -18.2526$$

The estimate of the coefficient b_5 of the variable x_5 is given in the row "X variable 3" and in the column "Coefficients" of the computer output:

$$b_5 = 0.1550$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned}\hat{y} &= a + b_1 x_1 + b_4 x_4 + b_5 x_5 \\ &= 7.2235 + 482.0952 x_1 - 18.2526 x_4 + 0.1550 x_5\end{aligned}$$

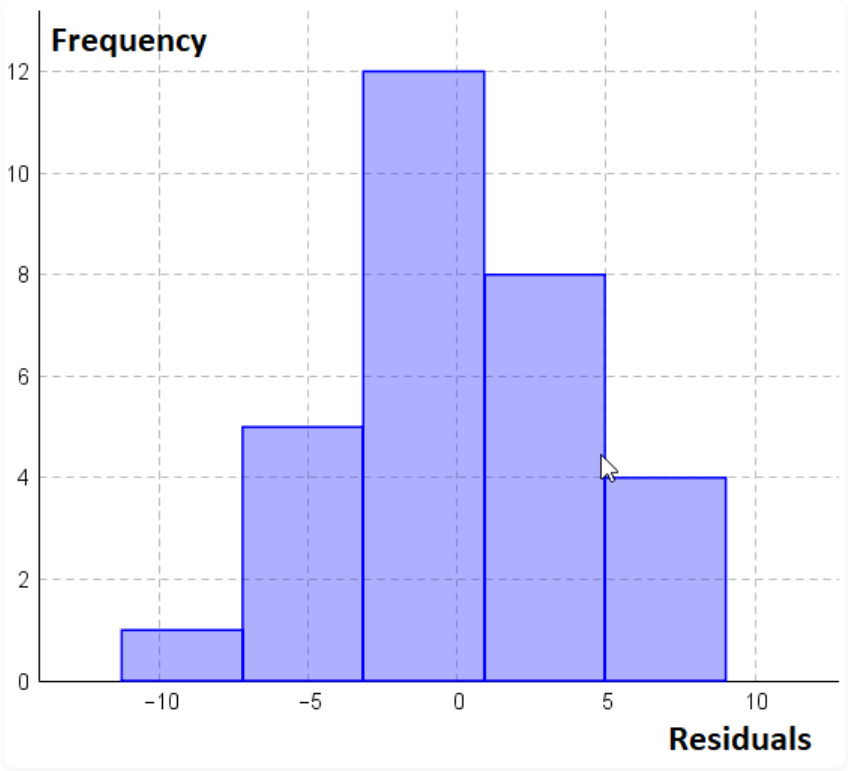
We note that the P-values of all coefficients are now small (less than 0.05) and thus we finally found an appropriate regression model.

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.9192502341					
R Square	0.8450209928					
Adjusted R Square	0.8271387997					
Standard Error	4.754270935					
Observations	30					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	3	3204.319605	1068.106535	47.2548857	0.000000000115	
Residual	26	587.6803953	22.60309213			
Total	29	3792				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	7.22354436	26.88181216	0.2687149333	0.7902698273	-48.03281135	62.47990007
X Variable 1	482.0952026	102.2786963	4.713544656	0.000071592503	271.8583334	692.3320717
X Variable 2	-18.25259064	2.053758943	-8.88740653	0.000000002328	-22.47415257	-14.03102872
X Variable 3	0.1549890334	0.02792917009	5.549360503	0.000007933918	0.09757980262	0.2123982641

Step 10

(g) The highest bars in the histogram are roughly in the middle of the histogram, which implies that the histogram is approximately normally distributed.

This then implies that is it reasonable to conclude that the normality assumption has been met.

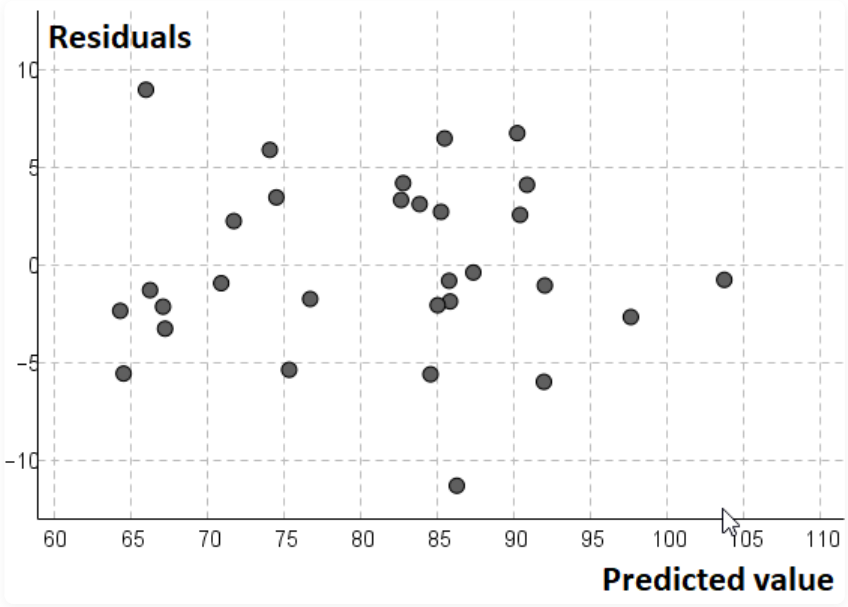


Step 11

(h) The residuals appear to be randomly scattered in the residual plot and there is no noticeable pattern in the points.

Moreover, the vertical spread appears to be approximately the same everywhere in the residual plot.

This then indicates that none of the regression assumptions appear to be violated.



Result

(a) $\hat{y} = 11.4643 + 467.9614x_1 + 0.0266x_2 - 0.0963x_3 - 16.7134x_4 + 0.1583x_5 - 2.5693x_6$, No, The number of wins does not appear to be affect by the league in which the team plays.

(b) 87.17%

(c) Two are strong while others are weak, No

(d) There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

(e) Yes

(f) $\hat{y} = 7.2235 + 482.0952x_1 - 18.2526x_4 + 0.1550x_5$

(g) Yes

(h) None of the regression assumptions appear to be violated.

Exercise 35

PROBLEM STATEMENT

Chapter 14, Page 565

Solution A

Step 1

Let

Y=Maintenance cost

X_1=Age

X_2=Miles

X_3=Bus type

Step 2

(a) I will use Excel to determine the regression output, which is given in the image below.

(b) The value of R^2 is given to the right of "R Square" in the regression output:

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.4863987633
R Square	0.236583757
Adjusted R Square	0.2064489053
Standard Error	47.82424884
Observations	80

$$R^2 \approx 0.2366 = 23.66\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

23.66% of the variation in the maintenance cost is explained by the variation in the independent variables.

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	3	53868.32047	17956.10682	7.850835283	0.000124594547
Residual	76	173824.067	2287.158777		
Total	79	227692.3875			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	358.723936	32.20368356	11.13922062	0	294.5847425	422.8631295
X Variable 1	8.377872137	2.024782267	4.137665701	0.000089769715	4.345168819	12.41057545
X Variable 2	0.04597432725	0.0403737946	1.138717024	0.2583979013	-0.03443705234	0.1263857068
X Variable 3	-9.258761041	10.34318814	-0.8948249758	0.3736431271	-29.85306585	11.33554377

Step 3

(c) I will use Excel to determine the correlation matrix of the given data, which results in the following table:

	Y	X_1	X_2	X_3
Y	1			
X_1	0.4648	1		
X_2	0.2315	0.2535	1	
X_3	-0.1178	-0.0677	-0.0567	1

$$b_1 = 8.3779$$

We note that the largest correlation of 0.4648 between the dependent variable and an independent variable belongs to the variable X_1 (age). However, all correlations with the dependent variable are weak as they are all less than 0.5.

The estimate of the coefficient b_2 of the variable x_2 is given in the row "X variable 2" and in the column "Coefficients" of the computer output:

$$b_2 = 0.0460$$

Moreover, we note that none of the correlation between two independent variables (X_i) are large (less than -0.7 or larger than 0.7) and thus it is safe to assume that there are no problems with multicollinearity.

The estimate of the coefficient b_3 of the variable x_3 is given in the row "X variable 3" and in the column "Coefficients" of the computer output:

$$b_3 = -9.2588$$

Combining these estimates, we then obtain the multiple regression equation:

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned}\hat{y} &= a + b_1 x_1 + b_2 x_2 + b_3 x_3 \\ &= 358.7239 + 8.3779 x_1 + 0.0460 x_2 - 9.2588 x_3\end{aligned}$$

Step 4

(d)

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$H_0 : \text{All coefficients are zero}$$
$$H_1 : \text{At least one of the coefficients is nonzero}$$

The P-value of the global test is given in the row "Regression" and in the column "Significance F" of the ANOVA table:

$$P \approx 0.0001$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.0001 < 0.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

Step 5

(e)

$$\alpha = \text{Significance level} = 0.05$$

The P-values of the coefficients of the variables are given in the column "P-value" of the regression output.

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$x_1 \quad P = 0.00009 < 0.05 \Rightarrow \text{ Reject } H_0$$
$$x_2 \quad P = 0.2584 > 0.05 \Rightarrow \text{ Fail to reject } H_0$$
$$x_3 \quad P = 0.4169 > 0.05 \Rightarrow \text{ Fail to reject } H_0$$

There is sufficient evidence to support the claim that the coefficient of age is nonzero. However, there is not sufficient evidence to support the claim that the coefficients of miles and bus type are nonzero.

Since two of the coefficients appear to be zero, we would considering deleting the two corresponding variables (miles and bus type).

Step 6

(f) I will use Excel to determine the regression output, which is given in the image below. We start by deleting the variable with the largest P-value, which is variable x_3 (bus type).

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.4794697932					
R Square	0.2298912826					
Adjusted R Square	0.2098884588					
Standard Error	47.720492					
Observations	80					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	2	52344.49501	26172.2475	11.49294143	0.000042879875	
Residual	77	175347.8925	2277.245357			
Total	79	227692.3875				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	353.9029297	31.58874056	11.20345172	0	291.0017122	416.8041472
X Variable 1	8.469119739	2.017307758	4.198228904	0.000071523818	4.452146863	12.48609262
X Variable 2	0.04732311689	0.04025244525	1.175658189	0.2433538297	-0.03282974036	0.1274759741

The estimate of the constant a is given in the row "Intercept" and in the column "Coefficients" of the computer output:

$$a = 353.9029$$

The estimate of the coefficient b_1 of the variable x_1 is given in the row "X Variable 1" and in the column "Coefficients" of the computer output:

$$b_1 = 8.4691$$

The estimate of the coefficient b_2 of the variable x_2 is given in the row "X variable 2" and in the column "Coefficients" of the computer output:

$$b_2 = 0.0473$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned}\hat{y} &= a + b_1 x_1 + b_2 x_2 \\ &= 353.9029 + 8.4691 x_1 + 0.0473 x_2\end{aligned}$$

We note that the P-value of the variable x_2 is still large (0.2424) and thus we will delete variable x_2 (miles) from the model as well.

Step 7

I will use Excel to determine the regression output, which is given in the image below. We delete the variables x_3 (bus type) and x_2 (miles) from the model.

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.4648307319					
R Square	0.2160676093					
Adjusted R Square	0.2060171941					
Standard Error	47.83725585					
Observations	80					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	1	49196.94983	49196.94983	21.4983763	0.000014001845	
Residual	78	178495.4377	2288.403047			
Total	79	227692.3875				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	386.7958612	14.70089537	26.31104103	0	357.5286272	416.0630952
X Variable 1	9.070234114	1.956210865	4.636634156	0.000014001845	5.175717511	12.96475072

The estimate of the constant a is given in the row "Intercept" and in the column "Coefficients" of the computer output:

$$a = 386.7959$$

The estimate of the coefficient b_1 of the variable x_1 is given in the row "X Variable 1" and in the column "Coefficients" of the computer output:

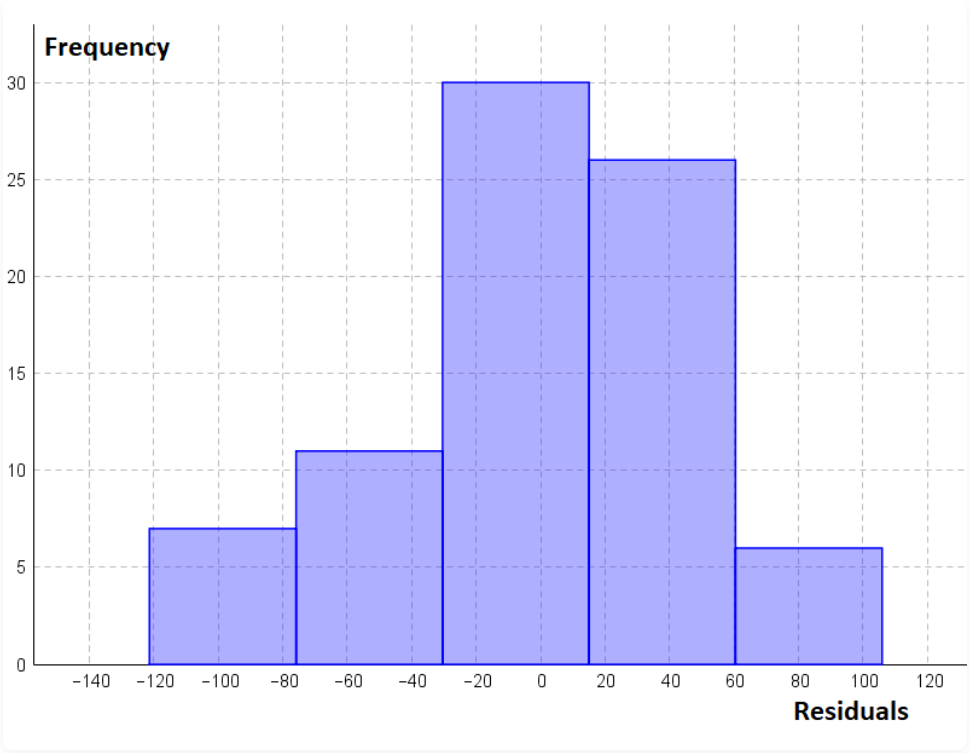
$$b_1 = 9.0702$$

Combining these estimates, we then obtain the multiple regression equation:

$$\begin{aligned} \hat{y} &= a + b_1 x_1 \\ &= 386.7959 + 9.0702 x_1 \end{aligned}$$

Step 8

(g)

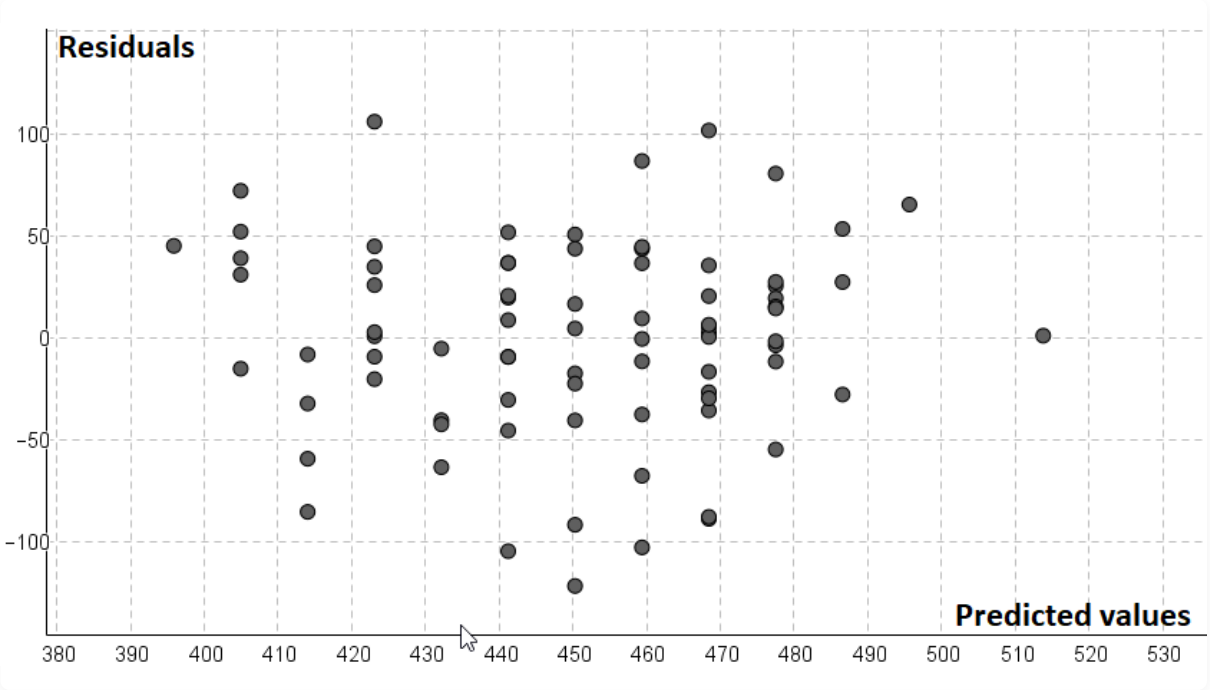


The highest bars in the histogram are roughly in the middle of the histogram, which implies that the histogram is approximately normally distributed.

This then implies that is it reasonable to conclude that the normality assumption has been met.

Step 9

(h)



The residuals appear to be randomly scattered in the residual plot and there is no noticeable pattern in the points.

Moreover, the vertical spread appears to be approximately the same everywhere in the residual plot.

This then indicates that none of the regression assumptions appear to be violated.

Result

(a) $\hat{y} = 358.7239 + 8.3779x_1 + 0.0460x_2 - 9.2588x_3$

(b) 23.66%

(c) All are weak, No

(d) There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

(e) Yes, miles and bus tyope

(f) $\hat{y} = 386.7959 + 9.0702x_1$

(g) Yes

(h) None of the regression assumptions appear to be violated.

Statistics

Chapter 14 — Instructor Solutions Manual

October 24, 2026 • Total Exercises: 3

PROBLEM STATEMENT

Chapter 14, Page 569

Solution A

Step 1

Given:

$$\begin{aligned}\hat{y} &= 67 + 8x_1 - 10x_2 + 0.004x_3 - 3x_4 \\ x_1 &= \text{number of employees} \\ x_2 &= \text{overhead cost} \\ x_3 &= \text{average markup} \\ x_4 &= \text{theft loss} \\ y &= \text{net profit}\end{aligned}$$

(a) A **response variable** (or dependent variable) is a variable that measures an outcome or result of a study, while an **explanatory variable** (or independent variable) is a variable that we think explains or causes changes in the response variables.

We are interested in estimating the profit in this case, which implies that the dependent variable is the net profit.

Step 2

(b) The full name of the given equation is the **multiple regression equation**.

$$\hat{y} = a + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4$$

Step 3

(c) Evaluate the multiple regression equation at $x_1 = 112$, $x_2 = 65$ (thousand), $x_3 = 50$, and $x_4 = 50$ (thousand).

$$\begin{aligned}\hat{y} &= 67 + 8x_1 - 10x_2 + 0.004x_3 - 3x_4 \\ &= 67 + 8(112) - 10(65) + 0.004(50) - 3(50) \\ &= 163.2\end{aligned}$$

Thus the predicted net sales is 163.2 thousand dollars or $\$163,200$.

Step 4

(d) Given:

$$R^2 \approx 0.86 = 86\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

86% of the variation in the net profit is explained by the variation in the independent variables.

Step 5

(e) Given:

$$s = 3$$

The standard error of the estimate is the average deviation between the actual y-value and the predicted y-value.

Thus the estimated net profit deviates on average by 3 thousand dollars from the actual net profit.

🎯 Result

(a) Net profit

(b) $\hat{y} = a + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4$

(c) 163.2 thousand dollars

(d) 86% of the variation in the net profit is explained by the variation in the independent variables.

(e) The estimated net profit deviates on average by 3 thousand dollars from the actual net profit.

PROBLEM STATEMENT

Chapter 14, Page 569

Solution A

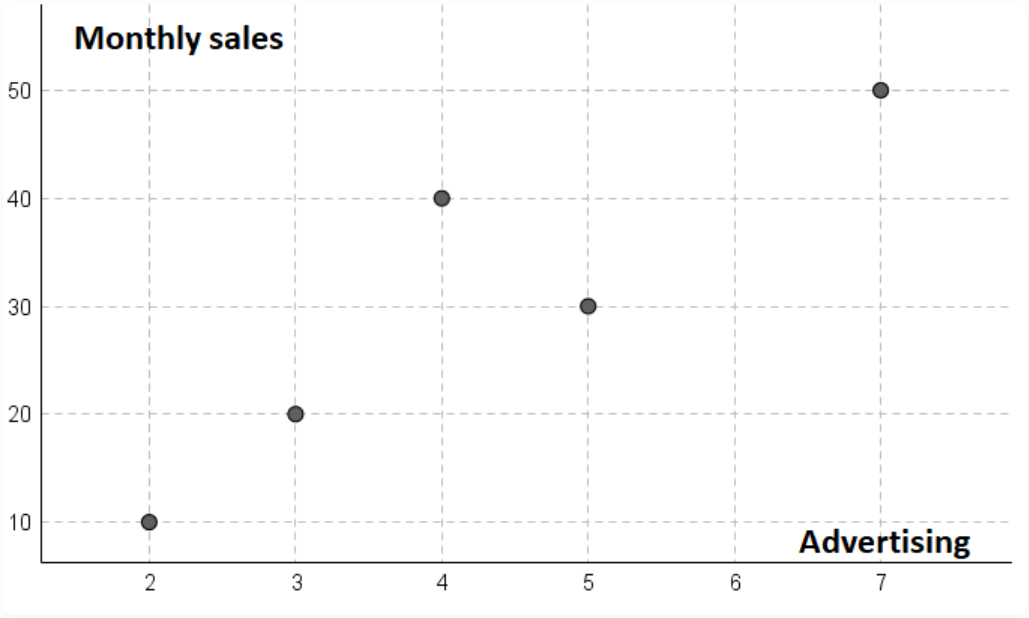
Step 1

Given:

$$n = \text{Sample size} = 5$$

(a) **Scatter diagram**

Annual Bus Bench Advertising is on the horizontal axis and Monthly Sales is on the vertical axis.



Step 2

(b) **Correlation coefficient**

Formula correlation coefficient:

$$r = \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y}$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{2+4+5+7+3}{5} = 4.2$$
$$\overline{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{10+40+30+50+20}{5} = 30$$

The variance is the sum of squared deviations from the mean divided by $n-1$. The standard deviation is the square root of the variance:

$$s_x = \sqrt{\frac{(2-4.2)^2 + ... + (3-4.2)^2}{5-1}} \approx 1.9235$$
$$s_y = \sqrt{\frac{(10-30)^2 + ... + (20-30)^2}{5-1}} \approx 15.8114$$

Step 3

Next, we determine the deviation from the mean for each data value and the product of these deviations for each pair (x, y).

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$
2	10	-2.2	-20	44
4	40	-0.2	10	-2
5	30	0.8	0	0
7	50	2.8	20	56
3	20	-1.2	-10	12
			Total	110

Finally, we evaluate the formula of the correlation coefficient:

$$\begin{aligned} r &= \frac{\sum(X - \overline{X})(Y - \overline{Y})}{(n-1)s_x s_y} \\ &= \frac{110}{(5-1)(1.9235)(15.8114)} \\ &\approx 0.9042 \end{aligned}$$

Step 4

(c) **Coefficient of determination**

The coefficient of determination is the square of the correlation coefficient r:

$$r^2 = 0.9042^2 \approx 0.8176 = 81.76\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

81.76% of the variation in the monthly sales is explained by the variation in the annual bus bench advertising.

Step 5

(d) **Regression equation**

The estimate b of the slope is the product of the correlation coefficient and the ratio of the standard deviations.

$$b = r\frac{s_y}{s_x} = 0.9042\frac{15.8114}{1.9235} = 7.4324$$

The estimate a of the intercept is the average of y decreased by the product of the estimate of the slope and the average of x.

$$a = \overline{y} - b\overline{x} = 30 - 7.4324 \cdot 4.2 = -1.2162$$

General least-squares equation: $\hat{y} = a + bx$

$$\hat{y} = a + bx = -1.2162 + 7.4324x$$

Step 6

(e) Evaluate the least-squares equation found in part (d) at $x = 4.5$ (thousand):

$$\hat{y} = a + bx = -1.2162 + 7.4324(4.5) = 32.2297$$

Thus the estimated monthly sales is 32.2297 thousand dollars or $\$32,229.70$.

Step 7

(f) 81.76% of the variation in the monthly sales is explained by the variation in the annual bus bench advertising.

Since 81.76% is high, the regression equation is appropriate to make predictions.

Result

(a) See Scatter diagram

(b) $r = 0.9042$

(c) 81.76%

(d) $\hat{y} = -1.2162 + 7.4324x$

(e) 32.2297 thousand dollars

(f) Summarize part (a) to (e)

PROBLEM STATEMENT

Chapter 14, Page 569

Solution A

Step 1

(a) The regression sum of squares is given in the row "Regression" and in the column "SS" of the ANOVA table:

$$SSR = 1050.8$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table:

$$SS\ total = 1134.6$$

The coefficient of multiple determination is the regression sum of squares divided by the total sum of squares:

$$r^2 = \frac{SSR}{SS\ total} = \frac{1050.8}{1134.6} \approx 0.9261 = 92.61\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

92.61% of the variation in the dependent variable is explained by the variation in the independent variables.

Step 2

(b) The error sum of squares is given in the row "Error" and in the column "SS" of the ANOVA table:

$$SSE = 83.8$$

The error degrees of freedom is given in the row "Error" and in the column "DF" of the ANOVA table:

$$df_E = 20$$

The standard error of the estimate is the square root of the error sum of squares divided by the error degrees of freedom:

$$s = \sqrt{\frac{SSE}{n-(k+1)}} = \sqrt{\frac{SSE}{df_E}} = \sqrt{\frac{83.8}{20}} \approx 2.0469$$

Step 3

(c)

$$\alpha = \text{Significance level} = 0.05$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$H_0 : \text{All coefficients are zero}$$
$$H_1 : \text{At least one of the coefficients is nonzero}$$

The F- critical value is given in the row with $dfd = df_E = 20$ and in the column with $dfn = df_R = 4$ of the F distribution-table $\alpha = 0.05$ in the appendix :

$$F_\alpha = 2.87$$

Thus we reject the null hypothesis H_0 when the value of the test statistic is greater than 2.87.

The value of the test statistic is the ratio of the regression mean square and the mean square error, which are the values given in the column "MS" of the ANOVA table.

$$F = \frac{MSR}{MSE} = \frac{262.70}{4.19} \approx 62.6969$$

Since 62.6969 is greater than 2.87, we reject the null hypothesis and thus there is sufficient evidence to support the claim that at least one of the coefficients is nonzero.

Step 4

(d) Determine the critical values from the Student's T distribution table in the appendix in the row with $df = df_E = 20$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.086$$

The rejection region then contains all values smaller than -2.086 and all values greater than 2.086, thus we reject H_0 when $t < -2.086$ or $t > 2.086$.

When the t-ratio is then between -2.086 and 2.086, then we fail to reject the null hypothesis of a zero coefficient and thus the corresponding variable can then be deleted from the model. In this case, we note that variable X_2 has a t-ratio between -2.086 and 2.086, thus variable X_2 can be deleted from the model.

🎯 **Result**

(a) 92.61%

(b) $s = 2.0469$

(c) There is sufficient evidence to support the claim that at least one of the coefficients is nonzero.

(d) Yes

Statistics

Chapter 14 — Instructor Solutions Manual

October 24, 2026 • Total Exercises: 2

PROBLEM STATEMENT

Chapter 14, Page 571

Solution A

Step 1

(a) The total degrees of freedom is given in the row "Total" and in the column "DF" of the ANOVA table:

$$df_{total} = 29$$

The total degrees of freedom is the sample size decreased by 1, thus the sample size is obtained by increasing the total degrees of freedom increased by 1.

$$n = df_{total} + 1 = 29 + 1 = 65$$

Step 2

(b) The estimate of the constant a is given in the row "Constant" and in the column "Coefficients" of the computer output:

$$a = 90.6190$$

The estimate of the coefficient b_1 of the variable x is given in the row "Slope" and in the column "Coefficients" of the computer output:

$$b = -0.9401$$

Combining these estimates, we then obtain the multiple regression equation:

$$\hat{y} = a + bx = 90.6190 - 0.9401x$$

The estimated value of the dependent variable is 90.6190 when the independent value is 0.

The dependent value decreases on average by 0.9401 per unit of the dependent variable.

Step 3

(c) Evaluate the multiple regression equation from part (b) at $x = 10$:

$$\hat{y} = a + bx = 90.6190 - 0.9401(10) = 81.2180$$

Thus the value of the dependent variable is estimated to be 81.2180.

Step 4

(d) The regression sum of squares is given in the row "Regression" and in the column "SS" of the ANOVA table:

$$SSR = 129.7275$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table:

$$SS\ total = 380.1667$$

The coefficient of multiple determination is the regression sum of squares divided by the total sum of squares:

$$r^2 = \frac{SSR}{SS\ total} = \frac{129.7275}{380.1667} \approx 0.3412 = 34.12\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

34.12% of the variation in the dependent variable is explained by the variation in the independent variables.

Step 5

(e) The estimate of the coefficient b_1 of the variable x is given in the row "Slope" and in the column "Coefficients" of the computer output:

$$b = -0.9401$$

Since the slope $b = -0.9401$ is negative, there is a negative relationship between the variables and thus the correlation coefficient needs to be negative as well.

The correlation coefficient is then the negative square root of the coefficient of determination.

$$r = -\sqrt{r^2} = -\sqrt{0.3412} \approx$$

Step 6

Given claim: Negative association

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes that the correlation coefficient is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \rho \geq 0 \text{ (There is no linear correlation)}$$
$$H_a : \rho < 0 \text{ (There is a negative linear correlation)}$$

Step 7

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Left-tailed}$$

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all t-value below the t-value $-t_0$ that has a probability of 0.05 to its left.

$$P(t < -t_0) = 0.10$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 2 = 30 - 2 = 28$ and in the column with $\alpha = 0.05$ (which is the negation of the critical value in the table).

$$t = -1.701$$

The rejection region then contains all values smaller than -1.701 and thus we reject H_0 when the t-value is smaller than -1.701.

The value of the test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{-0.5841\sqrt{28-2}}{\sqrt{1-(-0.5841)^2}} \approx -3.669$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$t = --3.669 < -1.701 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a negative association between the variables.

🎯 Result

(a) $n = 30$

(b) $\hat{y} = -0.6190 - 0.9401x$

(c) 81.2180

(d) 34.12% of the variation in the dependent variable is explained by the variation in the independent variables.

(e) $r = -0.5841$, There is sufficient evidence to support the claim of a negative association between the variables.

PROBLEM STATEMENT

Chapter 14, Page 572

Solution A

Step 1

(a) The total degrees of freedom is given in the row "Total" and in the column "DF" of the ANOVA table:

$$df_{total} = 29$$

The total degrees of freedom is the sample size decreased by 1, thus the sample size is obtained by increasing the total degrees of freedom increased by 1.

$$n = df_{total} + 1 = 29 + 1 = 30$$

Step 2

(b) The regression degrees of freedom is given in the row "Regression" and in the column "DF" of the ANOVA table:

$$df_R = 4$$

The regression degrees of freedom represent the number of independent variables in the study and thus the study contains 4 independent variables in this case.

Step 3

(c) The regression sum of squares is given in the row "Regression" and in the column "SS" of the ANOVA table:

$$SSR = 227.0928$$

The total sum of squares is given in the row "Total" and in the column "SS" of the ANOVA table:

$$SS\ total = 380.1667$$

The coefficient of multiple determination is the regression sum of squares divided by the total sum of squares:

$$r^2 = \frac{SSR}{SS\ total} = \frac{227.0928}{380.1667} \approx 0.5974 = 59.74\%$$

The coefficient of determination measures the proportion of variation in the y-variable that is by the variation in the x-variable.

59.74% of the variation in the dependent variable is explained by the variation in the independent variables.

Step 4

(d)

$$\alpha = \text{Significance level} = 0.01$$

The null hypothesis states that all coefficients are equal to 0, while the alternative hypothesis states that at least one of the coefficients is nonzero.

$$H_0 : \text{All coefficients are zero}$$
$$H_1 : \text{At least one of the coefficients is nonzero}$$

The P-value of the global test is given in the row "Regression" and in the column "p-value" of the ANOVA table:

$$P = 0.000$$

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$P = 0.000 < 0.01 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

Step 5

(e)

$$\alpha = \text{Significance level} = 0.05$$

The P-values are given in the column "p-value" of the regression output.

If the P-value is less than the significance level, then we reject the null hypothesis H_0:

$$\begin{aligned} x_1 \quad & P = 0.010 < 0.05 \Rightarrow \text{ Reject } H_0 \\ x_2 \quad & P = 0.690 > 0.05 \Rightarrow \text{ Fail to reject } H_0 \\ x_3 \quad & P = 0.006 < 0.05 \Rightarrow \text{ Reject } H_0 \\ x_4 \quad & P = 0.030 < 0.05 \Rightarrow \text{ Reject } H_0 \end{aligned}$$

There is not sufficient evidence to support the claim that the coefficient of X_2 is nonzero, while there is sufficient evidence to support the claim that all other coefficients are nonzero.

Since the coefficient of X_2 appears to be approximately zero, we would consider eliminating variable X_2.

Step 6

(f) We start by deleting the independent variable with the highest P-value, which is the independent variable X_1.

We determine the regression output for the model with independent variable X_1 deleted. If any of the independent variables still has a P-value greater than 0.05 in the new regression model, then we proceed with deleting this variable from the model as well.

Result

(a) $n = 30$

(b) 4 independent variables

(c) 59.74%

(d) There is sufficient evidence to support the claim that at least of the regression coefficients is different from 0.

(e) Yes, X_2

Statistics

Chapter 14 — Instructor Solutions Manual

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Chapter 14 **Exercise 1** Page 571

PROBLEM STATEMENT

Chapter 14, Page 571

Solution A

Step 1

A **response variable** (or dependent variable) is a variable that measures an outcome or result of a study, while an **explanatory variable** (or independent variable) is a variable that we think explains or causes changes in the response variables.

The independent variable (x-variable) is always scaled on the horizontal axis, while the dependent variable (y-variable) is always scaled on the $vertical$ axis.

Result

Vertical axis

PROBLEM STATEMENT

Chapter 14, Page 571

Solution A

Step 1

Level of measurement

The level of measurement is **nominal**, if the values of the variables can only be named and not ordered.

The level of measurement is **ordinal**, if the values of the variables can be named and ordered, but the difference between two values has no meaning.

The level of measurement is **interval**, if the difference between two values has a meaning, but the variable contains no absolute zero.

The level of measurement is **ratio**, if the variable contains an absolute zero.

The correlation coefficient can only be calculated when the data values are numerical (with numerical meaning). This then also implies that the difference between two data values needs to have meaning as well (as we use differences in the calculation of the correlation coefficient) and thus we require at least the *interval* level of measurement for the data (that is, the level of measurement of both variables is either interval and/or ratio).

Result

Interval

PROBLEM STATEMENT

Chapter 14, Page 571

Solution A

Step 1

The correlation coefficient measures the strength of the linear relationship between two variables.

When there is no correlation between two variables, then this implies that there is no linear relationship between the variables and thus the correlation coefficient is approximately **zero**.

Result

0

PROBLEM STATEMENT

Chapter 14, Page 571

Solution A

Step 1

The correlation coefficient measures the strength of the linear relationship between two variables.

Moreover, the correlation coefficient can take on values from -1 to $+1$.

The strongest correlation between two variables is then the correlation that is the largest in absolute value (that is, the value that is closest to -1 or $+1$).

In this case, we note that the correlation coefficient -0.77 is the largest in absolute value and thus the value -0.77 indicates the strongest correlation between two variables.

Result

$$-0.77$$

PROBLEM STATEMENT

Chapter 14, Page 571

Solution A

Step 1

The correlation coefficient r can take on values from -1 to $+1$.

Since the coefficient of determination r^2 is the square of the correlation coefficient r, the coefficient of determination can take on values from 0 to 1.

This then implies that it is $\mathbf{impossible}$ that the coefficient of determination is greater than 1.

Result

Impossible

PROBLEM STATEMENT

Chapter 14, Page 571

Solution A

Step 1

Given:

$$\hat{y} = 7 - 0.5x$$
$$r^2 = \text{coefficient of determination} = 0.81 = 81\%$$

The point at which the regression equation crosses the y-axis always has x-coordinate 0 (as the y-axis is the line $x = 0$).

The y-coordinate of this point is the y-intercept, which is the constant in the regression equation and thus the y-coordinate of this point is 7.

This then implies that the point where the regression equation crosses the y-axis is $(0, 7)$.

Result

$$(0, 7)$$

PROBLEM STATEMENT

Chapter 14, Page 571

Solution A

Step 1

Given:

$$\hat{y} = 7 - 0.5x$$
$$r^2 = \text{coefficient of determination} = 0.81 = 81\%$$

The slope of the regression equation is the coefficient of x:

$$b = -0.5$$

The slope of the regression equation represents the average increase in the dependent y-variable per unit of the independent x-variable.

Thus we expect the dependent variable to decrease by 0.5 units when the independent variable increases by 1 unit.

Result

Decrease by 0.5 units

PROBLEM STATEMENT

Chapter 14, Page 571

Solution A

Step 1

Given:

$$\hat{y} = 7 - 0.5x$$
$$r^2 = \text{coefficient of determination} = 0.81 = 81\%$$

The slope of the regression equation is the coefficient of x:

$$b = -0.5$$

Since the slope $b = -0.5$ is negative, there is a negative relationship between the variables and thus the correlation coefficient needs to be negative as well.

The correlation coefficient r is then the negative square root of the coefficient of determination.

$$r = -\sqrt{r^2} = -\sqrt{0.81} = -0.9$$

Result

$$r = -0.9$$

PROBLEM STATEMENT

Chapter 14, Page 571

Solution A

Step 1

Given:

$$\hat{y} = 7 - 0.5x$$
$$r^2 = \text{coefficient of determination} = 0.81 = 81\%$$

The standard error of the estimate measures the average deviation between the data points and the regression line.

When all data points are on the regression line, then there are no deviations between the data points and the regression line, which implies that the standard error of the estimate then needs to be 0.

Result

0

Chapter 14 Exercise 10

Page 571

PROBLEM STATEMENT

Chapter 14, Page 571

Solution A

Step 1

There is no limitation on the number of independent variables that you are allowed to use in a multiple regression equation and thus the maximum number of independent variables allowed is **unlimited**.

Result

Unlimited

Chapter 14 Exercise 11

Page 571

PROBLEM STATEMENT

Chapter 14, Page 571

Solution A

Step 1

We always assume that there exists a **linear** relationship between the dependent variable and the set of independent variables within a multiple regression analysis.

Result

Linear

Chapter 14 Exercise 12

Page 571

PROBLEM STATEMENT

Chapter 14, Page 571

Solution A

Step 1

A **residual** is the difference between the observed Y-value and the predicted Y-value $\hat{Y}$.

Result

Residual

Chapter 14 Exercise 13

Page 571

PROBLEM STATEMENT

Chapter 14, Page 571

Solution A

Step 1

A dummy variable always has **two** possible outcomes: 0 and 1.

1 represents the occurrence of some event, while 0 represents that the event does not occur.

Result

Two

Chapter 14 Exercise 14

Page 571

PROBLEM STATEMENT

Chapter 14, Page 571

Solution A

Step 1

A **correlation matrix** is a table that shows all possible correlation coefficients between the dependent variable and each independent variable, and between each pair of independent variables.

Result

Correlation matrix

Chapter 14 Exercise 15

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PROBLEM STATEMENT

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Solution A

Step 1

A graph of the residuals will have approximately a **normal distribution** when there exists a linear relationship between the dependent variable and the set of independent variables.

Result

Normal distribution